11. The value of potential barrier across a depletion region 2·0 μm wide is 0·4 V. The value of the electric field in this region is : *1*

(A) $5 \times 10^{6}\ \text{Vm}^{-1}$

(B) $5 \times 10^{-6}\ \text{Vm}^{-1}$

(C) $0{\cdot}8 \times 10^{6}\ \text{Vm}^{-1}$

(D) $2{\cdot}0 \times 10^{5}\ \text{Vm}^{-1}$

CBSE 2026C

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(D) $2{\cdot}0 \times 10^{5}\ \text{Vm}^{-1}$

CBSE 2026C

(D) $2.0 \times 10^{5}\ V\,m^{-1}$

12. Which of the following element can be used as dopant in Si crystal to obtain a p-type semiconductor ? *1*

(A) Phosphorus (B) Arsenic

(C) Boron (D) Antimony

CBSE 2026C

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(A) Phosphorus

(B) Arsenic

(C) Boron

(D) Antimony

CBSE 2026C

(C) Boron

28. Explain the working of a full-wave rectifier. Differentiate between a half-wave rectifier and a full-wave rectifier. *3*

CBSE 2026C

28. Explain the working of a full-wave rectifier. Differentiate between a half-wave rectifier and a full-wave rectifier. *3*

CBSE 2026C

In full wave rectifier two diodes are joined in the circuit with a centre tap. During the positive half cycle, one of the diodes become forward biased and conducts, whereas the other one being reverse biased does not conduct. During the negative half cycle, first one get reverse biased and does not conduct, whereas the other one being forward biased conducts. **Half Wave Rectifier** / **Full Wave Rectifier** Rectified output is obtained only during one half of the AC cycle. / Output voltage is obtained during both positive and negative half cycles.	½ ½ 1+1

11. In forward and reverse biased silicon p-n junction, the dominant mechanisms for the motion of charge carriers are :

(A) diffusion in forward bias, drift in reverse bias

(B) drift in both forward bias and in reverse bias

(C) drift in forward and diffusion in reverse bias

(D) diffusion in both forward and reverse bias

CBSE 2026C

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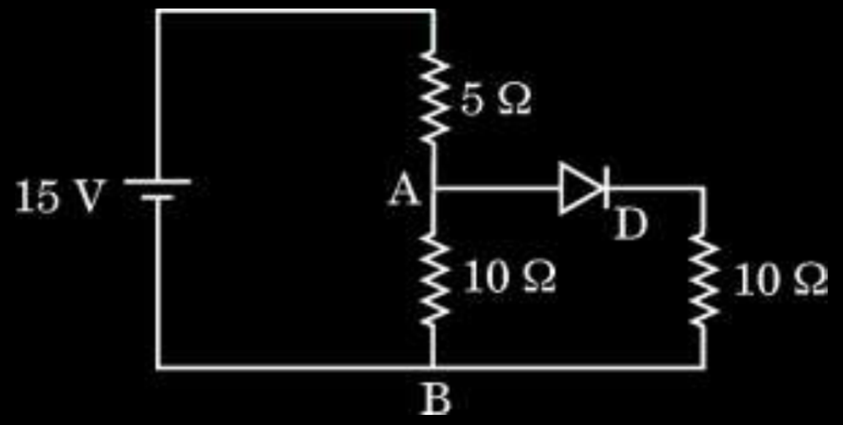
CBSE 2026C

(A) diffusion in forward bias, drift in reverse bias

22. (a) A diode is connected to 220 V (rms) ac source in series with a capacitor as shown in the diagram.

Find the voltage (rms) across the capacitor. *1*

(b) Find V_{AB} in the circuit given below. *2*



CBSE 2026C

(a)	The diode will conduct for the positive half cycle of AC supply because it is forward biased. Negative half cycle of AC supply, this is reversed biased and does not conduct. So the output would be half wave rectified output. $V_{rms} = \frac{V_0}{2} = \frac{220\sqrt{2}}{2} = \frac{220}{\sqrt{2}}$ volt	1
	Note : 1 mark should be awarded on attempting this part of the question.	
(b)	$\frac{1}{R_{AB}} = \frac{1}{R_1} + \frac{1}{R_2} = \frac{1}{10} + \frac{1}{10}$ $R_{AB} = 5\ \Omega$ $R_{\text{net}} = 5 + 5 = 10\ \Omega$ $I = \frac{15}{10} = 1.5$ A $V_{AB} = IR_{AB}$ $= 1.5 \times 5$ $= 7.5$ V	½ ½ ½ ½

25. Draw a labelled diagram of a full wave rectifier circuit. State its working principle and show the input-output waveforms. CBSE 2026C *3*

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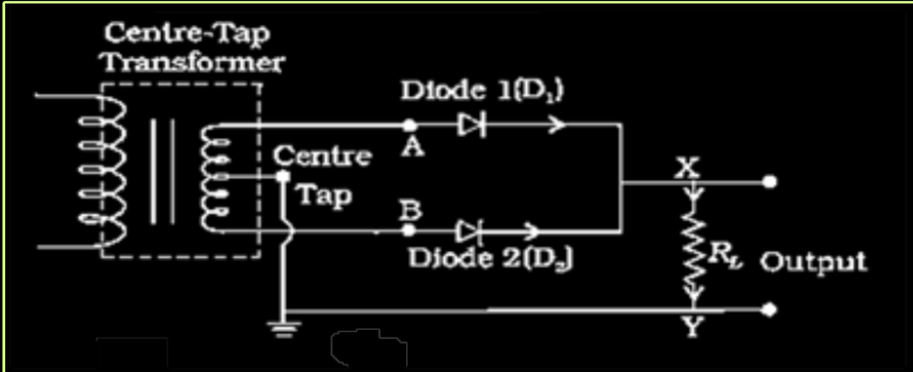
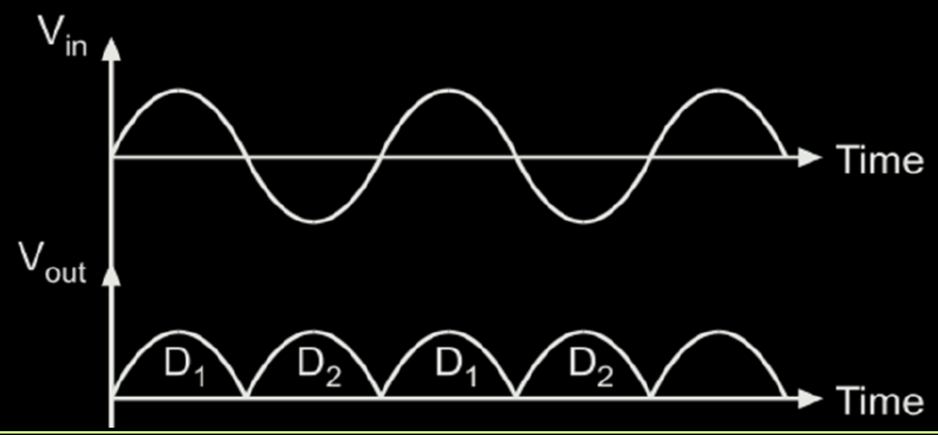


1

Working principle : A p-n junction diode allows current to pass only when it is forward biased.

1

Alternatively: A p-n junction diode offers low resistance when forward biased.



½

½

11. A good diode checked by a multimeter should indicate :

(A) high resistance in reverse bias and a low resistance in forward bias

(B) high resistance in both forward bias and reverse bias

(C) low resistance in both reverse bias and forward bias

(D) high resistance in forward bias and low resistance in reverse bias

CBSE 2026

11. A good diode checked by a multimeter should indicate :

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(B) high resistance in both forward bias and reverse bias

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CBSE 2026

(A) high resistance in reverse bias and low resistance in forward bias

21. Suppose a pure Si crystal has 5×10^{28} atoms per m^3. It is doped with 5×10^{22} atoms per m^3 of Arsenic. Calculate the majority and minority carrier concentration in the doped silicon. (Given : $n_i = 1{\cdot}5 \times 10^{16}\ m^{-3}$) *2*

CBSE 2026

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CBSE 2026

Majority carrier concentration = Doping concentration	
∴ Majority carrier concentration (n_e)=5 x10^{22} atoms per m^3	½
$n_i^2 = n_e n_h$	½
$n_h = \frac{n_i^2}{n_e}$	
$n_h = \frac{(1.5 \times 10^{16})^2}{5 \times 10^{22}}$	½
$=4.5 \times 10^9$ atoms per m^3	
Minority carrier concentration = 4.5 x 10^9 atoms per m^3	½

28. Draw a circuit diagram of a full-wave rectifier using p-n junction diodes. Explain its working and show the input-output waveforms. *3*

CBSE 2026

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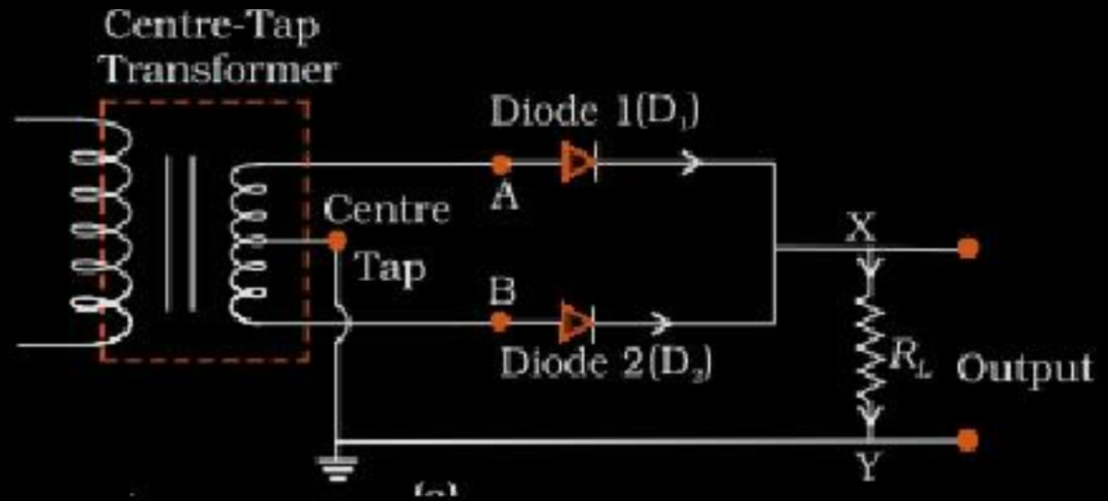
CBSE 2026



1

Working:

Suppose the input voltage to A w.r.t centre top at any instant is positive.

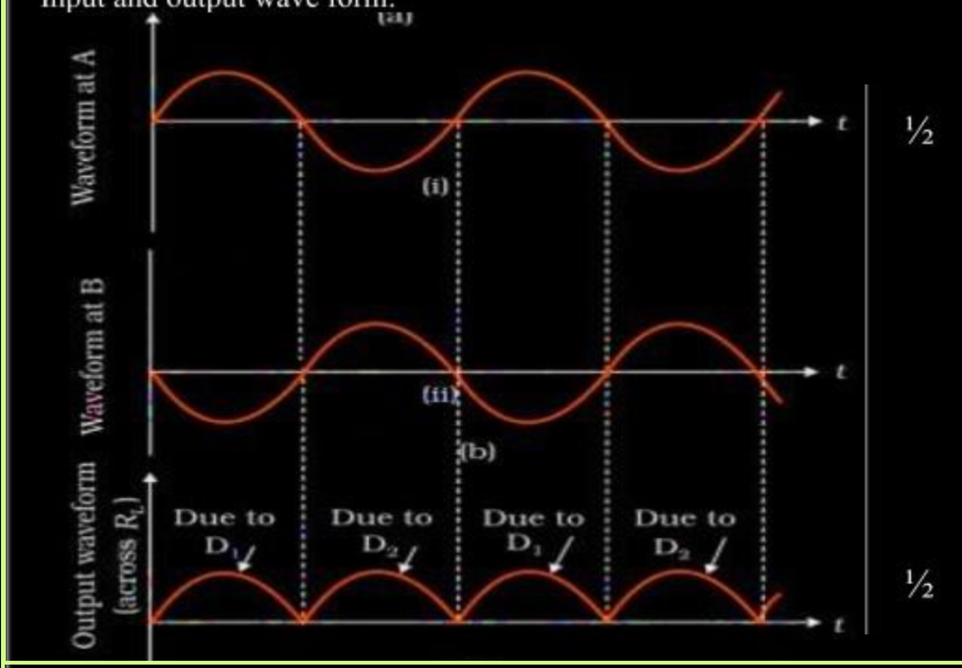
At that instant voltage at B being out of phase will be negative.

So, diode D_1 gets forward biased and conducts while D_2 being reverse biased is not conducting. Hence, during this positive half cycle we get an output current and output voltage across the load.

1

When the voltage at A becomes negative w.r.t centre top, the voltage at B would be positive, the diode D_1, would not conduct but D_2 conduct and giving an output current and voltage across the load

Input and output wave form:



½

½

12. In a reversed-biased p-n junction diode, the applied voltage mostly drops across :

(A) p-region only

(B) n-region only

(C) depletion region

(D) the diode

CBSE 2026

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(A) p-region only

(B) n-region only

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(D) the diode

CBSE 2026

(C) Depletion region

16. *Assertion (A) :* The conductivity of an n-type semiconductor is higher than that of a p-type semiconductor at a given temperature.

Reason (R) : The electrons being in the conduction band in n-type semiconductor are more mobile than the holes in the valence band in p-type semiconductor. CBSE 2026

16. *Assertion (A) :* The conductivity of an n-type semiconductor is higher than that of a p-type semiconductor at a given temperature.

Reason (R) : The electrons being in the conduction band in n-type semiconductor are more mobile than the holes in the valence band in p-type semiconductor. CBSE 2026

(A) Both Assertion (A) and Reason (R) are true, and Reason (R) is the correct explanation of Assertion (A).

21. The hole concentration in an intrinsic semiconductor is 5×10^8 m^{-3}. When it is doped with certain impurity, the electron concentration becomes 4×10^{12} m^{-3}. Find the new value of the hole concentration. Also identify the type of new semiconductor formed after doping. CBSE 2026 *2*

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$n_e n_h = n_i^2$	½
Hole concentration $(n_h) = \frac{n_i^2}{n_e}$	½
$= \frac{5\times10^8\times5\times10^8}{4\times10^{12}}$	
$n_h = 6.25 \times 10^4$ m^{-3}	½
The new semiconductor formed is n-type semiconductor.	½

28. With the help of a circuit diagram, explain the working of a full wave rectifier. Depict the input and output waveforms. CBSE 2026 3

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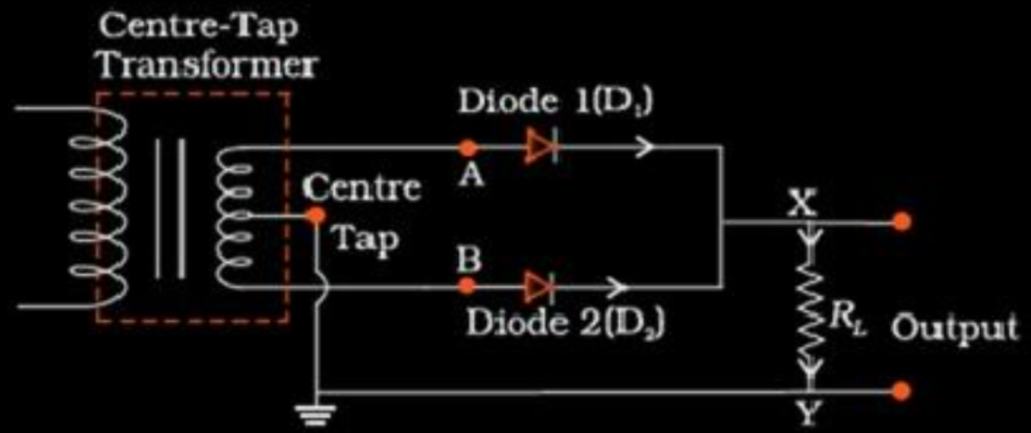
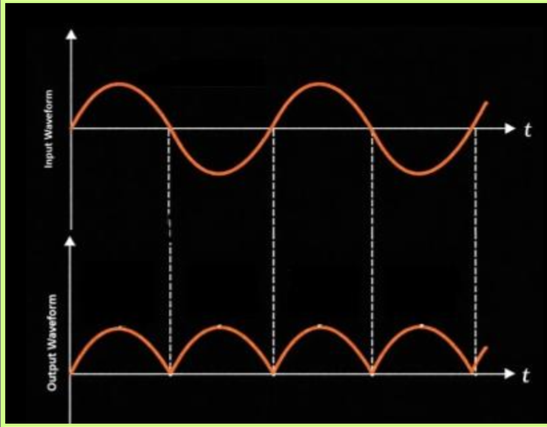


1

Let the input voltage A with respect to the centre tap at any instant is positive. At that instant, voltage at B being out of phase will be negative. So, diode D_1 gets forward biased and conducts (while D_2 being reverse biased is not conducting). Hence, during this positive half cycle we get an output current. When voltage at A becomes negative w.r.t. centre tap, the voltage at B would be positive. In this part of the cycle diode D_1 would not conduct but diode D_2 would be giving output current.

1



½

½

11. When the forward bias voltage in a semiconductor diode is changed from 0·8 V to 1·0 V, the forward current changes by 2·0 mA. The forward bias resistance of the diode will be :

(A) 200 Ω (B) 175 Ω

(C) 100 Ω (D) 125 Ω

CBSE 2026

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(A) 200 Ω (B) 175 Ω

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CBSE 2026

(C) 100 Ω

12. The process named 'minority carrier injection' in a p-n junction diode occurs during :

(A) forward biasing

(B) reverse biasing

(C) no biasing at low temperature

(D) no biasing at high temperature

CBSE 2026

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CBSE 2026

(A) forward biasing

21. How are charge carriers created in an intrinsic semiconductor ? Explain. *2*

CBSE 2026

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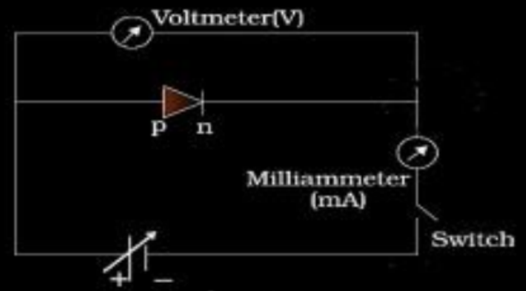
CBSE 2026

At low temperatures, all the covalent bonds in an intrinsic semiconductor remain intact. As temperature increases, thermal energy becomes available and electrons may break away, becoming free electrons available for conduction.	1
The neighbourhood from which the free electron has come out leaves a vacancy with an effective positive charge. This vacancy is called as hole. Hence charge carriers are created.	1

28. With the help of circuit diagrams, briefly explain the forward biasing and the reverse biasing of a p-n junction diode. CBSE 2026 3

28. With the help of circuit diagrams, briefly explain the forward biasing and the reverse biasing of a p-n junction diode. CBSE 2026 *3*

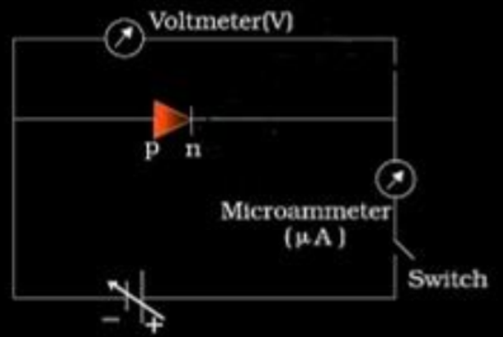
Forward biasing



1

The p-side of the diode is connected to positive terminal and the n-side is connected to the negative terminal of the battery. ½

Reverse biasing



1

The p-side of the diode is connected to negative terminal and n-side is connected to the positive terminal of the battery. ½

8. Which of the following p-n junction diodes is forward biased ?

(A)

(B)

(C)

(D)

CBSE 2026

8. Which of the following p-n junction diodes is forward biased ?

(A)

(B)

(C)

(D)

CBSE 2026

(B)

18. Explain, with the help of suitable diagram, the two important processes that occur during the formation of a p-n junction. *2*

CBSE 2026

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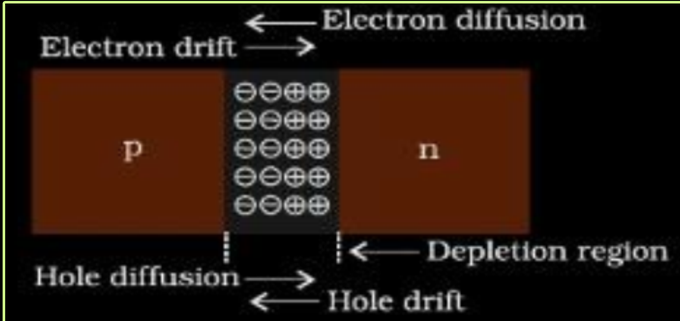
CBSE 2026

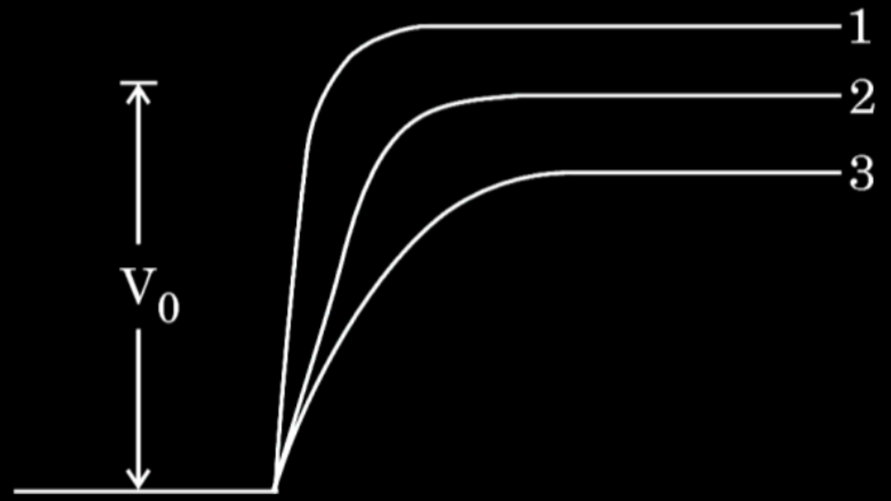


1

Diffusion: During the formation of p-n junction, due to concentration gradient across p and n sides, holes diffuse from p-side to n-side and electrons diffuse from n-side to p-side. This process is called diffusion. ½

Drift: As diffusion continues, a depletion region consisting of the immobile ions get created. Due to its field, an electron moves from p-side to n-side and a hole moves from n-side to p-side. This motion is called drift which is opposite to diffusion process. ½

7. If the given figure, V_0 is the potential barrier across a p-n junction in an unbiased condition. Which of the following statements is correct ?



(A) Curves 1 and 3 both correspond to forward biasing of p-n junction.
(B) Curve 3 corresponds to forward biasing and Curve 1 corresponds to reverse biasing.
(C) Curve 1 corresponds to forward biasing and Curve 3 corresponds to reverse biasing.
(D) Curves 1 and 3 both correspond to reverse biasing.

CBSE 2026

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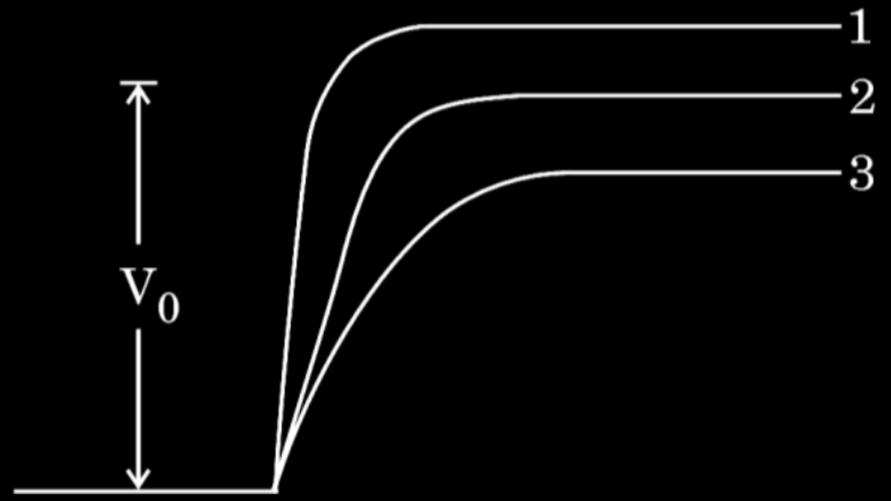

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(C) Curve 1 corresponds to forward biasing and Curve 3 corresponds to reverse biasing.
(D) Curves 1 and 3 both correspond to reverse biasing.

CBSE 2026

(B) Curve 3 correspondence to forward biasing and Curve 1 corresponds to reverse biasing.

17. Write any two points of difference between intrinsic and extrinsic semiconductors. CBSE 2026 *2*

17. Write any two points of difference between intrinsic and extrinsic semiconductors. CBSE 2026 *2*

Intrinsic Semiconductor	Extrinsic Semiconductor
1. Pure Semiconductor	1. Doped with impurity atoms
2.Low electrical conductivity	2.High electrical conductivity
3.Number of free electrons is equal to number of holes	3.Number of free electrons is not equal to number of holes
4.Conductivity depends on temperature only.	4.Conductivity depends on temperature and concentration of doping.

(Any two) 1+1

11. A n-type semiconductor is obtained by doping silicon with **1**

(A) Boron
(B) Arsenic
(C) Indium
(D) Aluminium

CBSE 2026

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CBSE 2026

(B) Arsenic

12. When a forward biasing is applied across a junction diode, it – **1**

(A) raises the potential barrier height.

(B) decreases the depletion layer width.

(C) increases the resistance of the p-n junction.

(D) reduces the majority carrier current to zero.

CBSE 2026

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CBSE 2026

(B) decreases the depletion layer width

21. Differentiate between n-type and p-type semiconductors. **2**

CBSE 2026

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CBSE 2026

n-type	p-type	
Impurity/dopant atoms are pentavalent	Impurity/dopant atoms are trivalent	½ ×4
Electrons are the majority charge carriers and holes are the minority charge carriers. OR $n_e >> n_h$	Holes are the majority charge carriers and electrons are the minority charge carriers. OR $n_h >> n_e$	

28. (a) Explain the working of a junction diode as a half-wave rectifier.

(b) The efficiency of a half-wave rectifier is 40%. What happens to the remaining 60% power ? CBSE 2026 **3**

28. (a) Explain the working of a junction diode as a half-wave rectifier.

(b) The efficiency of a half-wave rectifier is 40%. What happens to the remaining 60% power ? CBSE 2026 **3**

(a) If an alternating voltage is applied across a diode in series with a load, only during the half cycles of the a.c input the diode is forward biased. Therefore, in the positive half cycle of a.c input, there is a current through the load resistor and we get an output voltage, whereas there is no output current in the negative half cycle of a.c input. Thus, the output voltage, though still varying but is restricted to only one direction and is said to be rectified. Since the rectified output of this circuit is only for half of the input a.c wave, it is called a half-wave rectifier.	**2**
(b) A half wave rectifier utilizes only one half of the a.c cycle, resulting in a pulsating d.c output. The other half of a.c is discarded or blocked. Discarded power during half cycle is 60%, which is not converted into d.c.	**1**

14. **Assertion (A)** : On forward biasing a p-n junction diode, the height of the barrier potential increases. **1**

Reason (R) : In forward biasing of a p-n junction diode, the direction of the applied voltage is in the same direction as the built-in potential.

CBSE 2026

14. **Assertion (A)** : On forward biasing a p-n junction diode, the height of the barrier potential increases. **1**

Reason (R) : In forward biasing of a p-n junction diode, the direction of the applied voltage is in the same direction as the built-in potential.

CBSE 2026

(D) Assertion (A) is false and Reason (R) is also false.

17. Write two points of difference between intrinsic and extrinsic semiconductors. **2**

CBSE 2026

17. Write two points of difference between intrinsic and extrinsic semiconductors. **2**

CBSE 2026

Intrinsic semiconductors	Extrinsic semiconductors	
1. $n_e = n_h$	1. $n_e \neq n_h$ (Alternatively $n_e \gg n_h$ or $n_e \ll n_h$	
2. Low conductivity	2. high conductivity	
3. Semiconductor without any impurity atoms added/ Pure semiconductor.	3. Semiconductor with doping/ Impure Semiconductor.	
4. Its conductivity depends only on temperature	4. Its conductivity depends on temperature and concentration of dopants	
5. They are naturally occurring semiconductors	5. They are manufactured semiconductors.	1+1
(Any two) or (Any other relevant differences)		

29. A p-type or n-type semiconductor can be converted into a p-n junction by doping it with suitable impurity. The motion of majority charge carriers causes diffusion current across the junction while the barrier electric field causes motion of minority carriers for drift current. In case of unbiased diode, the diffusion and drift currents are equal. This equilibrium is disturbed by the biasing batteries. Diodes, therefore, allow currents in one direction. This property of diode is used in making rectifiers. CBSE 2026

(i) Silicon is doped with which of the following to obtain p-type semiconductor ? **1**

(A) Phosphorus (B) Arsenic

(C) Boron (D) Antimony

(ii) A semiconductor has an electron concentration of 5×10^{22} m^{-3}. The concentration of holes is (given $n_i = 1.5 \times 10^{16}$ m^{-3}) **1**

(A) 5×10^{22} m^{-3} (B) 1.5×10^{6} m^{-3}

(C) 9×10^{8} m^{-3} (D) 4.5×10^{9} m^{-3}

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(i)	(C) Boron
(ii)	(D) $4.5 \times 10^{9} m^{-3}$

(iii) During forward biasing of a p-n junction diode, the **1**

(A) current is mainly due to drifting of majority carriers.

(B) current is mainly due to drifting of minority carriers.

(C) diffusion and drift currents are equal.

(D) current is of the order of 1 A.

(iv) The threshold voltage for silicon diode is about **1**

(A) 0.2 V (B) 0.5 V

(C) 0.7 V (D) 1.5 V

(iv) When we dope Ge with a pentavalent element, four of its electrons bond with four germanium neighbours but fifth electron remains weakly bound. The ionisation energy for this electron is about **1**

(A) 0.01 eV (B) 0.05 eV

(C) 0.1 eV (D) 0.15 eV

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(C) 0.1 eV (D) 0.15 eV

(iii) Award 1 mark to each student, who has attempted this part.

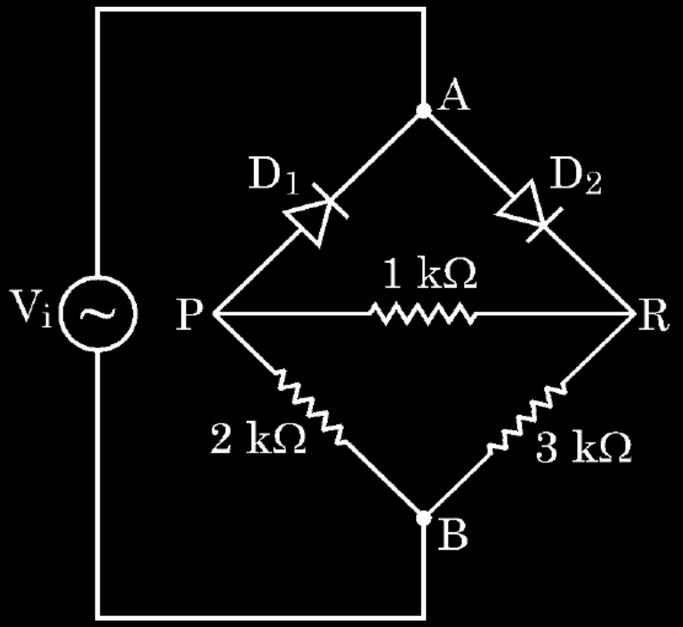
(iv) (C) 0.7V

OR

(A) 0.01 eV

25.

CBSE 2026



An ac voltage $V_i = 12 \sin (100 \pi t)V$ is applied between points A and B in a network of two ideal diodes and three resistors as shown in figure.

During the positive half-cycle of the input voltage V_i supplied to the network. **3**

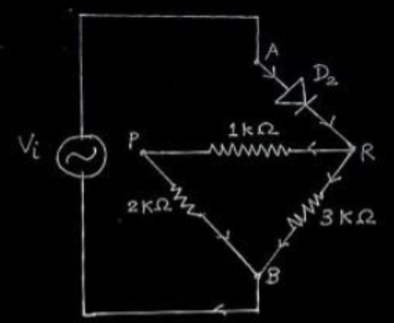
(a) Identify which of the two diodes will conduct and why ?

(b) Redraw an equivalent circuit diagram to show the flow of current.

(c) Calculate the output voltage drops V_0 across the three resistors when the input voltage attains its peak value.

(a) Diode D_2 is in forward bias and it conducts. ½

(b)



1

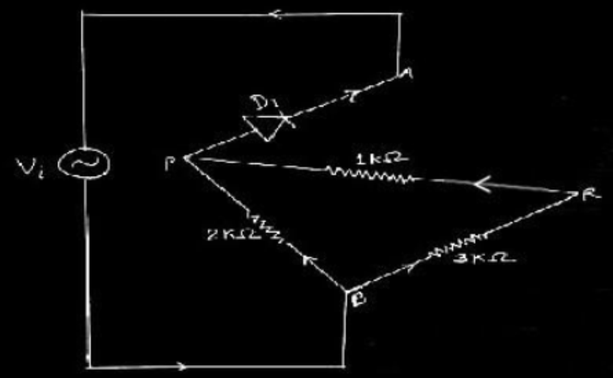
(c) Voltage drop across RP, PB and RB is

$V_{RP} = 4V$, $V_{PB} = 8V$, $V_{RB} = 12V$ ½+½+½

Alternatively:

If the student considers positive AC cycle reaching diode D_1
(a) Diode D_1 is in forward bias and it conducts.

(b)



(c) $V_{PB} = 12V, V_{PR} = 3V, V_{RB} = 9V$

26. Briefly explain the two important processes that occur during the formation of a p-n junction. CBSE 2026 **3**

26. Briefly explain the two important processes that occur during the formation of a p-n junction. CBSE 2026 **3**

Two important processes are in the formation of p-n junction. (i) Diffusion (ii) Drift	 ½ ½
Diffusion: During the formation of p-n junction, due to concentration gradient across p- and n-sides, holes diffuse from p-side to n-side $(p \rightarrow n)$ and electrons diffuse from n-side to p-side $(n \rightarrow p)$. This motion of charge carriers give rise to diffusion current across the junction.	1
Drift: Due to the positive space charge region on n-side of the junction and negative space charge region on p-side of the junction, an electric field directed from positive charge towards negative charge develops. Due to this field an electron on p-side of junction moves to n-side and a hole on n-side of the junction moves to p-side. The motion of charge carriers due to the electric field is called drift and constitutes drift current.	1

2. In an unbiased p-n junction, at equilibrium, which of the following statements is true ? 1

(A) Diffusion current is zero but drift current exists.

(B) Diffusion current exists but drift current is zero.

(C) Diffusion and drift currents are equal and opposite.

(D) Both the diffusion and drift currents exist but are unequal.

CBSE 2026

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CBSE 2026

(C) Diffusion and drift currents are equal and opposite.

27. A semiconductor has equal electron and hole concentration of 3×10^8 m^{-3}. On doping with a certain impurity, the hole concentration increases to 6×10^{10} m^{-3}. **3**

(a) What type of semiconductor is obtained on doping ?

(b) Calculate the new electron concentration of the semiconductor.

(c) How does the energy gap of semiconductor change with doping ? Draw the energy band diagram for it.

CBSE 2026

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CBSE 2026

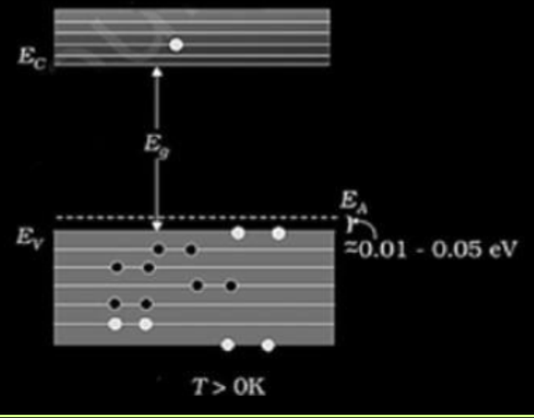
(a) p type semiconductor ½

(b) $\because n_e = \frac{n_i^2}{n_h}$ ½

$n_e = \frac{(3\times10^8)^2}{6\times10^{10}}$ ½

$n_e = 1.5\times10^6 \ m^{-3}$ ½

(c) • Energy gap effectively decreases



1

27. (a) Draw the V-I characteristics of silicon diode. **3**

(b) Explain the following terms :

(i) minority carrier injection in forward bias

(ii) breakdown voltage in reverse bias

CBSE 2026

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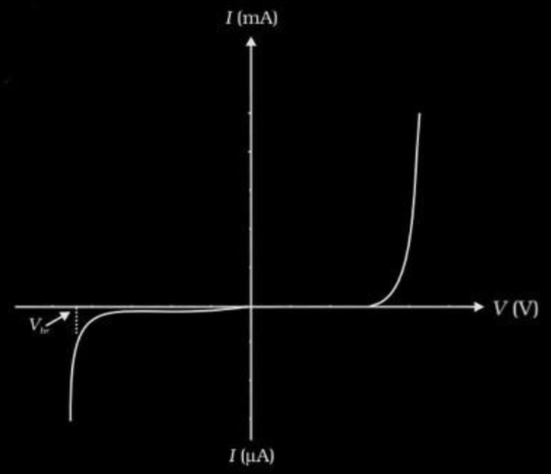
CBSE 2026

(a)



1

(b)

(i) Due to applied voltage, electrons from n-side cross the depletion region and reach p-side and holes from p-side cross the junction reach the n-side. This process under forward bias is called as minority charge injection. 1

(ii) The current under reverse bias is essentially voltage independent up to a critical reverse bias voltage known as breakdown voltage. 1

Alternatively:
The voltage in Reverse bias after which the value of reverse current starts increasing sharply is called breakdown voltage.

14. *Assertion (A) :* The electrical conductivity of a pure Ge crystal increases with increase in its temperature.

Reason (R) : The number of electrons excited by thermal excitation from the valence band to the conduction band, in a semiconductor, increases with increase in temperature.

CBSE 2025 C

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Reason (R) : The number of electrons excited by thermal excitation from the valence band to the conduction band, in a semiconductor, increases with increase in temperature.

CBSE 2025 C

(A) Both Assertion (A) and Reason (R) are true and Reason (R) is correct explanation of Assertion (A).

18. With the help of a circuit diagram, describe the method to obtain p-n junction diode characteristic when it is forward biased. Draw the shape of the characteristic. CBSE 2025 C *2*

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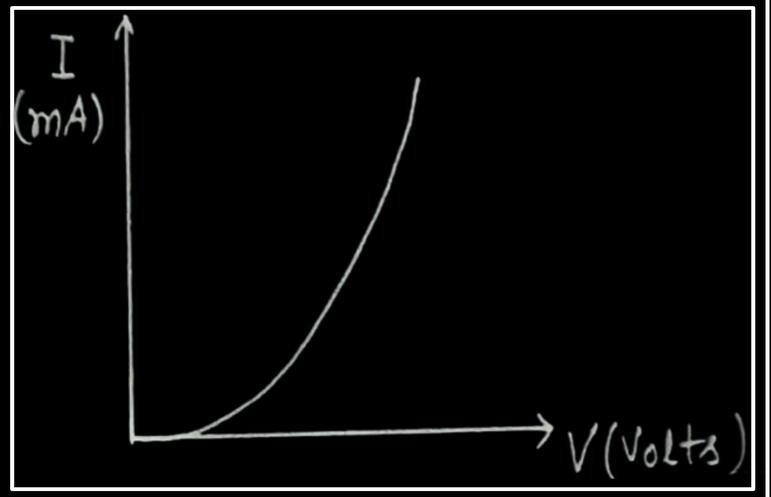
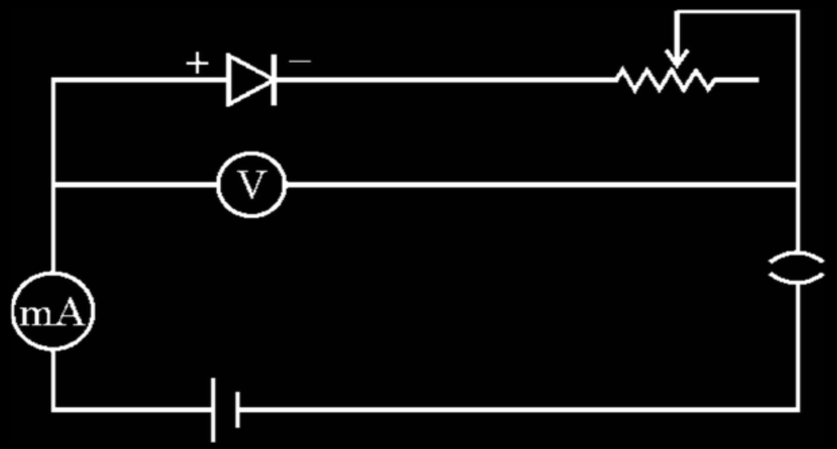


• 	1
• A very small current flows till the applied voltage reaches the barrier potential in the forward bias p-n junction diode. Diode current increases significantly when applied voltage becomes greater than barrier potential.	½

29. According to the band theory of solids, a semiconductor has a valence band and a conduction band separated by a gap, known as energy band gap. Pure semiconductors are called intrinsic semiconductors. At room temperature, some electrons from the valence band can acquire enough energy to cross the band gap and enter the conduction band. The number of conduction electrons is equal to the number of holes is an intrinsic semiconductor. The number of charge carriers can be changed by doping of a suitable impurity in a pure semiconductor. Such semiconductors are known as extrinsic semiconductors. These are of two types (n-type and p-type).

A p-n junction is the basic building block of semiconductor devices. Two important processes occur during formation of a p-n junction : diffusion and drift. A 'depletion layer' is formed in a p-n junction. This is responsible for a junction potential barrier. A semiconductor diode is basically a p-n junction with metallic contacts provided at the ends for the application of an external voltage. A diode can be forward-biased or reverse-biased. The barrier height and the depletion layer width in a p-n junction changes depending on the nature of the biasing.

CBSE 2025 C

(i) (a) Which of the following statements is ***not*** true ? *1*

(A) The resistance of an intrinsic semiconductor decreases with the increase of temperature.

(B) Doping pure Si with trivalent impurities gives p-type semiconductors.

(C) The majority charge carriers in n-type semiconductors are holes.

(D) A p-n junction can act as a semiconductor diode.

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(D) A p-n junction can act as a semiconductor diode.

(C) Majority charge carrier is n type semiconductors are holes.

(b) In a unbiased p-n junction : *1*

(A) the diffusion current is zero everywhere.

(B) the drift current is zero everywhere.

(C) the electric potential is zero everywhere.

(D) the drift current and the diffusion current cancel each other.

(b) In a unbiased p-n junction : *1*

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(ii) The impurity atoms with which pure Ge should be doped to convert it into an n-type semiconductor, is : *1*

(A) Boron (B) Phosphorous

(C) Aluminium (D) Indium

(ii) The impurity atoms with which pure Ge should be doped to convert it into an n-type semiconductor, is : *1*

(A) Boron (B) Phosphorous

(C) Aluminium (D) Indium

(B) Phosphorus

(iii) The energy band gap in Ge at 0 K is about : *1*

(A) 0·72 eV (B) 1·1 eV

(C) 3·0 eV (D) 5·4 eV

(iii) The energy band gap in Ge at 0 K is about : *1*

(A) 0·72 eV (B) 1·1 eV

(C) 3·0 eV (D) 5·4 eV

(A) 0.72 eV

(iv) In a p-n junction diode under reverse bias, the barrier height : *1*

(A) is reduced and the depletion layer width decreases.

(B) is reduced and the depletion layer width increases.

(C) increases and the depletion layer width also increases.

(D) increases and the depletion layer width decreases.

(iv) In a p-n junction diode under reverse bias, the barrier height : *1*

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11. Which of the following impurity atoms, when doped in silicon, would produce an n-type semiconductor ?

(A) Arsenic

(B) Aluminium

(C) Boron

(D) Indium

CBSE 2025 C

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CBSE 2025 C

(A) Arsenic

12. Which of the following statements is correct during formation of a p-n junction ?

(A) Initially the diffusion current is small and the drift current is large.

(B) Initially the diffusion current and the drift current are equal to each other.

(C) Finally, the diffusion current and the drift current are equal to each other.

(D) Electrons move from p-region to n-region.

CBSE 2025 C

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CBSE 2025 C

(C) Finally, diffusion current and the drift current are equal to each other.

21. In a p-type Si semiconductor sample the hole concentration is $2{\cdot}25 \times 10^{15}\ \text{cm}^{-3}$. If the intrinsic carrier concentration is $1{\cdot}5 \times 10^{10}\ \text{cm}^{-3}$, calculate the electron concentration in the sample.

CBSE 2025 C *2*

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CBSE 2025 C *2*

$n_e n_h = n_i^2$	½
$= \dfrac{2 \times 1.5 \times 10^{-2} \times 540 \times 10^{-9}}{0.09 \times 10^{-2}}$	½
$= \dfrac{1.5 \times 10^{10} \times 1.5 \times 10^{10}}{2.25 \times 10^{15}}$	½
$= 10^5\ \text{cm}^{-3}$	½

28. (a) Differentiate between the two types of extrinsic semiconductors.

(b) The mobility of an electron in the conduction band of a semiconductor is more than the mobility of a hole (or electron) in the valance band. Explain why. CBSE 2025 C *3*

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(b) The mobility of an electron in the conduction band of a semiconductor is more than the mobility of a hole (or electron) in the valance band. Explain why. CBSE 2025 C *3*

(a)

n- type of semiconductor	p-type of semiconductor
1. It is formed by doping with group 15 elements. 2. Majority charge carriers are electrons.	1. It is formed by doping with group 13 elements. 2. Majority charge carriers are holes.

1

(b) Electron in conduction band are almost free and get easily accelerated by an E-field. Electrons in V.B. are bound between the atoms of a semiconductor, less accelerated by an E-field. So acquire lesser drift velocity than that of electrons in the C.B. The mobility of electrons in the V.B. is less than the mobility of the electrons in CB. As the motion of an e^- in the VB is equivalent to the motion of a hole in opposite direction, hence the mobility of hole in the V.B. is smaller than mobility of an electrons in CB.

2

2. Germanium crystal is doped at room temperature with a minute quantity of boron. The charge carriers in the doped semiconductors will be :

(A) electrons only

(B) holes only

(C) holes and few electrons

(D) electrons and few holes

CBSE 2025

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CBSE 2025

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12. A p-n junction diode is forward biased. As a result,

(A) both the potential barrier height and the width of depletion layer decrease.

(B) both the potential barrier height and the width of depletion layer increase.

(C) the potential barrier height decreases and the width of depletion layer increases.

(D) the potential barrier height increases and the width of depletion layer decreases.

CBSE 2025

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CBSE 2025

(A) Both the potential barrier height and width of depletion layer decrease.

21. Draw energy band diagrams of n-type and p-type semiconductors at temperature T > 0 K. Show the donor/acceptor energy levels with the order of difference of their energies from the bands. CBSE 2025 *2*

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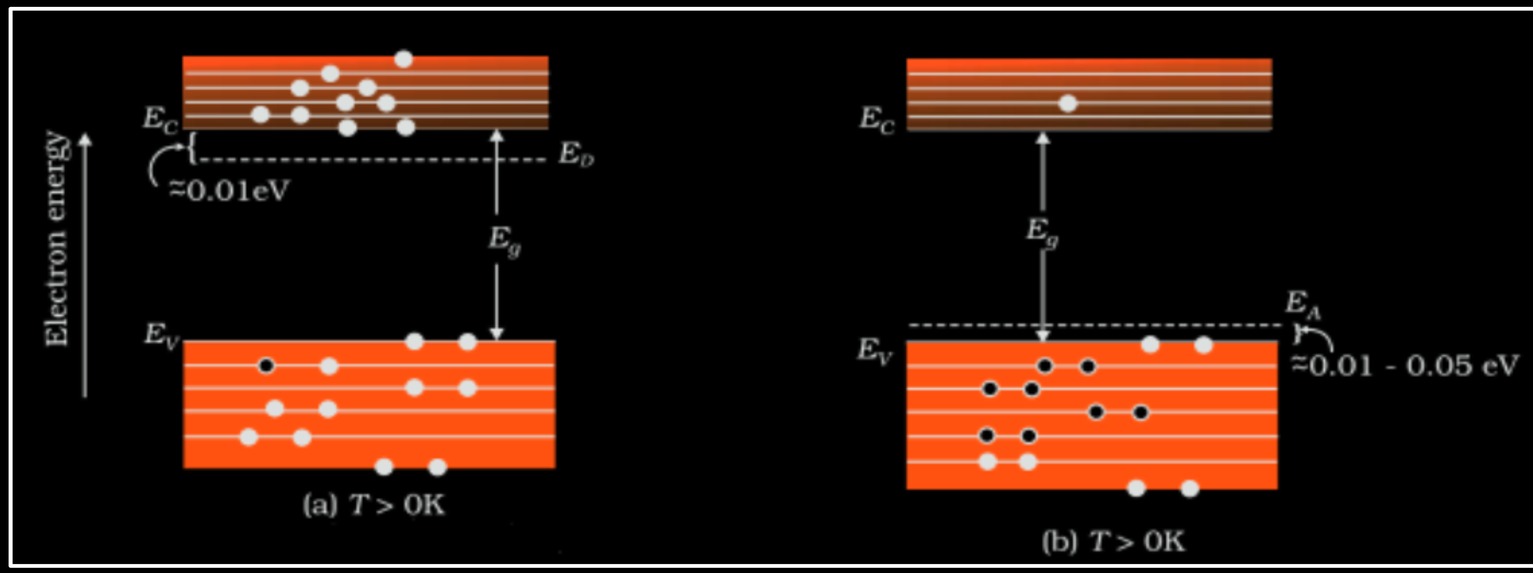

1+1

22. Explain the process of formation of 'depletion layer' and 'potential barrier' in a p-n junction region of a diode, with the help of a suitable diagram. Which feature of junction diode makes it suitable for its use as a rectifier ? CBSE 2025 *3*

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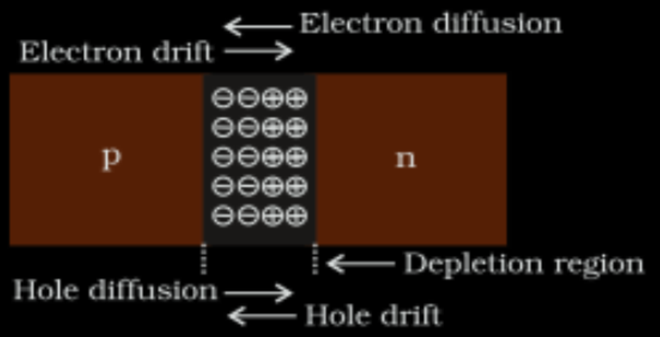
When an electron diffuses from n-side to p-side, it leaves behind an ionized donor on n side. Similarly when a hole diffuses from p-side to n-side, it leaves behind an ionized acceptor on p side. This space charge region consisting of immobile ions on either side of the junction is known as depletion layer.	1
As diffusion process continues width of depletion layer increases and consequently strength of electric field increases across the junction and thus the drift current. The potential that prevents the movement of electron from n region into p region is called potential barrier.	1
 **Note : Award 1 mark if formation of depletion layer is shown with the help of above diagram.**	
Diode allows current to pass only when it is forward biased as resistance is small whereas in reverse bias resistance is very large. **Alternatively**: Diode is unidirectional.	1

16. *Assertion (A) :* A hole is an apparent free particle with effective positive electronic charge.

Reason (R) : A hole is not necessarily a vacancy left behind by an electron in the valence band.

CBSE 2025

16. *Assertion (A) :* A hole is an apparent free particle with effective positive electronic charge.

Reason (R) : A hole is not necessarily a vacancy left behind by an electron in the valence band.

CBSE 2025

(C) Assertion (A) is true, But Reason (R) is false.

17. In an n-type semiconductor electron-hole combination is a continuous process at room temperature. Yet the electron concentration is always greater than the hole concentration in it. Explain. CBSE 2025 *2*

17. In an n-type semiconductor electron-hole combination is a continuous process at room temperature. Yet the electron concentration is always greater than the hole concentration in it. Explain. CBSE 2025 *2*

In a doped semiconductor the total number of conduction electrons is due to the electrons contributed by donors and those generated intrinsically, while the total number of holes is only due to the holes from the intrinsic sources.	2

Extrinsic semiconductors are made by doping pure or intrinsic semiconductors with suitable impurity. There are two type of dopants used in doping, Si or Ge, and using them p-type and n-type semiconductors can be obtained. A p-n junction is the basic building block of many semiconductor devices. Two important processes occur during the formation of a p-n junction : diffusion and drift. When such a junction is formed, a ‘depletion layer’ is created consisting of immobile ion-cores. This is responsible for a junction potential barrier. The width of a depletion layer and the height of potential barrier changes when a junction is forward-biased or reverse-biased. A semiconductor diode is basically a p-n junction with metallic contacts provided at the ends for application of an external voltage. Using diodes, alternating voltages can be rectified.

CBSE 2025

(a) In reverse-biased p-n junction : *1*

(A) the drift current is of the order of few mA.

(B) the applied voltage mostly drops across the depletion region.

(C) the depletion region width decreases.

(D) the current increases with increase in applied voltage.

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(B) the applied voltage mostly drops across the depletion region.

(C) the depletion region width decreases.

(D) the current increases with increase in applied voltage.

(B) The applied voltage mostly drops across the depletion region

(b) The output frequency of a full-wave rectifier with 50 Hz as input frequency is : *1*

(A) 25 Hz (B) 50 Hz

(C) 100 Hz (D) 200 Hz

(b) The output frequency of a full-wave rectifier with 50 Hz as input frequency is : *1*

(A) 25 Hz (B) 50 Hz

(C) 100 Hz (D) 200 Hz

(C) 100 Hz

(ii) During formation of a p-n junction : *1*

(A) a layer of negative charge on n-side and a layer of positive charge on p-side appear.

(B) a layer of positive charge on n-side and a layer of negative charge on p-side appear.

(C) the electrons on p-side of the junction move to n-side initially.

(D) initially diffusion current is small and drift current is large.

(ii) During formation of a p-n junction : *1*

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(B) A layer of positive charge on n side and a layer of negative charge on p side appear

(iii) When a pentavalent atom occupies the position of an atom in the crystal lattice of Si, four of its electrons form covalent bonds with four silicon neighbours, while the fifth remains bound to the parent atom. The energy required to set this electron free is about : *1*

(A) 0·5 eV (B) 0·1 eV

(C) 0·05 eV (D) 0·01 eV

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(A) 0·5 eV (B) 0·1 eV

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(C) 0.05 eV

(iv) Which of the following is a donor impurity atom for Ge ? *1*

(A) Boron (B) Antimony

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(B) Antimony

11. The general purpose diode is normally used in – **1**

(A) beyond cut-in voltage region

(B) before cut-in voltage region.

(C) beyond the reverse saturation current region.

(D) before the reverse saturation current region.

CBSE 2025

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CBSE 2025

(D) before the the reverse saturation current.

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CBSE 2025

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CBSE 2025

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21. When the voltage across a p-n junction diode is increased from 0.90 V to 0.95 V, the diode current changes by 0.005 A. Find the dynamic resistance of the diode. CBSE 2025 **2**

21. When the voltage across a p-n junction diode is increased from 0.90 V to 0.95 V, the diode current changes by 0.005 A. Find the dynamic resistance of the diode. CBSE 2025 **2**

Dynamic resistance $r_d = \frac{\Delta V}{\Delta I}$	½
$r_d = \frac{0.95 - 0.90}{0.005} = \frac{0.05}{0.005}$	1
$r_d = 10\,\Omega$	½

15. *Assertion (A) :* The impurities in p-type Si are not pentavalent atoms.

Reason (R) : The hole density in valance band in p-type semiconductor is almost equal to the acceptor density.

CBSE 2025

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Reason (R) : The hole density in valance band in p-type semiconductor is almost equal to the acceptor density. CBSE 2025

(B) Both Assertion (A) and Reason (R) are true, but Reason (R) is the not the correct explanation of the Assertion (A)

27. (a) Draw the energy-band diagrams for conductors, semiconductors and insulators at T = 0 K. How is an electron-hole pair formed in a semiconductor at room temperature ?

(b) Carbon and silicon both, are members of IV group of periodic table and have the same lattice structure. Carbon is an insulator whereas silicon is a semiconductor. Explain. CBSE 2025 *3*

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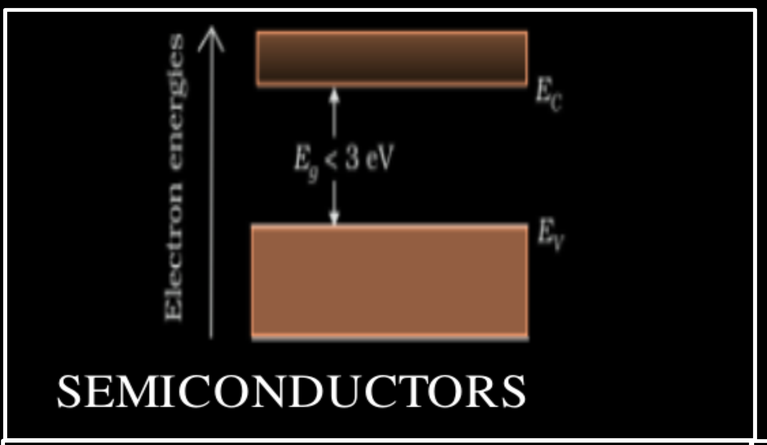
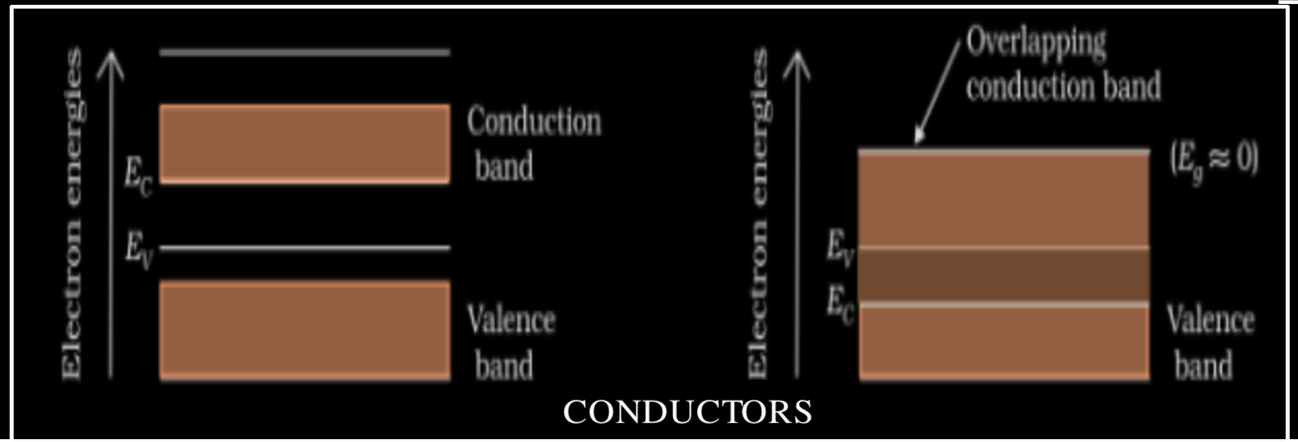
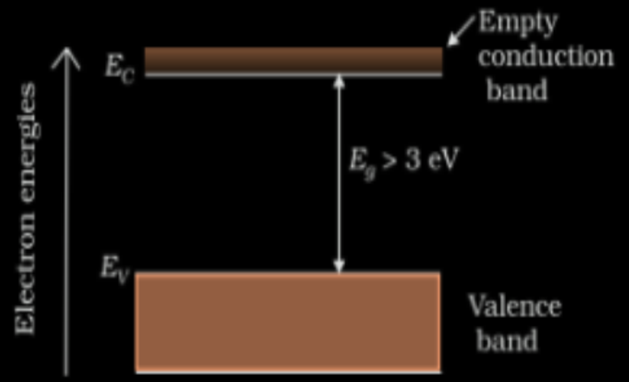


SEMICONDUCTORS



CONDUCTORS



INSULATORS

At room temperature, thermal energy is sufficient for electrons to make them free from the bonds and create a vacancy called hole. Hence electron hole pair is formed. ½

(b) The valence electron in carbon and silicon lie in the second and third orbit respectively. So, the energy required to take out an electron will be less for silicon as compared to carbon. Hence number of free electrons for conduction in silicon are significant but negligibly small for carbon. **1**

28. Differentiate between half-wave and full-wave rectification. With the help of a circuit diagram, explain the working of a full-wave rectifier. *3*

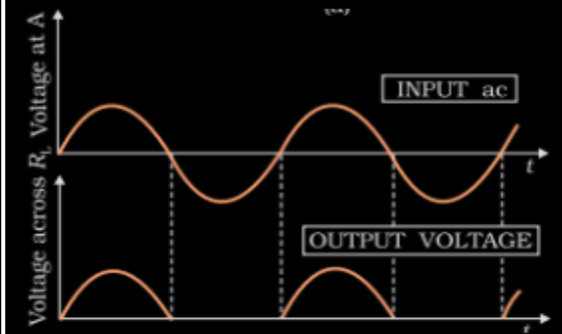
CBSE 2025

28. Differentiate between half-wave and full-wave rectification. With the help of a circuit diagram, explain the working of a full-wave rectifier. *3*

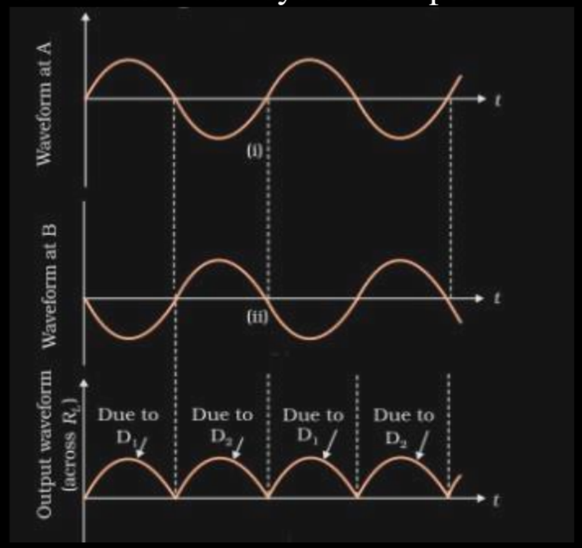
CBSE 2025

In half wave rectification there is output in one half of input cycle, whereas in full wave rectification, output is obtained for both half cycles of input (positive and negative)

Alternatively

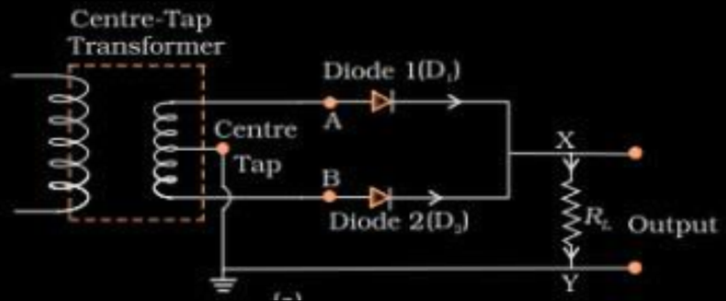


Half wave Rectification



Full wave Rectification

Working of full wave rectifier:



Suppose the input voltage to A with respect to the centre-tap at any instant is positive. At that instant, voltage at B being out of phase will be negative. So, diode D_1 gets forward biased and conducts (while D_2 being reverse biased is not conducting). Hence, during this positive half cycle we get an output current (and output voltage across the load resistor R_L). In the course of ac cycle when the voltage at A becomes negative with respect to centre tap, the voltage at B would be positive. In this part of the cycle diode D_1 would not conduct but diode D_2 would, giving an output current and output voltage (across R_L) during the negative half cycle of the input ac.

1

1

12. Which of the following is an electrical conductor at room temperature ?

(A) Sn

(B) Mica

(C) Si

(D) C

CBSE 2025

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(A) Sn

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CBSE 2025

(A) Sn

15. *Assertion (A) :* n-type semiconductor is not negatively charged.

Reason (R) : Neutral pentavalent impurity atom doped in intrinsic semiconductor (neutral) donates its fifth unpaired electron to the crystal lattice and becomes a positive donor.

CBSE 2025

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CBSE 2025

(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A)

---

17. In an intrinsic semiconductor, carrier's concentration is 5×10^8 m^{-3}. On doping with impurity atoms, the hole concentration becomes 8×10^{12} m^{-3}.

(a) Identify (i) the type of dopant and (ii) the extrinsic semiconductor so formed.

(b) Calculate the electron concentration in the extrinsic semiconductor. *2*

CBSE 2025

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(a) Identify (i) the type of dopant and (ii) the extrinsic semiconductor so formed.

(b) Calculate the electron concentration in the extrinsic semiconductor. *2*

CBSE 2025

(a) (i) Trivalent (ii) p – type semi conductor	½ ½
(b) Electron concentration $n_e = \frac{n_i^2}{n_h}$ $n_e = \frac{(5\times10^8)^2}{8\times10^{12}}$ $n_e = 3.125\times10^4 m^{-3}$	 ½ ½

22. (a) What are majority and minority charge carriers of p-type and n-type semiconductors ?

(b) Explain briefly the formation of diffusion current and drift current in a p-n junction diode. CBSE 2025 *3*

22. (a) What are majority and minority charge carriers of p-type and n-type semiconductors ?

(b) Explain briefly the formation of diffusion current and drift current in a p-n junction diode. CBSE 2025 *3*

(a)	In p-type semiconductor	½
	Majority charge carriers - holes	½
	Minority charge carriers - electrons	
	In n-type semiconductors	
	Majority charge carriers - electrons	½
	Minority charge carriers – holes	½
	(b) Diffusion current – during the formation of p n junction , and due to the concentration gradient across p and n – sides , holes diffuse from p side to n side (p $\rightarrow$ n) and electrons diffuse from n – side to p – side (n $\rightarrow$ p). This motion of charge carriers gives rise to diffusion current across the junction.	½
	Drift current –Due to electric field at junction, an electron on p – side of the junction moves to n- side and a hole on n – side of the junction moves to p-side. This motion of charge carriers due to electric field gives drift current.	½

10. When the resistance measured between p and n ends of a p-n junction diode is high, it can act as a/an – **1**

(A) resistor (B) inductor

(C) capacitor (D) switch

CBSE 2025

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(A) resistor (B) inductor

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CBSE 2025

(A) resistor / (C) capacitor

15. **Assertion (A)** : In a semiconductor diode the thickness of depletion layer is not fixed. **1**

Reason (R) : Thickness of depletion layer in a semiconductor device depends upon many factors such as biasing of the semiconductor.

CBSE 2025

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Reason (R) : Thickness of depletion layer in a semiconductor device depends upon many factors such as biasing of the semiconductor.

CBSE 2025

(A) If both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).

17. The threshold voltage of a silicon diode is 0.7 V. It is operated at this point by connecting the diode in series with a battery of V volt and a resistor of 1000 Ω. Find the value of V when the current drawn is 15 mA. **2**

CBSE 2025

17. The threshold voltage of a silicon diode is 0.7 V. It is operated at this point by connecting the diode in series with a battery of V volt and a resistor of 1000 Ω. Find the value of V when the current drawn is 15 mA. **2**

CBSE 2025

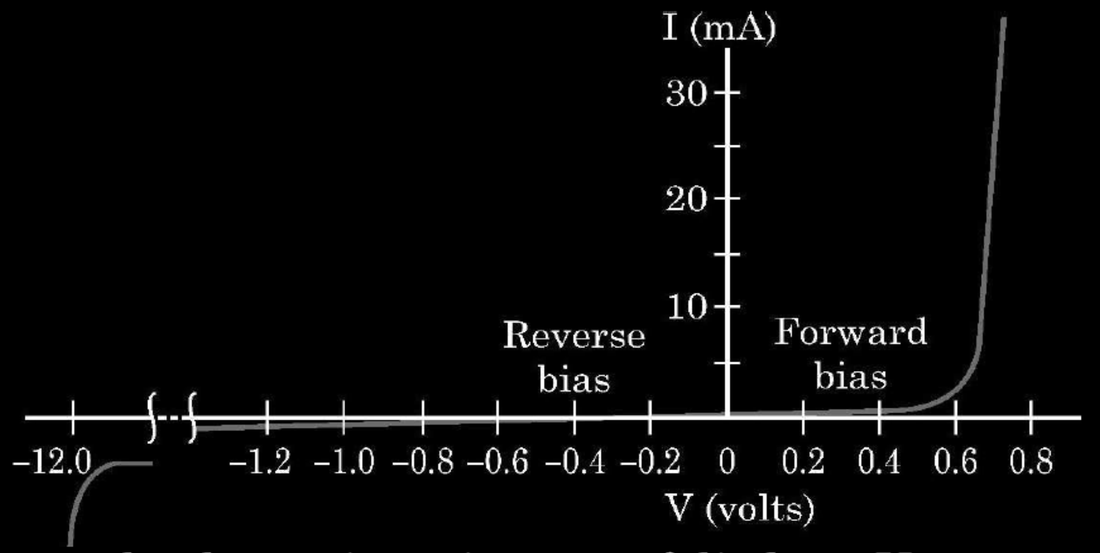
$V - V_o = IR$	½
$V - 0.7 = (15 \times 10^{-3}) \times 1000$	1
$V = 15.7$ volt	½

26. (a) What are majority and minority charge carriers in an extrinsic semiconductor ? **3**

(b) A p-n junction is forward biased. Describe the movement of the charge carriers which produce current in it.

(c) The graph shows the variation of current with voltage for a p-n junction diode.

CBSE 2025



Estimate the dynamic resistance of diode at V = –0.6 volt.

(a) In an extrinsic semiconductor, the charge carriers whose number density is large are known as majority charge carriers. In an extrinsic semiconductor, the charge carriers whose number density is small are known as minority charge carriers.	½+½
(b) Due to the applied forward voltage, electrons from n-side cross the depletion region and reach p-side. Similarly, holes from p-side cross the junction and reach the n-side. Due to the movement of these charge carriers current is produced.	1
(c) At V = -0.6 volt, I = 0, so dynamic resistance is infinite.	1

10. When a p-n junction diode is forward biased **1**

(A) the barrier height and the depletion layer width both increase.

(B) the barrier height increases and the depletion layer width decreases.

(C) the barrier height and the depletion layer width both decrease.

(D) the barrier height decreases and the depletion layer width increases.

CBSE 2025

10. When a p-n junction diode is forward biased **1**

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CBSE 2025

(C) The barrier height and the depletion layer width both decrease.

13. **Assertion (A)** : We cannot form a p-n junction diode by taking a slab of a p-type semiconductor and physically joining it to another slab of a n-type semiconductor. **1**

Reason (R) : In a p-type semiconductor $\eta_e >> \eta_h$ while in a n-type semiconductor $\eta_h >> \eta_e$.

CBSE 2025

13. **Assertion (A)** : We cannot form a p-n junction diode by taking a slab of a p-type semiconductor and physically joining it to another slab of a n-type semiconductor. **1**

Reason (R) : In a p-type semiconductor $\eta_e >> \eta_h$ while in a n-type semiconductor $\eta_h >> \eta_e$.

CBSE 2025

(C) Assertion (A) is true but Reason (R) is false.

21. A p-type Si semiconductor is made by doping an average of one dopant atom per 5×10^7 silicon atoms. If the number density of silicon atoms in the specimen is 5×10^{28} atoms m^{-3}, find the number of holes created per cubic centimetre in the specimen due to doping. Also give one example of such dopants.

CBSE 2025 **2**

21. A p-type Si semiconductor is made by doping an average of one dopant atom per 5×10^7 silicon atoms. If the number density of silicon atoms in the specimen is 5×10^{28} atoms m^{-3}, find the number of holes created per cubic centimetre in the specimen due to doping. Also give one example of such dopants. CBSE 2025 **2**

Doping - 1 dopant atom for 5×10^7 Si atoms and number density of Si atoms = $5\times10^{28}\ \frac{\text{atoms}}{\text{m}^3}$ (given)	
$\therefore$ No. of holes created per $m^3 = \frac{5\times10^{28}}{5\times10^7} = 10^{21}$	1
Number of holes created per cubic centimeter $= \frac{10^{21}}{10^6} = 10^{15}$	
Any one example of dopant - Aluminium / Indium / Gallium	1

27. (a) Draw circuit arrangement for studying V-I characteristics of a p-n junction diode. **3**

(b) Show the shape of the characteristics of a diode.

(c) Mention two information that you can get from these characteristics.

CBSE 2025

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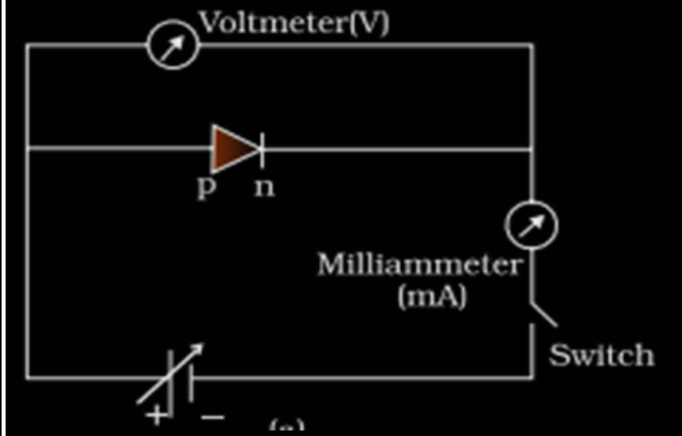
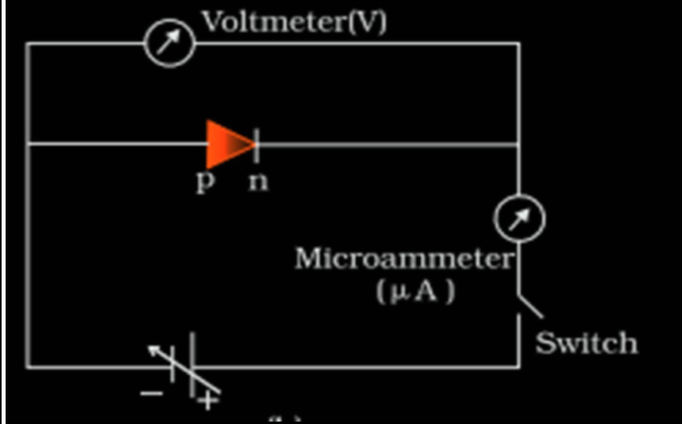
CBSE 2025

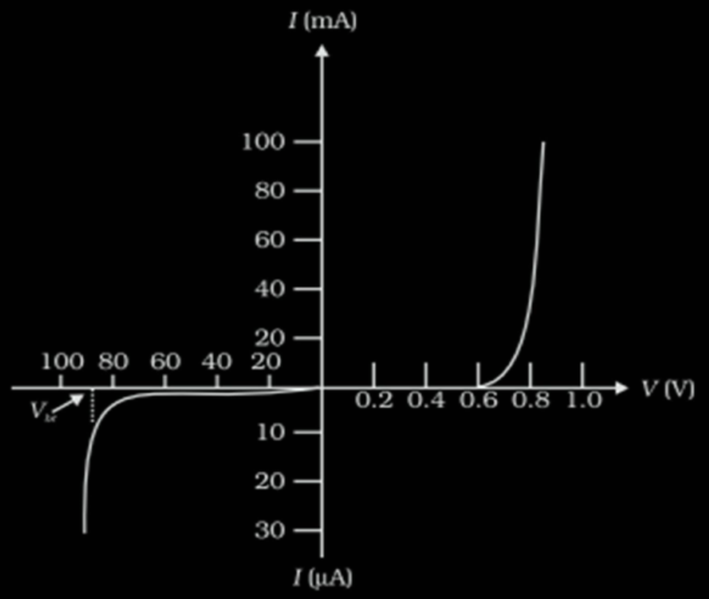


Circuit diagram for forward characteristics



Circuit diagram for Reverse characteristics

b)



1

Note : Please do not deduct marks for not writing values.

c) Any two informations

Knee voltage / reverse saturation current / Breakdown voltage / very low resistance in forward biasing / very high resistance in Reverse biasing.

½ + ½

11. When intrinsic silicon semiconductor is doped with Al atom, then it :

(A) decreases the number of holes in the conduction band.

(B) increases the number of holes in the valence band.

(C) increases the energy gap value.

(D) increases the number of electrons in the valence band.

CBSE 2024 C

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(C) increases the energy gap value.

(D) increases the number of electrons in the valence band.

CBSE 2024 C

(B) increase the number of holes in the valence bond.

17. Explain the terms depletion layer and potential barrier for a junction diode and their formation. CBSE 2024 C *2*

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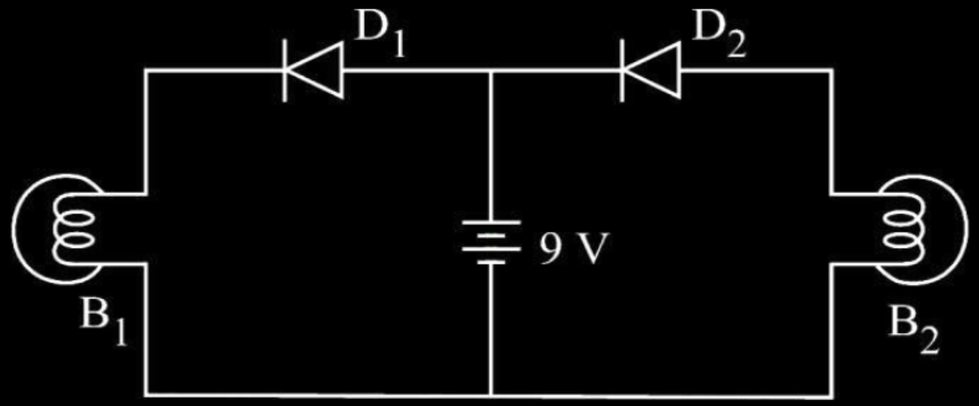
During the formation of p-n junction due to the concentration gradient holes diffuse from p- side to n-side leaving behind an immobile ionized acceptor (negative charge), hence forming a layer of negative charge (negative space-charge region) on p- side. Similarly, a layer of positive charge is developed on the n-side of the junction (positive space-charge region). This space – charge region on either side of the junction together is known as depletion layer. Due to the positive space- charge region on the n-side of the junction & negative space – charge region on the p- side of the junction a potential difference is developed across the junction of the two regions, which is called potential barrier.	1 1

29. The process of converting ac into dc is called rectification and the device used is called a rectifier. When ac signal is fed to a junction diode during positive half cycle, the diode is forward biased and current flows through it. During the negative half cycle, the diode is reverse biased and it does not conduct. Thus the ac signal is rectified. The p-n junction diodes can be used as half-wave and full-wave rectifiers.

CBSE 2024 C

(i) Which bulb/bulbs will glow in the given circuit ? *1*



(A) B_1 only

(B) B_2 only

(C) Both B_1 and B_2

(D) Neither B_1 nor B_2

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CBSE 2024 C

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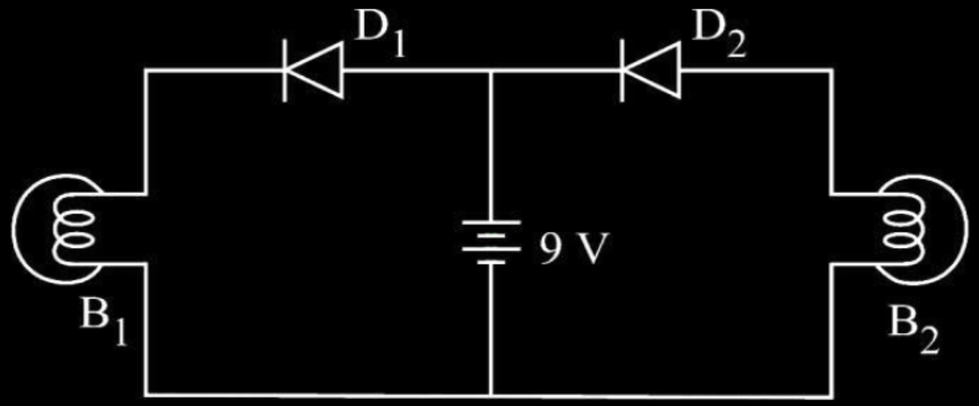


(A) B_1 only

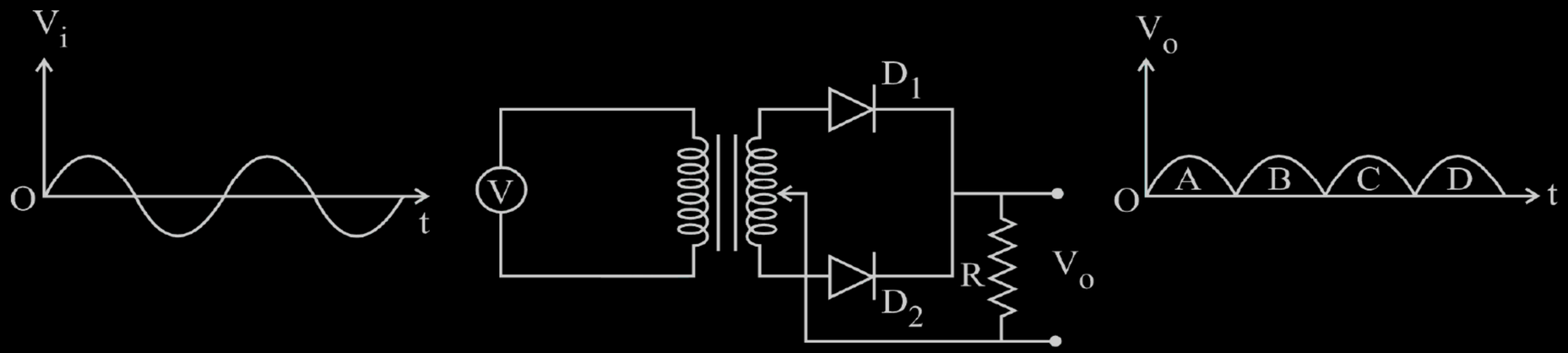
(B) B_2 only

(C) Both B_1 and B_2

(D) Neither B_1 nor B_2

(A) B_1 only

(ii) (a) A full-wave rectifier circuit is shown in the figure. The contribution in output waveform from junction diode D_1 is : *1*



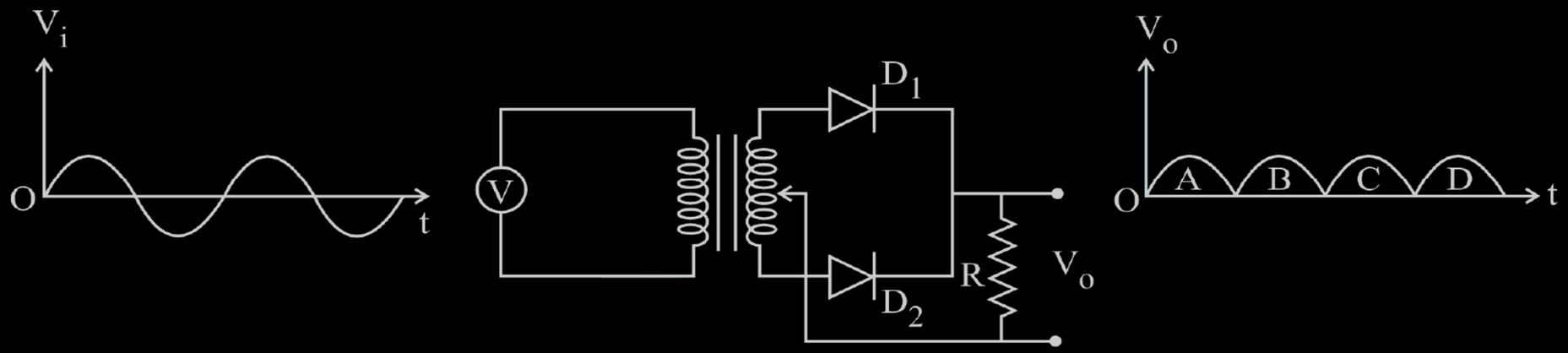
(A) A, D

(B) A, C

(C) B, D

(D) B, C

(ii) (a) A full-wave rectifier circuit is shown in the figure. The contribution in output waveform from junction diode D_1 is : *1*



(A) A, D (B) A, C

(C) B, D (D) B, C

(B) A, C

(b) The output in a half-wave rectifier is : *1*

(A) unidirectional without ripple (B) steady and continuous

(C) unidirectional with ripple (D) steady but discontinuous

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(A) unidirectional without ripple (B) steady and continuous

(C) unidirectional with ripple (D) steady but discontinuous

(C) unidirectional with ripple

(iii) In a p-n junction diode, the majority charge carriers on p-side and on n-side are, respectively : *1*

(A) electrons, electrons (B) electrons, holes

(C) holes, holes (D) holes, electrons

(iii) In a p-n junction diode, the majority charge carriers on p-side and on n-side are, respectively : *1*

(A) electrons, electrons (B) electrons, holes

(C) holes, holes (D) holes, electrons

(D) holes, electrons

(iv) If the frequency of the half-wave rectifier is 50 Hz, the frequency of full-wave rectifier is : *1*

(A) 25 Hz (B) 50 Hz

(C) 100 Hz (D) 200 Hz

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(A) 25 Hz (B) 50 Hz

(C) 100 Hz (D) 200 Hz

(C) 100 Hz

11. Which of the following statements is ***not*** true for a p-n junction diode under reverse bias ?

(A) The current is almost independent of the applied voltage.

(B) Holes flow from p-side to n-side.

(C) Electric field in the depletion region increases.

(D) n-side of the junction is connected to +ve terminal and p-side to –ve terminal of the battery.

CBSE 2024 C

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CBSE 2024 C

(B) Holes flow from p-side to n-side

20. Name the impurity atoms which are doped in an intrinsic semiconductor to convert it into (a) p-type, and (b) n-type semiconductor. Draw energy band diagrams of p-type and n-type semiconductors at temperature $T > 0$ K. Mark the donor and acceptor energy levels, showing the energy difference from the respective bands.

CBSE 2024 C

2

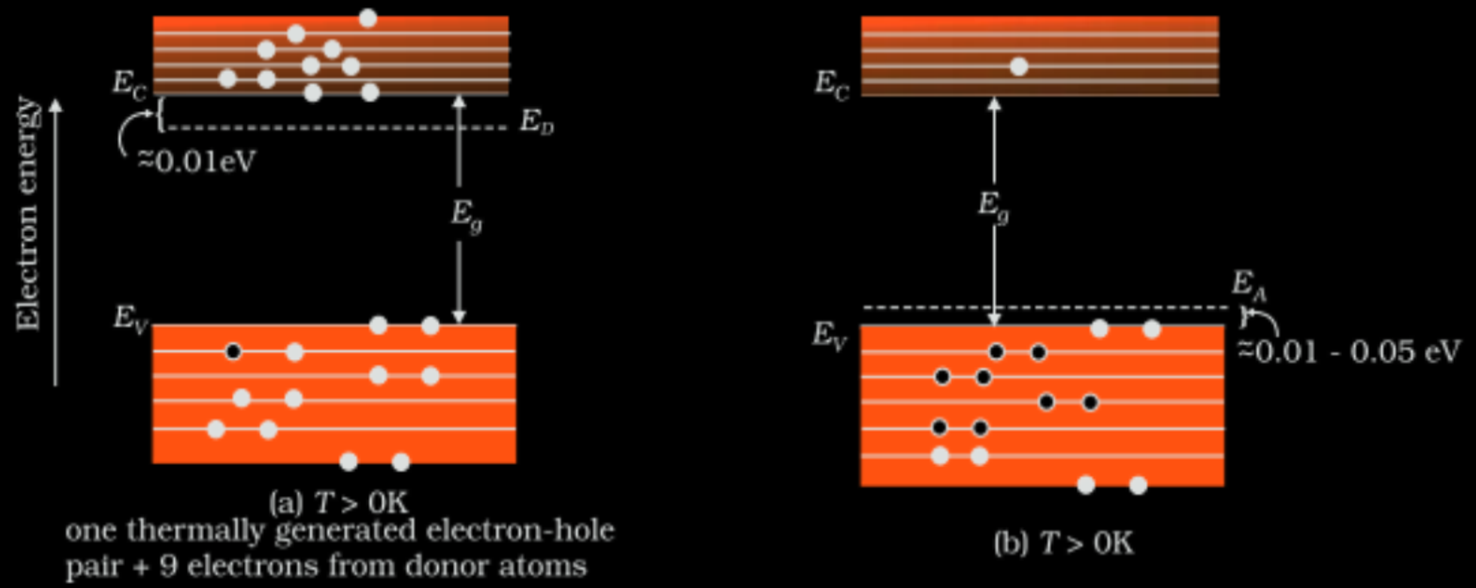
20. Name the impurity atoms which are doped in an intrinsic semiconductor to convert it into (a) p-type, and (b) n-type semiconductor. Draw energy band diagrams of p-type and n-type semiconductors at temperature T > 0 K. Mark the donor and acceptor energy levels, showing the energy difference from the respective bands.

CBSE 2024 C

2

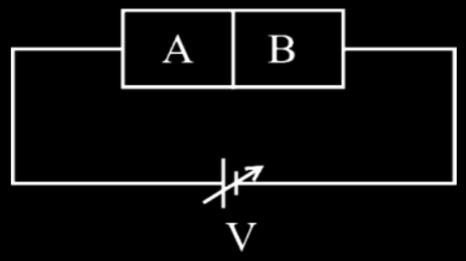
(a) Impurity atom of p- type is trivalent or group 13(acceptor impurity atom) ½

(b) Impurity atom of n- type is pentavalent or group 15(donor impurity atom) ½



½ + ½

17. Two halves of a silicon crystal (A and B) are doped with arsenic and boron respectively, forming a p-n junction in it. A battery is connected across it as shown in the figure.



(a) Will the junction be forward biased or reverse biased ? Give reason.

(b) Draw V-I graph for this arrangement.

CBSE 2024 C

2

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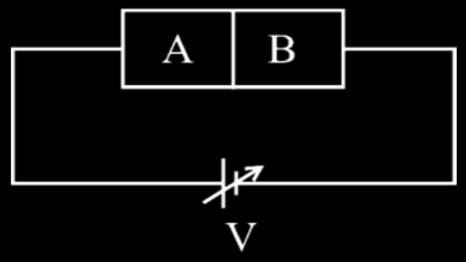
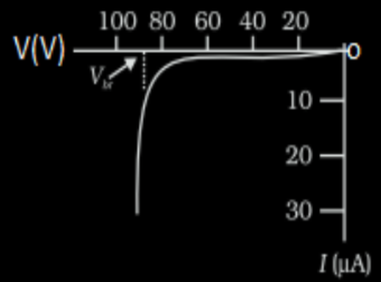


(a) Will the junction be forward biased or reverse biased ? Give reason.

(b) Draw V-I graph for this arrangement. *2*

CBSE 2024 C

(a) p-n junction is reverse biased. Crystal A is doped with arsenic which is pentavalent. Hence it is n-type. Crystal B is doped with boron which is trivalent. Hence it is p-type. (b)  Note : if the graph is drawn without values full credit will be given	½ ½ 1

11. The majority and the minority charge carriers inside a p-type semiconductor are respectively :

(A) electrons, holes

(B) holes, electrons

(C) holes, holes

(D) electrons, electrons

CBSE 2024 C

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CBSE 2024 C

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12. The output of a full-wave rectifier is :

(A) steady

(B) continuous with ripple

(C) continuous without any ripple

(D) zero during alternate half cycles of input

CBSE 2024 C

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(B) continuous with ripple

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(D) zero during alternate half cycles of input

CBSE 2024 C

(B) continuous with ripple

21. Explain the formation of potential barrier across an unbiased p-n junction. Write the effect on height of the barrier when a p-n junction is (a) forward biased, and (b) reverse biased. CBSE 2024 C *2*

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Due to the positive space- charge region on the n-side of the junction & negative space – charge region on the p- side of the junction a potential difference is developed across the junction of the two regions, which is called potential barrier.	1
(a) Barrier height reduces on forward biasing	½
(b) Barrier height increases on reverse biasing	½

28. (a) Explain the formation of valence and conduction band in a crystal.

(b) Discuss the difference between energy bands of (i) a metal, (ii) an insulator, and (iii) a semiconductor. CBSE 2024 C *3*

28. (a) Explain the formation of valence and conduction band in a crystal.

(b) Discuss the difference between energy bands of (i) a metal, (ii) an insulator, and (iii) a semiconductor. CBSE 2024 C *3*

(a) When the atoms come together to form a crystal, the filled energy levels having valence electrons come together to form the valence band and unfilled energy levels come together to form the conductor band above the energy gap.	1 ½
(b)	
(i) In case of metals the conduction band and valence band overlap.	½
(ii) In case of of insulators, a large band gap $E_g > 3eV$ exists between the conduction and valence bands.	½
(iii) In case of of semiconductors, a finite but small band gap $E_g < 3eV$ exists between the conduction and valence bands.	½

7. A pure Si crystal having 5×10^{28} atoms m^{-3} is dopped with 1 ppm concentration of antimony. If the concentration of holes in the doped crystal is found to be 4.5×10^{9} m^{-3}, the concentration (in m^{-3}) of intrinsic charge carriers in Si crystal is about

(A) 1.2×10^{15}

(B) 1.5×10^{16}

(C) 3.0×10^{15}

(D) 2.0×10^{16}

CBSE 2024

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(A) 1.2×10^{15} (B) 1.5×10^{16}

(C) 3.0×10^{15} (D) 2.0×10^{16}

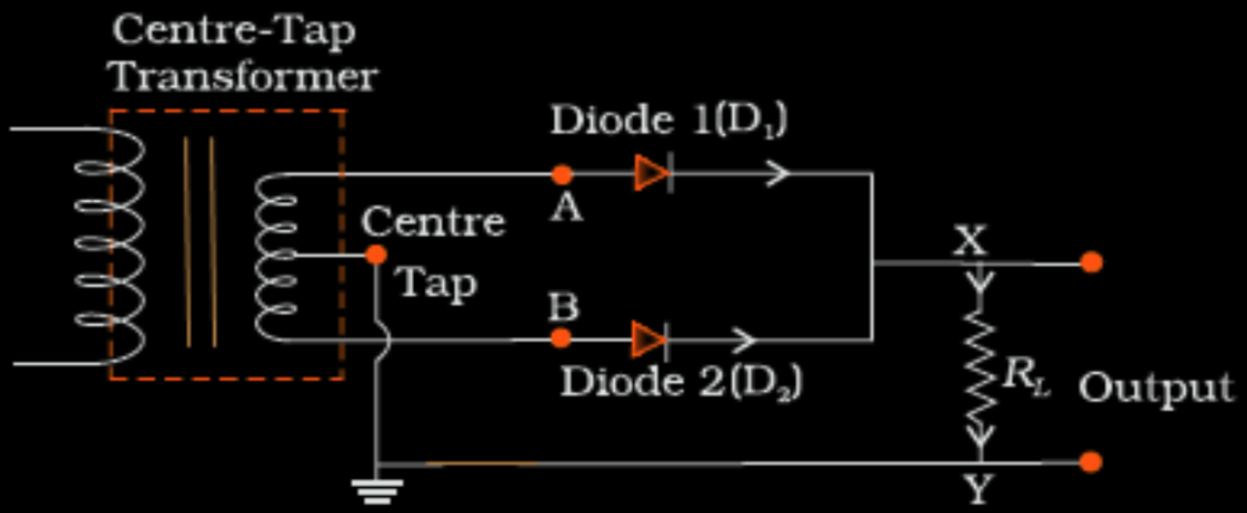
CBSE 2024

(B) 1.5×10^{16}

25. (a) Explain the characteristics of a p-n junction diode that makes it suitable for its use as a rectifier.

(b) With the help of a circuit diagram, explain the working of a full wave rectifier.

CBSE 2024

(a) p-n junction diode allows current to pass only when it is forward biased	1
(b) 	1
When input voltage to A, with respect to the centre tap at any instant is positive, at that instant voltage at B, being out of phase will be negative, diode D_1 gets forward biased and conducts while D_2 being reverse biased does not conduct. Hence during this half cycle an output current and output voltage across R_L is obtained. During second half of the cycle when voltage at A becomes negative with respect to centre tap, the voltage at B would be positive. Hence D_1 would not conduct but D_2 would be giving an output current and output voltage. Thus output voltage is obtained during both halves of the cycle.	1

26. Explain the following, giving reasons :

(a) A doped semiconductor is electrically neutral.

(b) In a p-n junction under equilibrium, there is no net current.

(c) In a diode, the reverse current is practically not dependent on the applied voltage.

CBSE 2024

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(b) In a p-n junction under equilibrium, there is no net current.

(c) In a diode, the reverse current is practically not dependent on the applied voltage.

CBSE 2024

(a) Charge of additional charge carriers is just equal and opposite to that of the ionised cores in the lattice.	1
(b) Under equilibrium, the diffusion current is equal to the drift current.	1
(c) Reverse current is limited due to concentration of minority charge carriers on either side of the junction.	1

15. **Assertion (A) :** The temperature coefficient of resistance is positive for metals and negative for p-type semiconductors.

Reason (R) : The charge carriers in metals are negatively charged, whereas the majority charge carriers in p-type semiconductors are positively charged.

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1

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(B) Both assertion (A) and Reason (R) are true and Reason (R)is not the correct explanation of Assertion(A).

21. Draw the circuit diagram of a p-n junction diode in (i) forward biasing and (ii) reverse biasing. Also draw its I-V characteristics in the two cases. **2**

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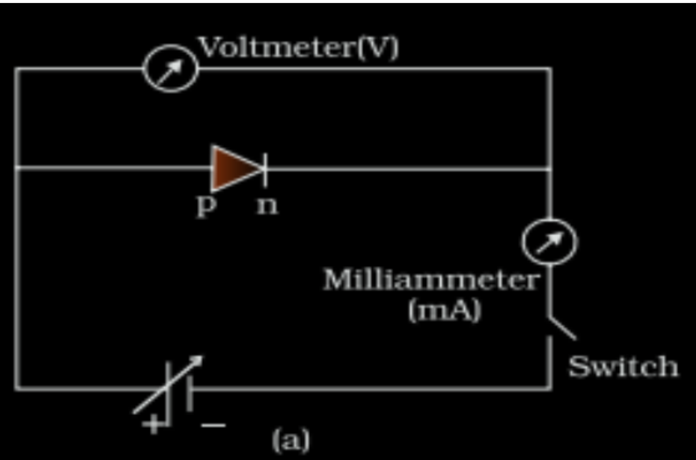
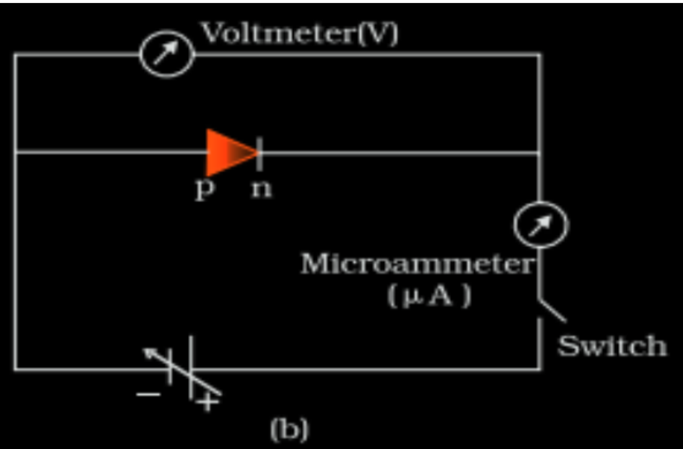
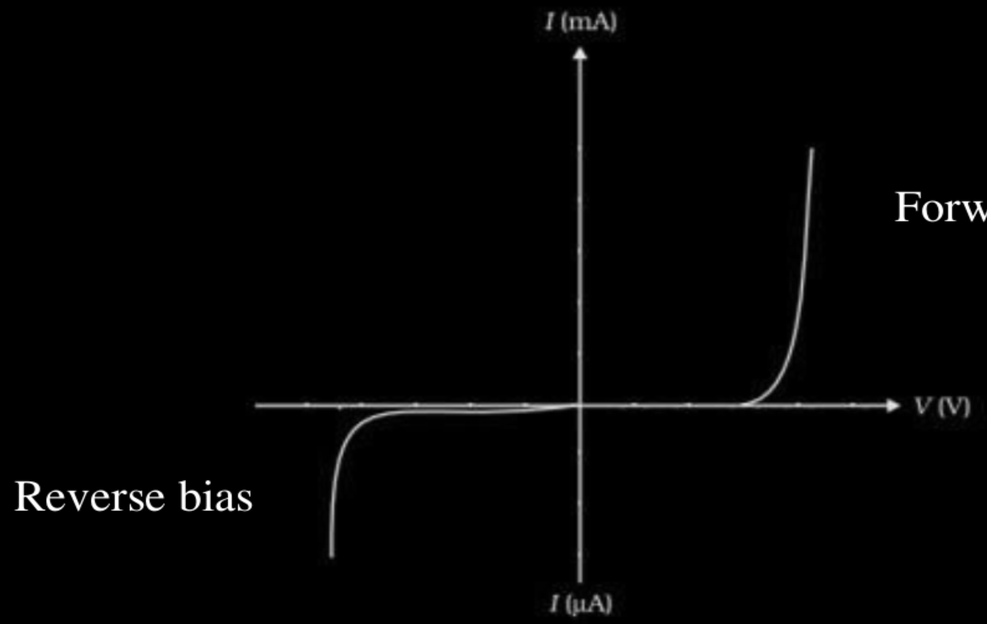




½ + ½

I-V characteristics in forward and reverse bias



½ + ½

29. A pure semiconductor like Ge or Si, when doped with a small amount of suitable impurity, becomes an extrinsic semiconductor. In thermal equilibrium, the electron and hole concentration in it are related to the concentration of intrinsic charge carriers. A p-type or n-type semiconductor can be converted into a p-n junction by doping it with suitable impurity. Two processes, diffusion and drift take place during formation of a p-n junction. A semiconductor diode is basically a p-n junction with metallic contacts provided at the ends for the application of an external voltage. A p-n junction diode allows currents to pass only in one direction when it is forward biased. Due to this property, a diode is widely used to rectify alternating voltages, in half-wave or full wave configuration. **4 × 1 = 4**

(i) When Ge is doped with pentavalent impurity, the energy required to free the weakly bound electron from the dopant is about

(A) 0.001 eV (B) 0.01 eV

(C) 0.72 eV (D) 1.1 eV

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(A) 0.001 eV (B) 0.01 eV

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(B) 0.01 eV

(ii) At a given temperature, the number of intrinsic charge carriers in a semiconductor is 2.0×10^{10} cm^{-3}. It is doped with pentavalent impurity atoms. As a result, the number of holes in it becomes 8×10^{3} cm^{-3}. The number of electrons in the semiconductor is

(A) 2×10^{24} m^{-3}

(B) 4×10^{23} m^{-3}

(C) 1×10^{22} m^{-3}

(D) 5×10^{22} m^{-3}

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(A) 2×10^{24} m^{-3} (B) 4×10^{23} m^{-3}

(C) 1×10^{22} m^{-3} (D) 5×10^{22} m^{-3}

(D) 5×10^{22} m^{-3}

(iii) (a) During the formation of a p-n junction –

(A) electrons diffuse from p-region into n-region and holes diffuse from n-region into p-region.

(B) both electrons and holes diffuse from n-region into p-region.

(C) electrons diffuse from n-region into p-region and holes diffuse from p-region into n-region.

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(D) both electrons and holes diffuse from p-region into n-region.

(C) Electrons diffuse from n-region into p-region and holes diffuse from p-region to n-region.

(iii) (b) Initially during the formation of a p-n junction –

(A) diffusion current is large and drift current is small.

(B) diffusion current is small and drift current is large.

(C) both the diffusion and the drift currents are large.

(D) both the diffusion and the drift currents are small.

(iii) (b) Initially during the formation of a p-n junction –

(A) diffusion current is large and drift current is small.

(B) diffusion current is small and drift current is large.

(C) both the diffusion and the drift currents are large.

(D) both the diffusion and the drift currents are small.

(A) Diffusion current is large and drift current is small.

(iv) An ac voltage $V = 0.5 \sin (100 \pi t)$ volt is applied, in turn, across a half-wave rectifier and a full-wave rectifier. The frequency of the output voltage across them respectively will be

(A) 25 Hz, 50 Hz

(B) 25 Hz, 100 Hz

(C) 50 Hz, 50 Hz

(D) 50 Hz, 100 Hz

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(D) 50 Hz , 100 Hz.

12. Si is doped with a pentavalent element. The energy required to set the additional electron free is about :

(A) 0·01 eV (B) 0·05 eV

(C) 0·72 eV (D) 1·1 eV

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(A) 0·01 eV (B) 0·05 eV

(C) 0·72 eV (D) 1·1 eV

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(B) 0.05 eV

13. *Assertion (A) :* In a semiconductor, the electrons in the conduction band have lesser energy than those in the valence band.

Reason (R) : Donor energy level is just above the valence band in a semiconductor.

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13. *Assertion (A) :* In a semiconductor, the electrons in the conduction band have lesser energy than those in the valence band.

Reason (R) : Donor energy level is just above the valence band in a semiconductor.

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(D) Assertion (A) is false and Reason (R) is also false.

21. Suppose a pure Si crystal has 5×10^{28} atoms m^{-3}. It is doped by 1 ppm concentration of boron. Calculate the concentration of holes and electrons, given that $n_i = 1{\cdot}5 \times 10^{16}\ m^{-3}$. Is the doped crystal n-type or p-type ? *2*

CBSE 2024

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$n_e n_h = n_i^2$	½
$n_h \approx 5\times10^{22} / m^3$	
$n_e = \frac{n_i^2}{n_h}$	
$n_e = \frac{(1.5\times10^{16})^2}{5\times10^{22}}$	½
$n_e = 4.5\times10^{9} / m^3$	½
$n_h > n_e$, it is a p- type crystal	½

28. (a) Draw the circuit diagrams for obtaining the V – I characteristics of a p-n junction diode. Explain briefly the salient features of the V – I characteristics in (i) forward biasing, and (ii) reverse biasing. *3*

CBSE 2024

CBSE 2024

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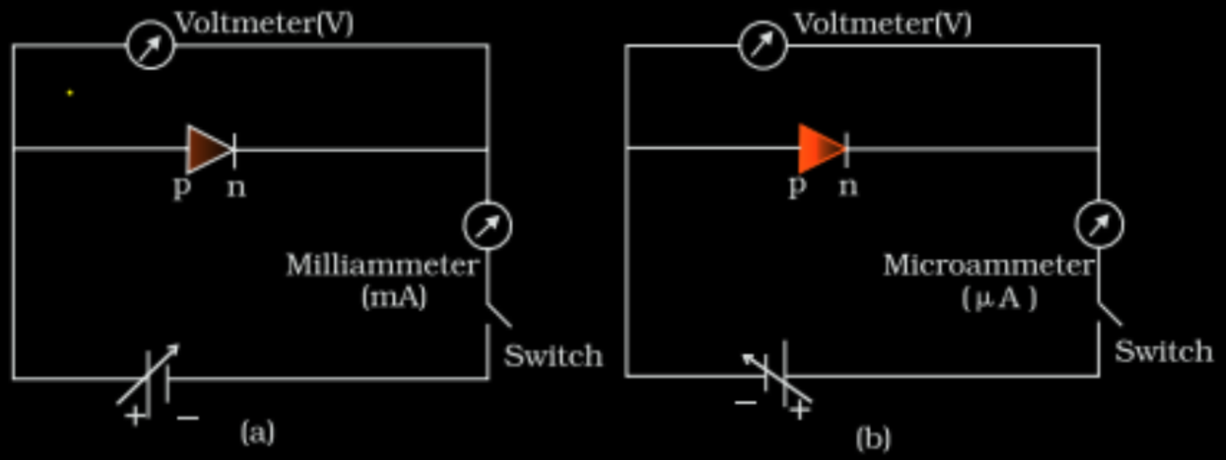

[any one circuit diagram] **1**

Salient features

(i) **Forward biasing**- After threshold voltage or cut in voltage diode current increases significantly (exponentially), even for a small increase in the bias voltage. **1**

(ii) **Reverse biasing**- Current is very small (~μA) and almost remains constant and it increases rapidly after breakdown voltage. **1**

(b) On the basis of energy band diagrams, distinguish between (i) an insulator, (ii) a semiconductor, and (iii) a conductor. *3*

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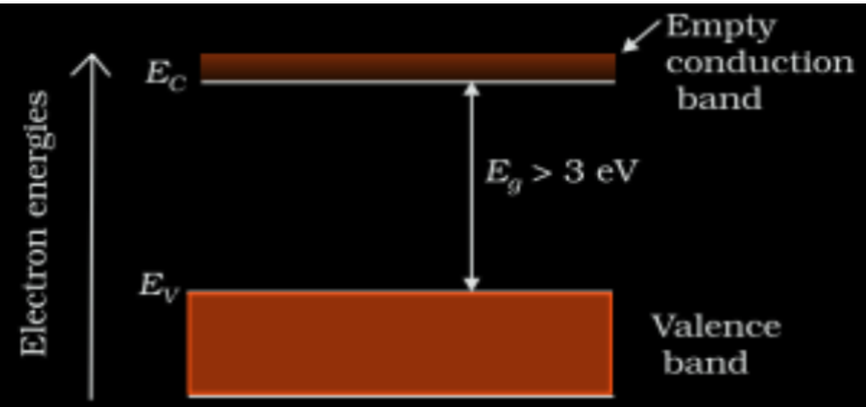
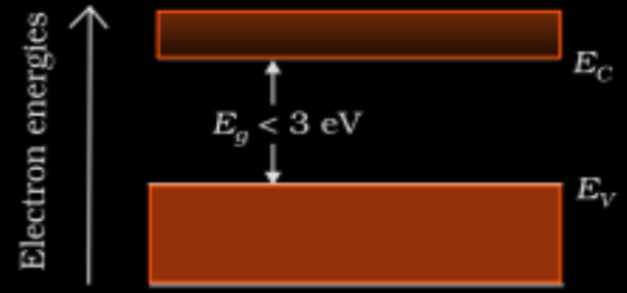
(i)



1

(ii)



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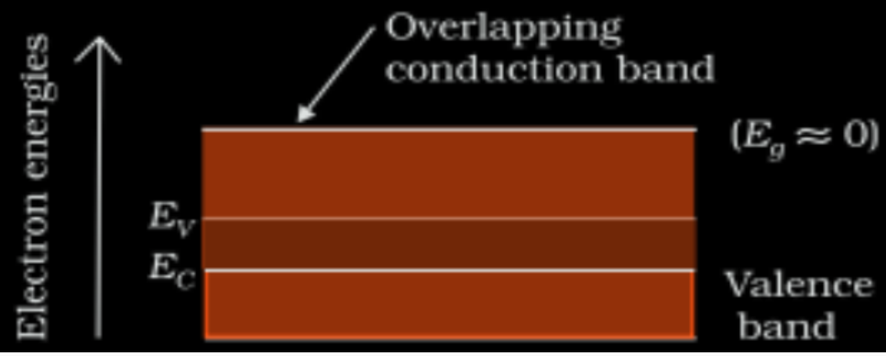
(iii)



1

9. An n-type semiconducting Si is obtained by doping intrinsic Si with :

(A) Al

(B) B

(C) P

(D) In

CBSE 2024

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(A) Al

(B) B

(C) P

(D) In

CBSE 2024

(C) P

11. When a p-n junction diode is subjected to reverse biasing :

(A) the barrier height decreases and the depletion region widens.

(B) the barrier height increases and the depletion region widens.

(C) the barrier height decreases and the depletion region shrinks.

(D) the barrier height increases and the depletion region shrinks.

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(A) the barrier height decreases and the depletion region widens.

(B) the barrier height increases and the depletion region widens.

(C) the barrier height decreases and the depletion region shrinks.

(D) the barrier height increases and the depletion region shrinks.

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(B) The barrier height increases and the depletion region widens.

18. How does the energy gap of an intrinsic semiconductor effectively change when doped with a (a) trivalent impurity, and (b) pentavalent impurity ? Justify your answer in each case. CBSE 2024 *2*

18. How does the energy gap of an intrinsic semiconductor effectively change when doped with a (a) trivalent impurity, and (b) pentavalent impurity ? Justify your answer in each case. CBSE 2024 *2*

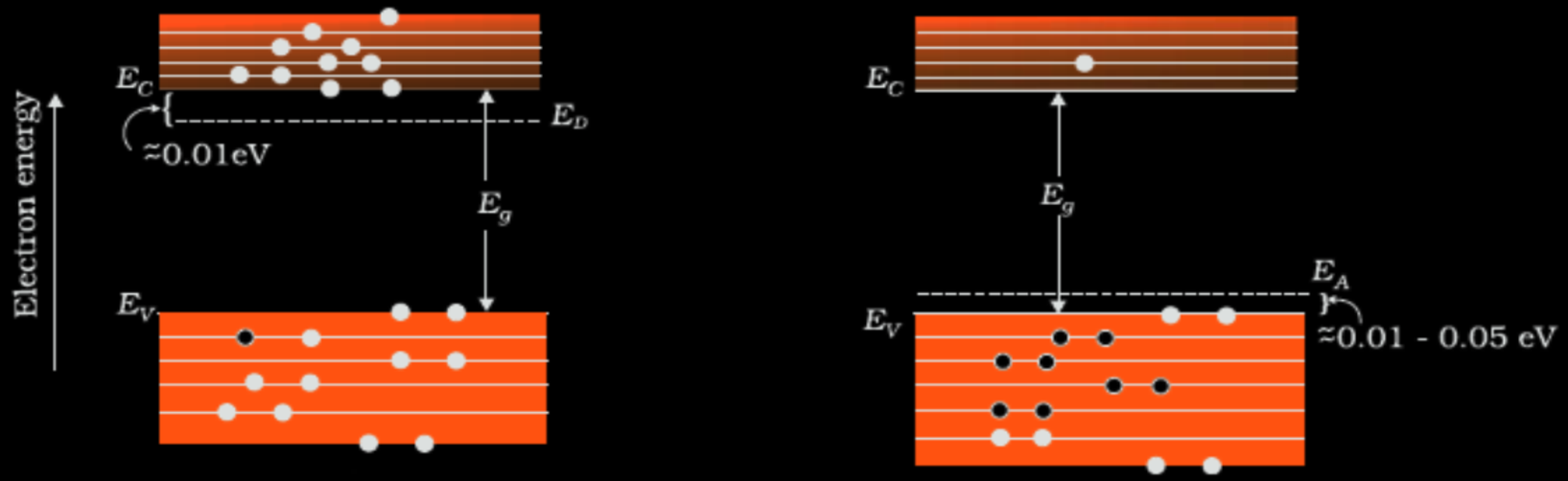
(i) Decreases ½

Justification: An acceptor energy level is formed just above the top of the valence band. ½

(ii) Decreases ½

Justification: A donor level is formed just below the bottom of conduction band. ½

Alternatively



(Note : Award the credit of justification if a student draws band diagram)

28. With the help of a circuit diagram, explain the working of a p-n junction diode as a full wave rectifier. Draw its input and output waveforms. *3*

CBSE 2024

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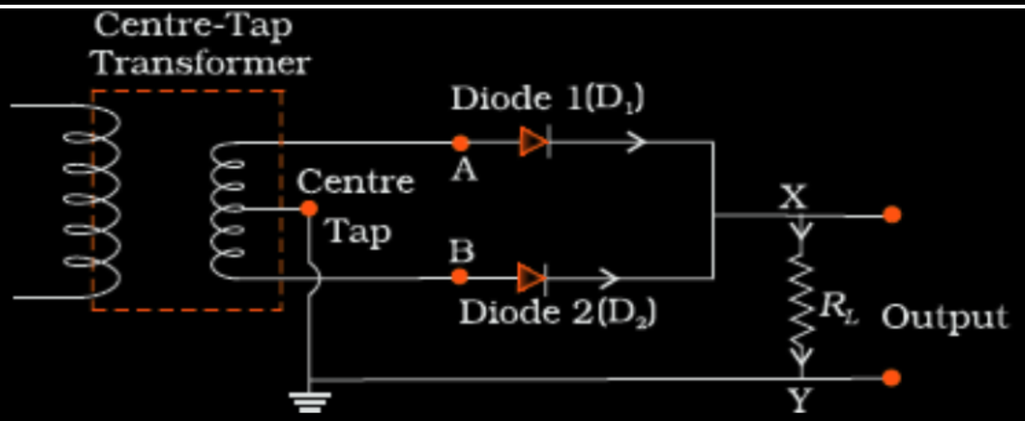

1

When input voltage at A with respect to the centre tap at any instant is positive, at that instant voltage at B, being out of phase will be negative, during the positive half cycle diode D_1 gets forward biased and conducts while diode D_2 gets reverse biased and does not conduct.
Hence during positive half cycle an output current and output voltage across R_L is obtained. **½**

During second half of the cycle when voltage at A becomes negative with respect to centre tap, the voltage at B would be positive hence D_1 would not conduct but D_2 would be giving an output current and output voltage.
We get output voltage in both positive and negative half cycles. **½**

12. Ge is doped with As. Due to doping,

(A) the structure of Ge lattice is distorted.

(B) the number of conduction electrons increases.

(C) the number of holes increases.

(D) the number of conduction electrons decreases.

CBSE 2024

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21. Differentiate between 'diffusion current' and 'drift current'. Explain their role in the formation of p-n junction. CBSE 2024 *2*

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Diffusion current: It is due to the flow of majority charge carriers. Drift current: It is due to the flow of minority charge carriers.	1
Due to the concentration gradient across p- , and n- sides, holes diffuse from p- side to n- side & electrons diffuse from n-side to p- side. This motion of charge carriers gives rise to diffusion current across the junction. Due to the positive space- charge region on n-side and negative space – charge region on p-side, an electric field directed from positive charge towards negative charge develops. Due to this field, motion of charge carriers comes into play giving rise to drift current.	1

Junction Diode as a Rectifier :

The process of conversion of an ac voltage into a dc voltage is called rectification and the device which performs this conversion is called a rectifier. The characteristics of a p-n junction diode reveal that when a p-n junction diode is forward biased, it offers a low resistance and when it is reverse biased, it offers a high resistance. Hence, a p-n junction diode conducts only when it is forward biased. This property of a p-n junction diode makes it suitable for its use as a rectifier.

Thus, when an ac voltage is applied across a p-n junction, it conducts only during those alternate half cycles for which it is forward biased. A rectifier which rectifies only half cycle of an ac voltage is called a half-wave rectifier and one that rectifies both the half cycles is known as a full-wave rectifier.

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(i) The root mean square value of an alternating voltage applied to a full-wave rectifier is $\frac{V_0}{\sqrt{2}}$. Then the root mean square value of the rectified output voltage is : *1*

(A) $\frac{V_0}{\sqrt{2}}$ (B) $\frac{V_0^2}{\sqrt{2}}$

(C) $\frac{2V_0}{\sqrt{2}}$ (D) $\frac{V_0}{2\sqrt{2}}$

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(A) $\frac{V_0}{\sqrt{2}}$ (B) $\frac{V_0^2}{\sqrt{2}}$

(C) $\frac{2V_0}{\sqrt{2}}$ (D) $\frac{V_0}{2\sqrt{2}}$

(A) $\frac{Vo}{\sqrt{2}}$

(ii) In a full-wave rectifier, the current in each of the diodes flows for : *1*

(A) Complete cycle of the input signal

(B) Half cycle of the input signal

(C) Less than half cycle of the input signal

(D) Only for the positive half cycle of the input signal

(ii) In a full-wave rectifier, the current in each of the diodes flows for : *1*

(A) Complete cycle of the input signal

(B) Half cycle of the input signal

(C) Less than half cycle of the input signal

(D) Only for the positive half cycle of the input signal

(B) half cycle of the input signal

(iii) In a full-wave rectifier : *1*

(A) Both diodes are forward biased at the same time.

(B) Both diodes are reverse biased at the same time.

(C) One is forward biased and the other is reverse biased at the same time.

(D) Both are forward biased in the first half of the cycle and reverse biased in the second half of the cycle.

(iii) In a full-wave rectifier : *1*

(A) Both diodes are forward biased at the same time.

(B) Both diodes are reverse biased at the same time.

(C) One is forward biased and the other is reverse biased at the same time.

(D) Both are forward biased in the first half of the cycle and reverse biased in the second half of the cycle.

(C) One is forward biased and the other is reverse biased at the

(iv) (a) An alternating voltage of frequency of 50 Hz is applied to a half-wave rectifier. Then the ripple frequency of the output will be : *1*

(A) 100 Hz (B) 50 Hz

(C) 25 Hz (D) 150 Hz

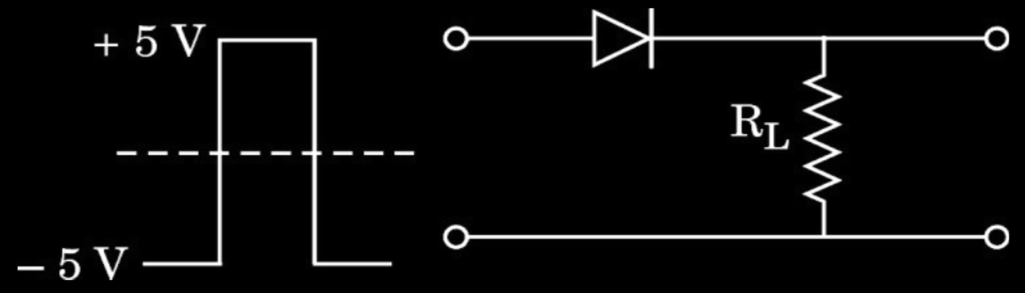
(iv) (a) An alternating voltage of frequency of 50 Hz is applied to a half-wave rectifier. Then the ripple frequency of the output will be : *1*

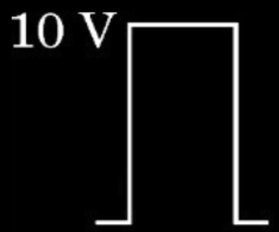
(A) 100 Hz (B) 50 Hz

(C) 25 Hz (D) 150 Hz

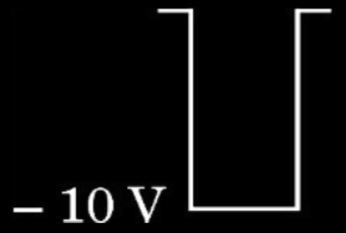
(B) 50 Hz

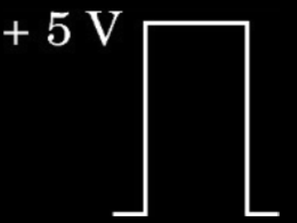
(b) A signal, as shown in the figure, is applied to a p-n junction diode. Identify the output across resistance R_L : *1*



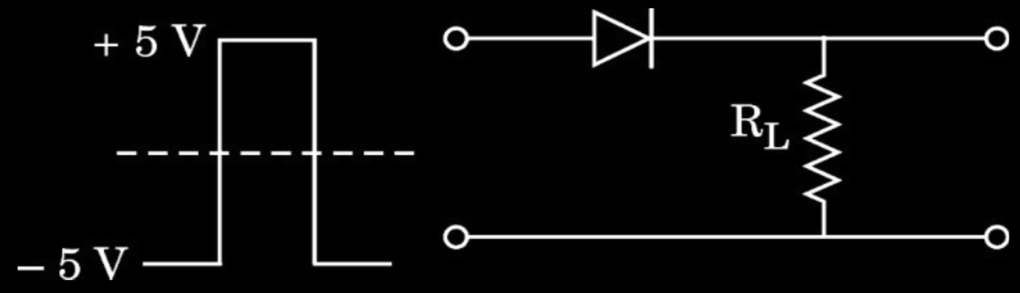
(A) 

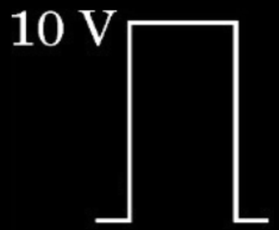
(C)

(B) 

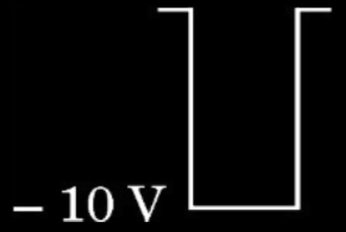
(D) 

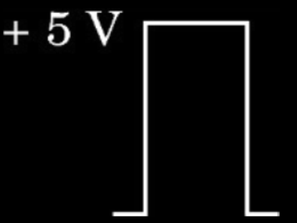
(b) A signal, as shown in the figure, is applied to a p-n junction diode. Identify the output across resistance R_L : *1*



(A) 

(C)

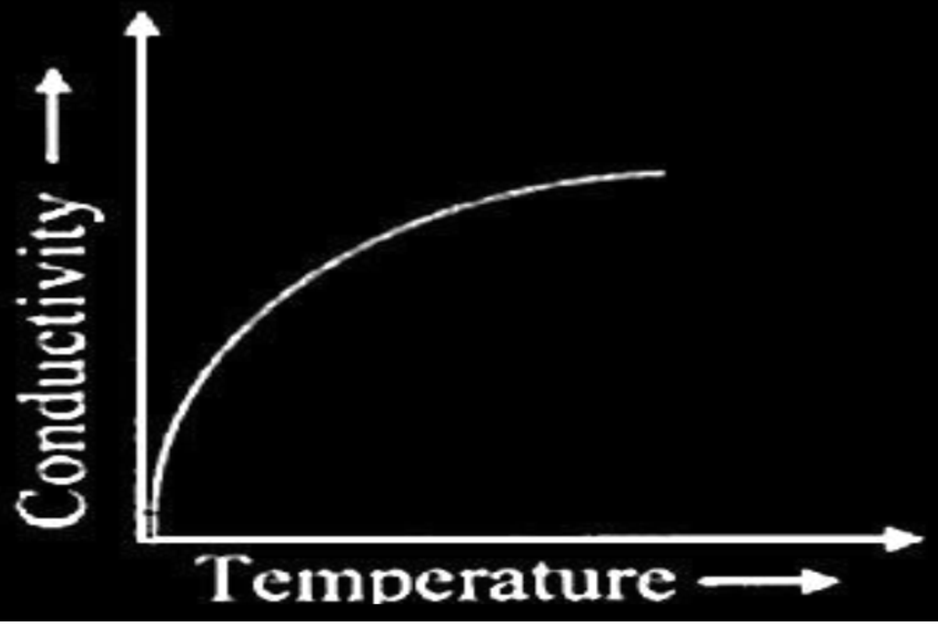
(B) 

(D) 

(D)

21. How does the conductivity of an intrinsic semiconductor vary with temperature ? Explain. Show the variation in a plot. CBSE 2024 *2*

21. How does the conductivity of an intrinsic semiconductor vary with temperature ? Explain. Show the variation in a plot. CBSE 2024 *2*

Conductivity will increase.	½
As the temperature increase , more thermal energy becomes available to these electrons and some of these electrons may break -away (becoming free electrons contributing to conduction)	½
	1

11. The effect on width of depletion layer and height of the potential barrier of a p-n junction, when it is forward biased, are respectively :

(A) decreases, decreases

(B) increases, increases

(C) decreases, increases

(D) increases, decreases

CBSE 2024

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CBSE 2024

(A) decreases, decreases

12. The doping of a semiconductor with donor impurity results in :

(A) formation of a p-type semiconductor

(B) formation of a new energy level just above the filled valence band

(C) decrease in the concentration of electrons

(D) formation of a new energy level just below the conduction band

CBSE 2024

12. The doping of a semiconductor with donor impurity results in :

(A) formation of a p-type semiconductor

(B) formation of a new energy level just above the filled valence band

(C) decrease in the concentration of electrons

(D) formation of a new energy level just below the conduction band

CBSE 2024

(D) formation of new energy level just below the conduction band

21. What are the important considerations for choosing dopants for silicon or germanium ? Justify your answer. CBSE 2024 *2*

21. What are the important considerations for choosing dopants for silicon or germanium ? Justify your answer. CBSE 2024 *2*

The important consideration for choosing dopants are (i) The dopants atoms must be either trivalent or pentavalent. (ii) The size of the dopants must be comparable to the size of the atom of the semiconductor.	1
<u>Justification</u>: (i) Trivalent or pentavalent dopants belong to nearby fourth group of periodic table to which Si and Ge belong. (ii) The size of the dopant should not distort the lattice structure of the semiconductor.	1

28. Explain the formation of energy bands in solids and hence define conduction band and valence band. CBSE 2024 *3*

28. Explain the formation of energy bands in solids and hence define conduction band and valence band. CBSE 2024 *3*

According to Bohr atomic model, in an isolated atom, the energy of any of the electrons is decided by the orbit in which it revolves. But when the atoms come together to form a solid, they are close to each other so outer orbits of electrons from neighbouring atoms would come close or even could overlap.	1
Because of this each electron will have unique position inside the crystal and have different energy level. This different energy level with continuous energy variations forms energy bands.	1
The energy band which includes the energy level of valence electrons & is completely occupied at 0K is called Valence Band.	½
The energy band which contains free electrons that participate in process of conduction (or conduction electrons) & is completely vacant at 0K is called Conduction Band.	½

13. Which of the following dopants in silicon will make it a p-type semiconductor ?

(a) As (b) Al

(c) Sb (d) P

CBSE 2023 C

13. Which of the following dopants in silicon will make it a p-type semiconductor ?

(a) As (b) Al

(c) Sb (d) P

CBSE 2023 C

(b) Al

15. When a p-n junction diode is forward biased :

(a) the depletion layer width increases and the barrier height is reduced.

(b) the depletion layer width decreases and the barrier height is reduced.

(c) both the depletion layer width and the barrier height increase.

(d) the depletion layer width decreases and the barrier height increases.

CBSE 2023 C

15. When a p-n junction diode is forward biased :

(a) the depletion layer width increases and the barrier height is reduced.

(b) the depletion layer width decreases and the barrier height is reduced.

(c) both the depletion layer width and the barrier height increase.

(d) the depletion layer width decreases and the barrier height increases.

CBSE 2023 C

(b) the depletion layer width decreases and barrier height is reduced.

---

24. Briefly explain how the valence band and conduction band are formed in a crystal. CBSE 2023 C *2*

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In a crystal, the atoms are close to each other and therefore the electrons interact with each other and also with neighboring atomic cores. This interaction of electrons causes the formation of energy bands. The energy band which includes the energy level of electrons is called valence band. The energy band above the valence band is called conduction band.	1½ ½

25. Explain how a depletion region is formed in a p-n junction. *2*

CBSE 2023 C

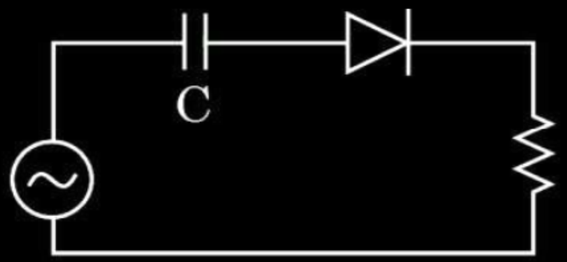
25. Explain how a depletion region is formed in a p-n junction. *2*

CBSE 2023 C

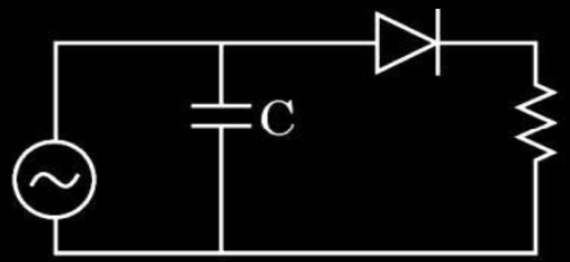
When a hole diffuses from p→n due to the concentration gradient, it leaves behind an ionized acceptor (negative charge) which is immobile. As the holes continue to diffuse,a layer of negative charge(or negative space charge region) on the p-side of the junction is developed. The space charge region on either side of the junction together is known as depletion region.	2

14. In which of the following diagrams is the capacitor 'C' connected correctly to provide smooth output of a half-wave rectifier ?

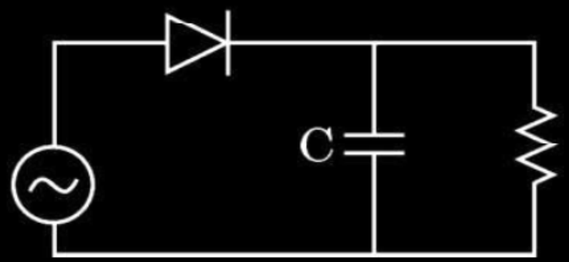
(a)



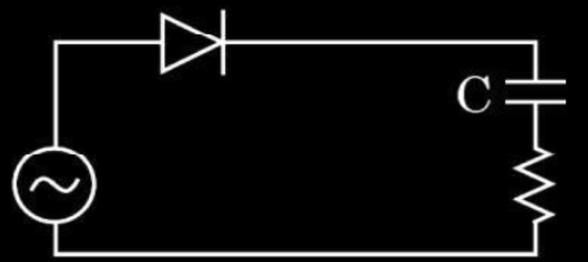
(b)



(c)



(d)



CBSE 2023 C

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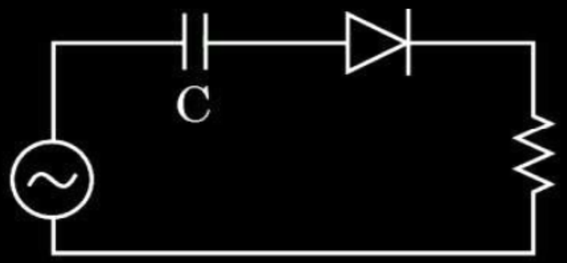
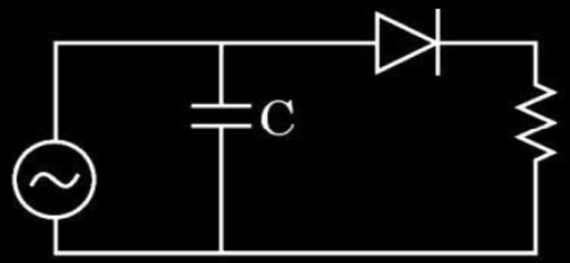
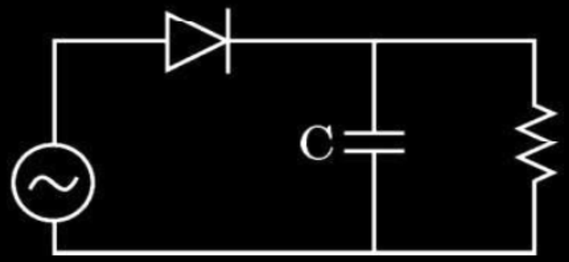


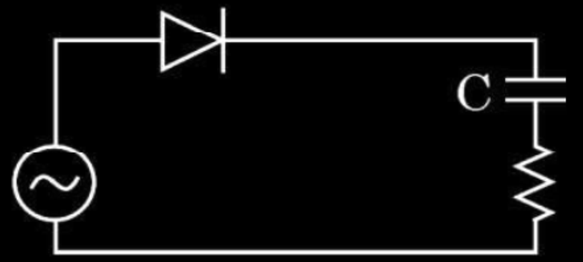
(b)



(c)

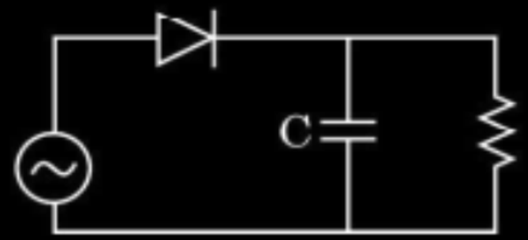


(d)



CBSE 2023 C

(c)



18. *Assertion (A) :* Silicon is preferred over germanium for making semiconductor devices.

Reason (R) : The energy gap for germanium is more than the energy gap for silicon.

CBSE 2023 C

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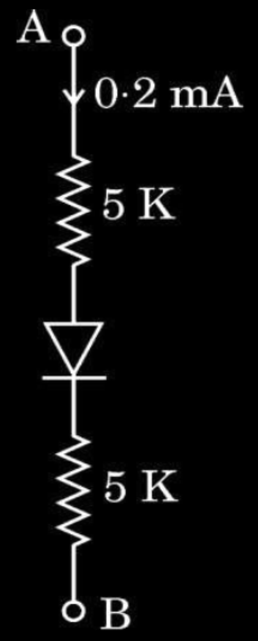
CBSE 2023 C

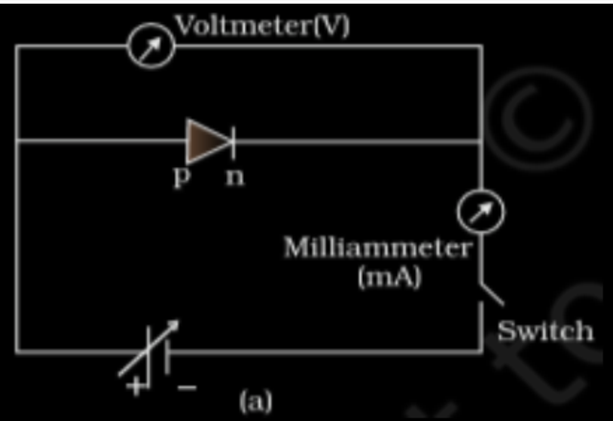
(c) Assertion (A) is true, but Reason (R) is false.

31. (a) (i) Draw the circuit diagram used to study I – V characteristics of a p-n junction diode in conducting mode. Mark on the graph the threshold voltage of the diode. Explain the significance of this voltage.

(ii) In the circuit shown in the figure, the forward voltage drop across the diode is 0·3 V. Find the voltage difference between A and B. *5*

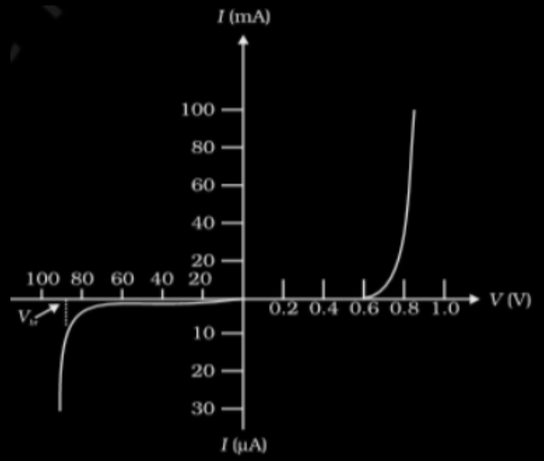


CBSE 2023 C



(a)

1



1

½

Beyond threshold voltage in forward bias diode current increases significantly even for very small increases in diode bias voltage.

½

(ii) $V_A - 5\times10^3\times0.2\times10^{-3} - 0.3 - 5\times10^3\times0.2\times10^{-3} - V_B = 0$

$V_A - V_B$ =2.3 volt

2

(b) (i) Briefly describe the classification of solids into metals, insulators and semi-conductors on the basis of energy level diagrams.

(ii) In a silicon diode, the current increases from 10 mA to 20 mA when the voltage changes from 0·6 V to 0·7 V. Calculate the dynamic resistance of the diode. CBSE 2023 C *5*

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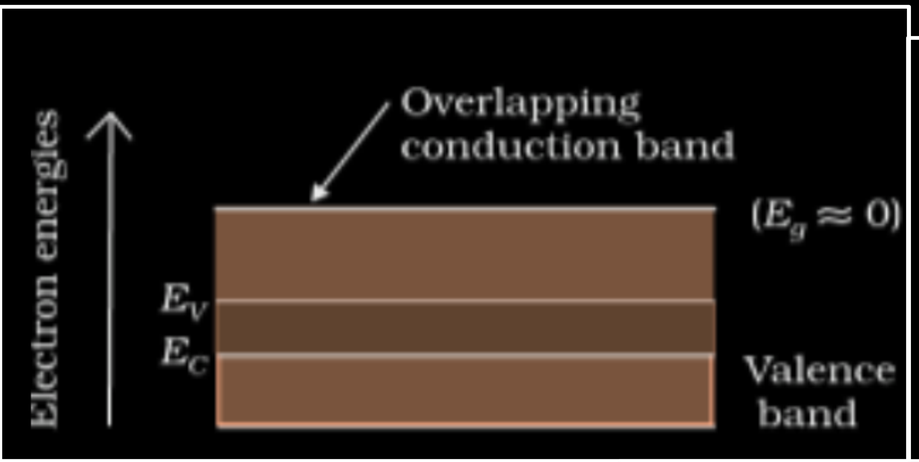
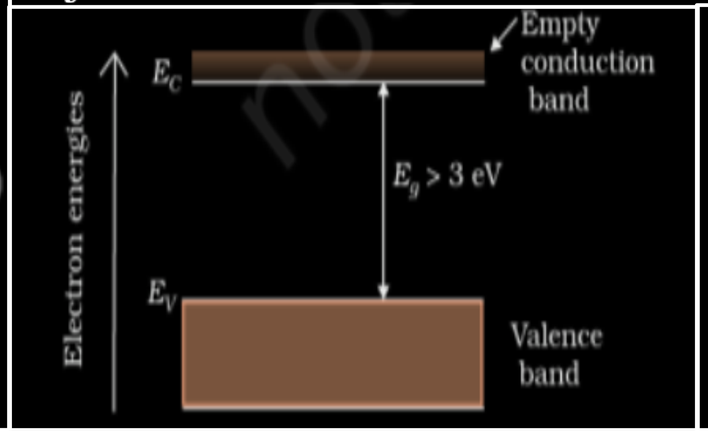
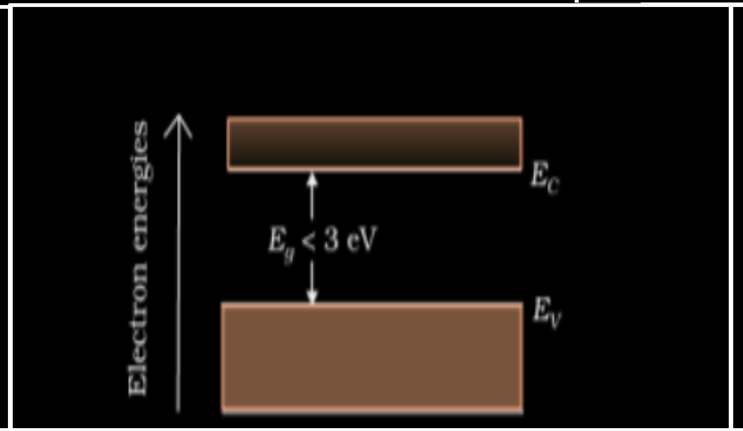
CBSE 2023 C

5







½ + ½+ ½

For Eg> 3eV material is insulate
For Eg < 3eV material is semiconductor
For Eg=0 or overlapping of conduction and valence band material is conductor.

½ + ½+ ½

(ii) $r_d = \frac{\triangle V}{\triangle I} = \frac{0.7 - 0.6}{(20-10)\times 10^{-3}}$

½ + ½

$r_d = 10\ \Omega$

1

14. Which of the following is ***not*** a semiconductor ? CBSE 2023

(a) Ge (b) Si (c) Sn (d) CdS

14. Which of the following is ***not*** a semiconductor ? CBSE 2023

(a) Ge (b) Si (c) Sn (d) CdS

(c) Sn

15. In an intrinsic semiconductor, the intrinsic carrier concentration (n_i), electron concentration (n_e) and hole concentration (n_h) are related as :

CBSE 2023

(a) $n_e + n_h = n_i$

(b) $n_i = \sqrt{n_e\, n_h}$

(c) $n_e = n_h = n_i$

(d) $n_e + n_h = \frac{n_i}{2}$

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Reason (R) : Silicon can be used at a higher temperature as compared to Germanium.

CBSE 2023

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(a) [A]- True [R]- True and correct explanation

24. Define the following terms in relation to the working of a p-n junction diode : *2*

(i) Knee voltage

(ii) Reverse saturation current

CBSE 2023

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(i) Knee voltage

(ii) Reverse saturation current

CBSE 2023

(i) In forward bias of a p-n junction diode the external applied voltage beyond which the current begins to flow rapidly is called Knee Voltage.	1
(ii) In reverse bias of a p-n junction diode the current is due to minority carriers which is not very much dependent on the applied voltage. This current is called reverse saturation current.	1

25. (a) How are the potential barrier and width of the depletion region affected when a p-n junction diode is (i) forward-biased, and (ii) reverse-biased ? CBSE 2023 *2*

25. (a) How are the potential barrier and width of the depletion region affected when a p-n junction diode is (i) forward-biased, and (ii) reverse-biased ? CBSE 2023 *2*

(i) In forward bias	
Potential barrier decreases	½
Width of depletion layer decreases	½
(ii) In reverse bias	
Potential barrier increases	½
Width of depletion layer increases	½

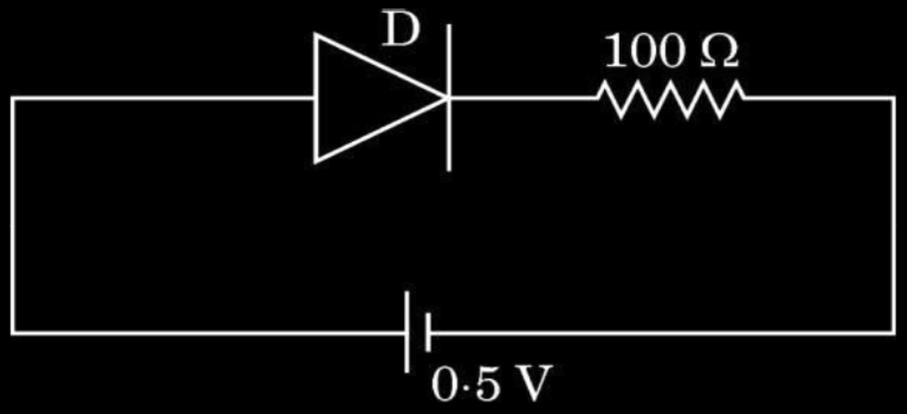
(b) Briefly explain doping of a semiconductor and its necessity. *2*

(b) Briefly explain doping of a semiconductor and its necessity. CBSE 2023 *2*

Doping is the process of adding impurity atom in an intrinsic semiconductor deliberately. An intrinsic semiconductor has low conductivity. By doping, conductivity of semiconductor is increased	1 1

13. The threshold voltage for a p-n junction diode used in the circuit is 0·7 V. The type of biasing and current in the circuit are :

CBSE 2023



(a) Forward biasing, 0 A

(b) Reverse biasing, 0 A

(c) Forward biasing, 5 mA

(d) Reverse biasing, 2 mA

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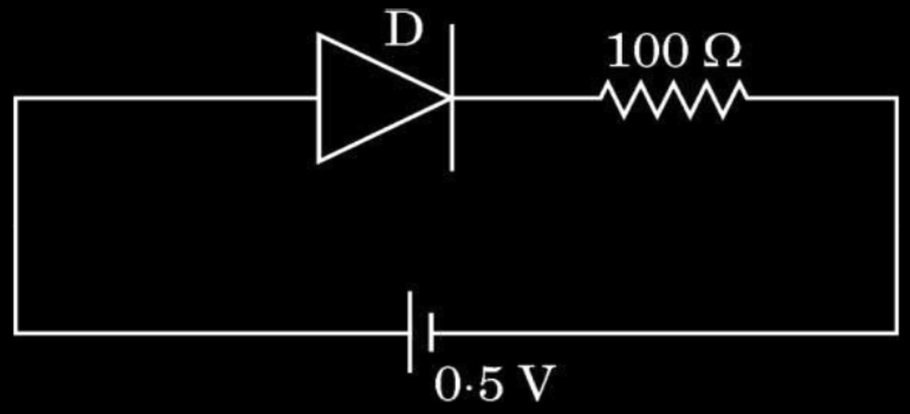

(a) Forward biasing, 0 A

(b) Reverse biasing, 0 A

(c) Forward biasing, 5 mA

(d) Reverse biasing, 2 mA

(a) Forward biasing, 0 A

16. *Assertion (A) :* In 'n' type semiconductor, number density of electrons is greater than the number density of holes but the crystal maintains an overall charge neutrality.

Reason (R) : The charge of electrons donated by donor atoms is just equal and opposite to that of the ionised donor.

CBSE 2023

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Reason (R) : The charge of electrons donated by donor atoms is just equal and opposite to that of the ionised donor.

CBSE 2023

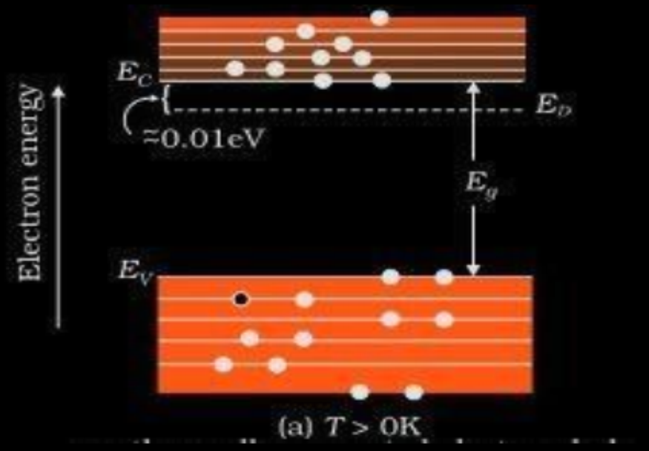
(a) Both Assertion (A) and Reason (R) are true, and Reason (R) is the correct explanation of the Assertion (A).

31. (a) (i) A germanium crystal is doped with antimony. With the help of energy-band diagram, explain how the conductivity of the doped crystal is affected.

(ii) Briefly explain the two processes involved in the formation of a p-n junction.

(iii) What will the effect of (1) forward biasing, and (2) reverse biasing be on the width of depletion layer in a p-n junction diode ?

CBSE 2023 *5*

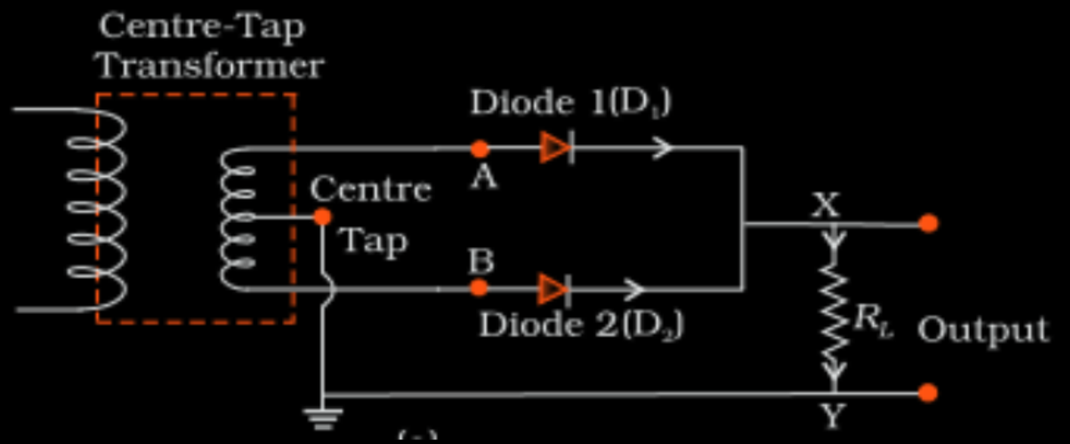
(i) With proper level of doping, the number of conduction electrons can be made much larger than the number of holes. Due to this conductivity of the doped crystal increases.	1
	
(ii)Two processes (a) Diffusion (b) drift	1
Diffusion: Due to concentration gradient majority charge carrier that is electron moves from $n \rightarrow p$ side and holes to $p \rightarrow n$ side. This movement of charges is called diffusion.	1
Drift: Due to the junction field, an electron on p-side of the junction moves to n- side and a hole on n- side of the junction moves to p- side. The motion of the charge carrier due to electric field is called drift.	1
(iii) (1) decreases (2) increases	½ ½

(b) (i) With the help of a circuit diagram, briefly explain the working of a full-wave rectifier using p-n junction diodes.

(ii) Draw V – I characteristics of a p-n junction diode. Explain how these characteristics make a diode suitable for rectification.

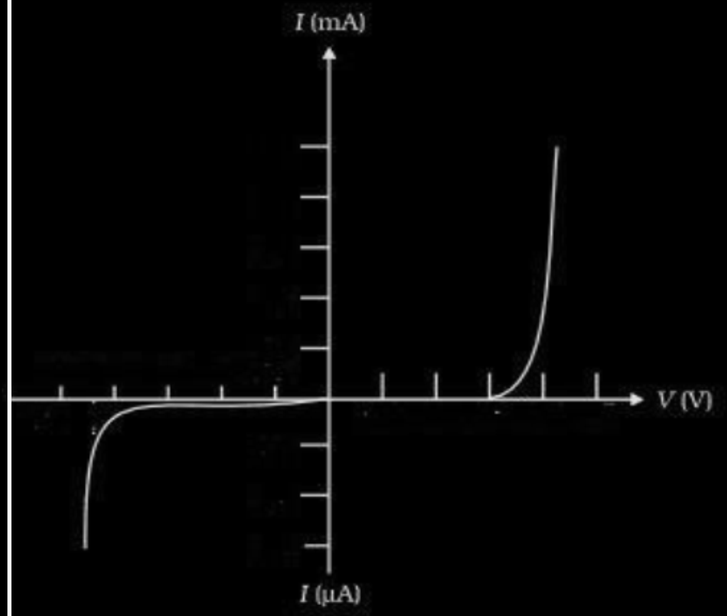
(iii) Carbon and silicon have the same lattice structure. Then why is carbon an insulator but silicon a semiconductor ? *5*

CBSE 2023



1

(ii)V-I characteristics



Working: Suppose the input voltage to A with respect to the centre tap at any instant is positive. At that instant voltage at B, being out of phase will be negative. So diode D_1 gets forward biased and conducts, while D_2 being reverse biased does not conduct. Similarly during second half of the cycle polarity get reversed so only D_2 will conduct. 1

This diagram shows that the diode conducts when forward biased and 1
does not conduct when reverse biased. This characteristics makes it 1
suitable for use for rectification.

(iii)The 4 bonding of electrons of C and Si lie respectively, in the second and third orbit. Hence energy required to take out an electron from their atoms will be much less than that for C. Hence number of free e^- for conduction in Si significant but negligibly small for C. 1

7. For the forward biasing of a p-n junction diode, which of the following statements is ***not*** correct ?

(a) The potential barrier decreases.

(b) Minority carrier injection occurs.

(c) Width of depletion layer increases.

(d) Forward current is due to the diffusion of both holes and electrons.

CBSE 2023

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CBSE 2023

(c) Width of depletion layer increases.

11. At a certain temperature in an intrinsic semiconductor, the electrons and holes concentration is $1{\cdot}5 \times 10^{16}\ m^{-3}$. When it is doped with a trivalent dopant, hole concentration increases to $4{\cdot}5 \times 10^{22}\ m^{-3}$. In the doped semiconductor, the concentration of electrons (n_e) will be :

(a) $3 \times 10^{6}\ m^{-3}$

(b) $5 \times 10^{7}\ m^{-3}$

(c) $5 \times 10^{9}\ m^{-3}$

(d) $6{\cdot}75 \times 10^{38}\ m^{-3}$

CBSE 2023

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CBSE 2023

(c) $5 \times 10^{9}\ m^{-3}$

19. Explain the property of a p-n junction which makes it suitable for rectifying alternating voltages. Differentiate between a half-wave and a full-wave rectifier. CBSE 2023 *2*

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pn-junction conducts in forward bias. **Alternatively:-**pn-junction is a uni-directional device. The half-wave rectifier gives output only for half of the input cycle .The full-wave rectifier gives output for both the halves of the input cycles. **Alternatively:-**If output waveform of both the rectifiers shown diagrammatically, then full credit to be given.	1 1

25. Draw a circuit diagram of a p-n junction diode in (a) forward biasing, and (b) reverse biasing. Draw the V – I characteristics for each case. CBSE 2023 *2*

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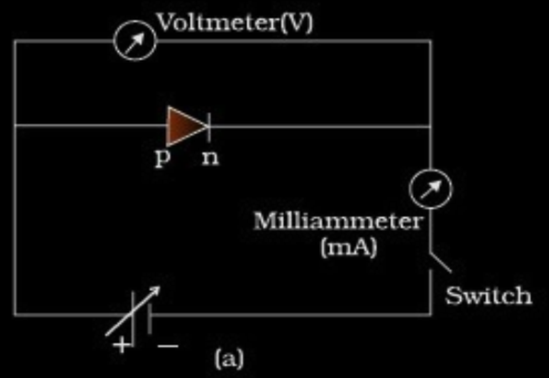
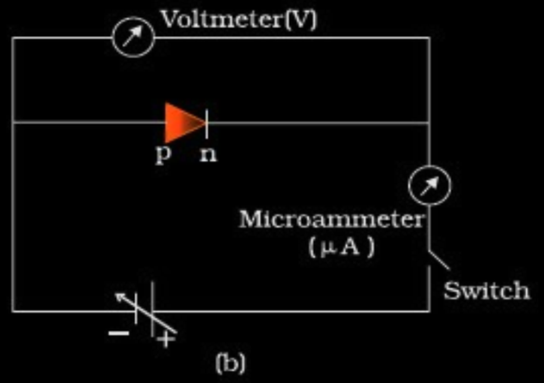
(a) Forward Bias



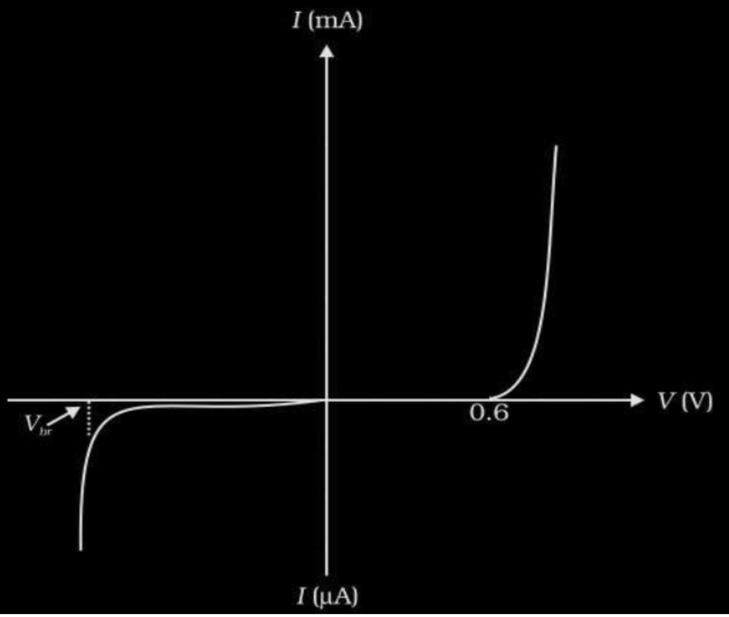
(b) Reverse Bias



½

½

VI Characteristics



½+½

5. At 0 K, the resistivity of an intrinsic semiconductor is :

(a) same as that at 0°C

(b) same as that at 300 K

(c) zero

(d) infinite

CBSE 2023

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(c) zero (d) infinite

CBSE 2023

(d) Infinite

19. How is the width of depletion layer of a p-n junction diode affected when it is (i) forward biased, and (ii) reverse biased ? Justify your answers. *2*

CBSE 2023

19. How is the width of depletion layer of a p-n junction diode affected when it is (i) forward biased, and (ii) reverse biased ? Justify your answers. *2*

CBSE 2023

(i) Forward Bias- Decreases The direction of the applied voltage (V) is opposite to the built-in potential V_o	½ ½
(ii) Reverse Bias- Increases The direction of the applied voltage (V) is same as the built-in potential V_o	½ ½

CBSE 2023

14. If a p-n junction diode is reverse biased,

(a) the potential barrier is lowered.

(b) the potential barrier remains unaffected.

(c) the potential barrier is raised.

(d) the current is mainly due to majority carriers.

CBSE 2023

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24. Explain the roles of diffusion current and drift current in the formation of the depletion layer in a p-n junction diode. CBSE 2023 *2*

24. Explain the roles of diffusion current and drift current in the formation of the depletion layer in a p-n junction diode. CBSE 2023 *2*

During the formation of p-n junction ,and due to the concentration gradient across p-,and n-sides, holes diffuse from p-side to n-side $(p \rightarrow n)$ and electrons diffuse from n-side to p-side$(n \rightarrow p)$.When an electron diffuses from $(n \rightarrow p)$, it leaves behind an ionized donor(positive charge) on n-side which is immobile. Similarly, when a hole diffuses from $(p \rightarrow n)$ due to the concentration gradient, it leaves behind an ionised acceptor (negative charge) which is immobile. This space–charge region on either side of the junction together is known as depletion region.	1
As a result, an electric field is developed across the junction. Due to this field, an electron on p-side of the junction moves to n-side and a hole on n-side of the junction moves to p-side. The motion of charge carriers due to the electric field is called drift .Initially, diffusion current is large and drift current is small. As the diffusion process continues, the electric field strength & hence drift current increases .This process continues till diffusion & drift current becomes equal.	1

1. In the energy-band diagram of n-type Si, the gap between the bottom of the conduction band E_C and the donor energy level E_D is of the order of :

(a) 10 eV (b) 1 eV

(c) 0·1 eV (d) 0·01 eV

CBSE 2023

1. In the energy-band diagram of n-type Si, the gap between the bottom of the conduction band E_C and the donor energy level E_D is of the order of :

(a) 10 eV (b) 1 eV

(c) 0·1 eV (d) 0·01 eV

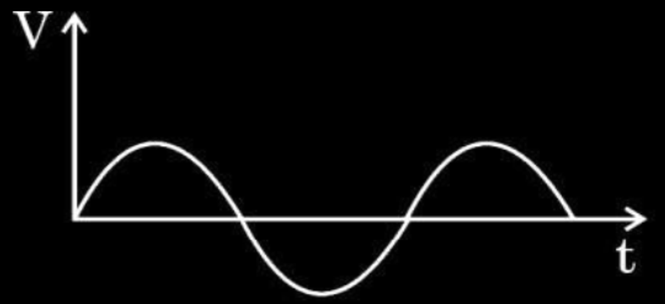
CBSE 2023

(d) 0.01 eV

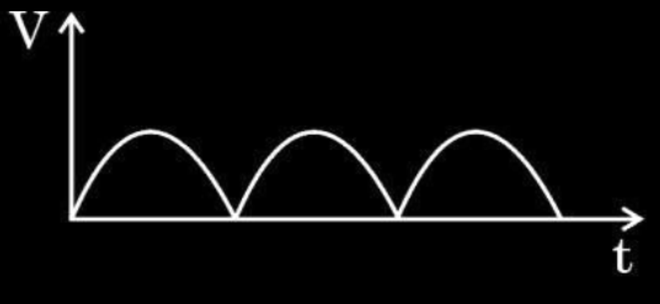
3. An ac source of voltage is connected in series with a p-n junction diode and a load resistor. The correct option for output voltage across load resistance will be :

CBSE 2023

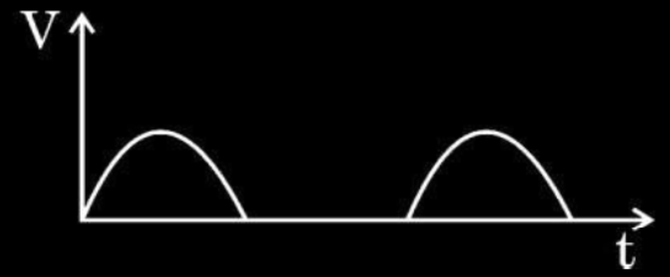
(a)



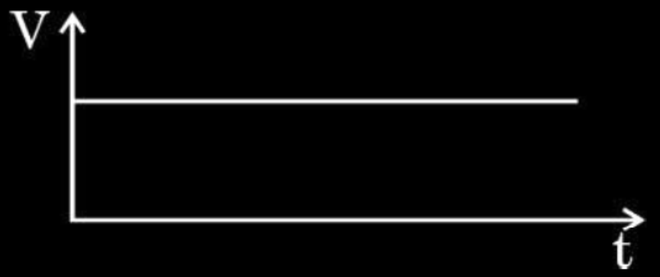
(b)



(c)



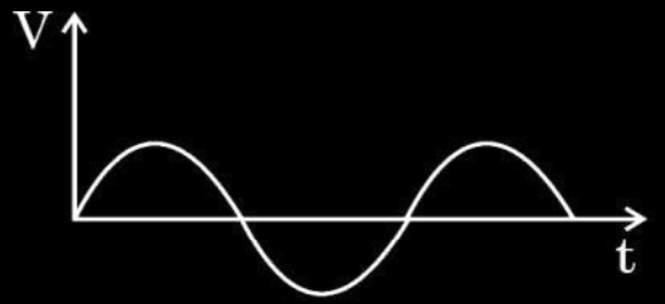
(d)



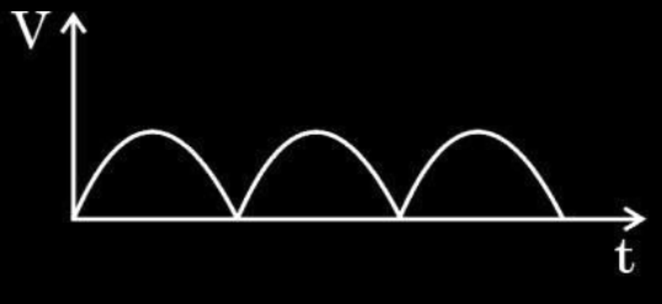
3. An ac source of voltage is connected in series with a p-n junction diode and a load resistor. The correct option for output voltage across load resistance will be :

CBSE 2023

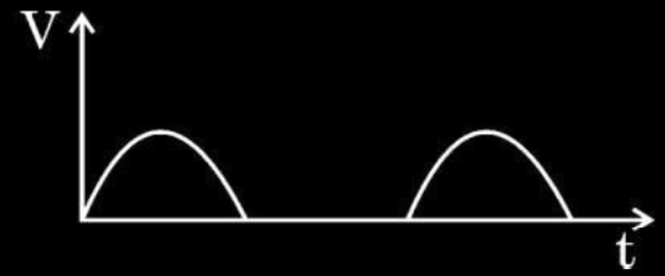
(a)



(b)



(c)



(d)

(c)

11. When an intrinsic semiconductor is doped with a small amount of trivalent impurity, then :

CBSE 2023

(a) its resistance increases.

(b) it becomes a p-type semiconductor.

(c) there will be more free electrons than holes in the semiconductor.

(d) dopant atoms become donor atoms.

11. When an intrinsic semiconductor is doped with a small amount of trivalent impurity, then :

CBSE 2023

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(b) it becomes a p-type semiconductor

20. Briefly explain how the diffusion and drift currents contribute to the formation of potential barrier in a p-n junction diode. CBSE 2023 *2*

20. Briefly explain how the diffusion and drift currents contribute to the formation of potential barrier in a p-n junction diode. CBSE 2023 *2*

The diffusion current due to concentration gradient at the junction forms a space charge region consisting of immobile charge carriers. Due to this an electric field is generated at the junction giving rise to drift current in a direction opposite to diffusion current. The potential at which diffusion current becomes equal to drift current is called potential barrier.	2

25. (a) Differentiate between intrinsic and extrinsic semiconductors. *2*

CBSE 2023

25. (a) Differentiate between intrinsic and extrinsic semiconductors. *2*

CBSE 2023

Intrinsic semiconductor	Extrinsic semiconductor	
1. Pure semiconductor.	Semiconductor is Doped with impurities.	
2. Low conductivity at room temperature.	High conductivity at room temperature.	
3. $n_e = n_h$	$n_e \neq n_h$	1+1

(b) Draw the circuit arrangement for studying the V – I characteristics of a p-n junction diode in forward bias and reverse bias. Show the plot of V – I characteristic of a silicon diode. CBSE 2023 *2*

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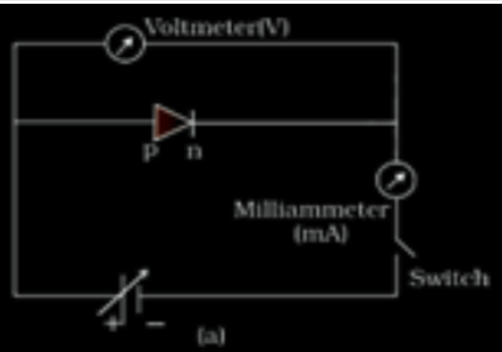
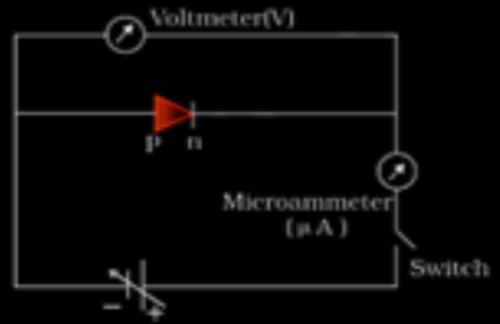
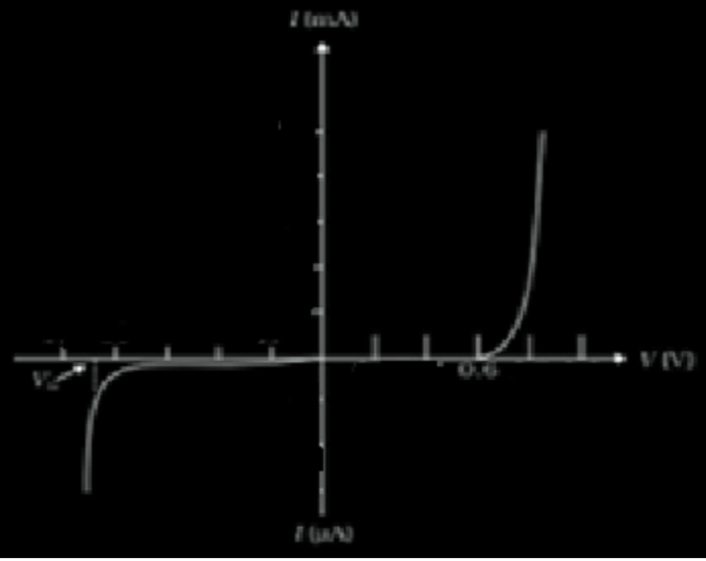


Forward Bias



Reverse Bias

½ + ½



½ + ½

4. A semiconductor device is connected in series with a battery, an ammeter and a resisitor. A current flows in the circuit. If the polarity of the battery is reversed, the current in the circuit almost becomes zero. The device is a/an **1**

(a) intrinsic semiconductor
(b) p-type semiconductor
(c) n-type semiconductor
(d) p-n junction diode

CBSE 2023

4. A semiconductor device is connected in series with a battery, an ammeter and a resisitor. A current flows in the circuit. If the polarity of the battery is reversed, the current in the circuit almost becomes zero. The device is a/an **1**

(a) intrinsic semiconductor (b) p-type semiconductor

(c) n-type semiconductor (d) p-n junction diode

CBSE 2023

(d) p-n junction diode

5. The formation of depletion region in a p-n junction diode is due to **1**

(a) movement of dopant atoms (b) diffusion of both electrons and holes

(c) drift of electrons only (d) drift of holes only

CBSE 2023

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(a) movement of dopant atoms (b) diffusion of both electrons and holes

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CBSE 2023

(b) Diffusion of both electrons and holes.

8. In an extrinsic semiconductor, the number density of holes is $4 \times 10^{20}\ m^{-3}$. If the number density of intrinsic carriers is $1.2 \times 10^{15}\ m^{-3}$, the number density of electrons in it is **1**

(a) $1.8 \times 10^{9}\ m^{-3}$ (b) $2.4 \times 10^{10}\ m^{-3}$

(c) $3.6 \times 10^{9}\ m^{-3}$ (d) $3.2 \times 10^{10}\ m^{-3}$

CBSE 2023

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(c) $3.6 \times 10^{9}\ m^{-3}$ (d) $3.2 \times 10^{10}\ m^{-3}$

CBSE 2023

(c) 3.6 x 10^{9} m^{-3}

21. With the help of a circuit diagram, explain how a full wave rectifier gives output rectified voltage corresponding to both halves of the input ac voltage. CBSE 2023 **2**

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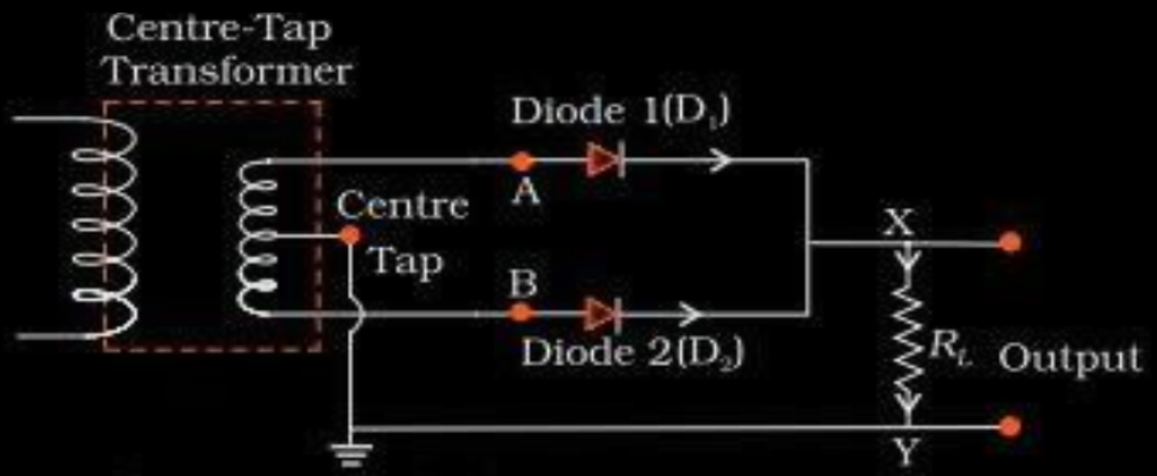


1

Alternatively

Give full credit for the circuit diagram if a candidate draws circuit diagram of bridge rectifier.

Explanation

Suppose the input voltage to A with respect to the centre tap at any instant is positive and B being out of phase will be negative so , diode D_1, gets forward biased and conducts while D_2 being reverse biased is not conducting. For the next half cycle of input voltage, the polarities are reversed and the diode D_2 conducts being forward biased.

1

22. Draw energy band diagram for an n-type and p-type semiconductor at $T > 0$ K.

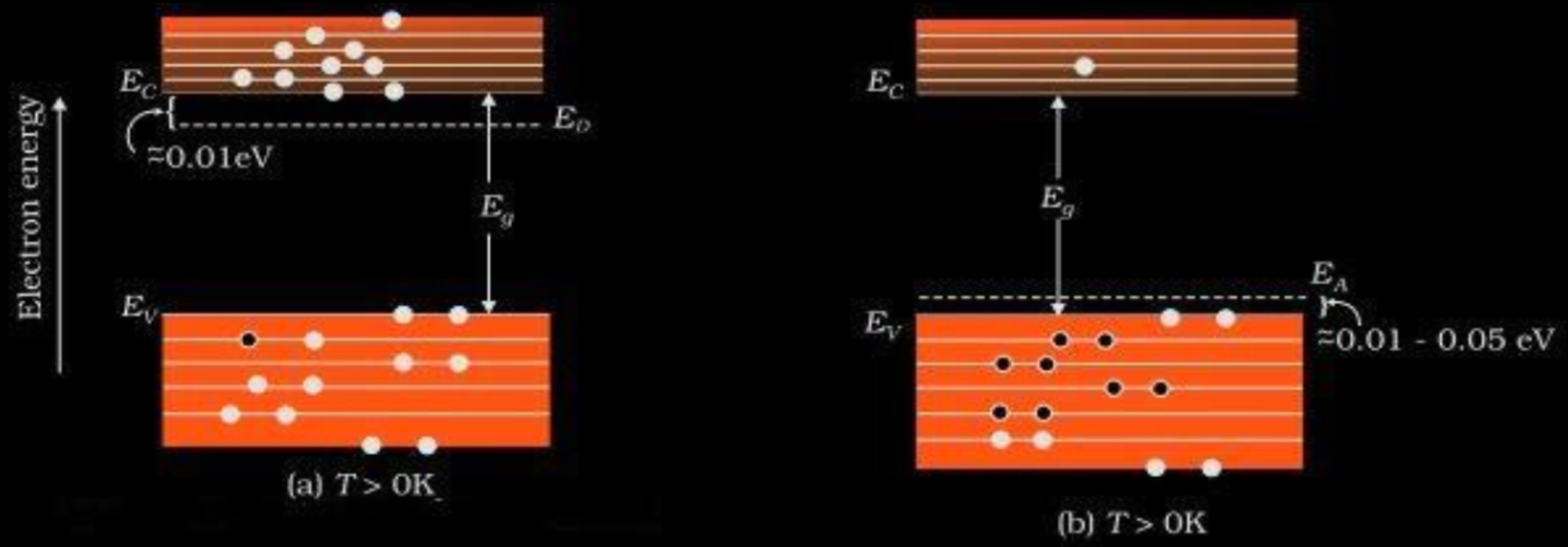
CBSE 2023

2

22. Draw energy band diagram for an n-type and p-type semiconductor at T > 0 K.

CBSE 2023

2



(a) n-type semiconductor (b) p-type semiconductor

1+1

8. Which one of the following elements will require the highest energy to take out an electron from them ? **1**

Pb, Ge, C and Si

(a) Ge
(b) C
(c) Si
(d) Pb

CBSE 2023

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Pb, Ge, C and Si

(a) Ge
(b) C
(c) Si
(d) Pb

CBSE 2023

(b) C

21. Answer the following giving reasons : **2**

(i) A p-n junction diode is damaged by a strong current.

(ii) Impurities are added in intrinsic semiconductors.

CBSE 2023

21. Answer the following giving reasons : **2**

(i) A p-n junction diode is damaged by a strong current.

(ii) Impurities are added in intrinsic semiconductors.

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i) Due to strong current, a junction diode gets heated, consequently large number of covalent bonds are broken and the junction is damaged.	1
ii) Deliberate addition of impurity atoms in intrinsic semiconductor increases its conductivity and is suitable for making electronic devices.	1
Alternatively Give full credit if a student writes that no electronic device can be developed using intrinsic semiconductor because of their low conductivity.	

14. Pieces of copper and of silicon are initially at room temperature. Both are heated to temperature T. The conductivity of **1**

(a) both increases.

(b) both decreases.

(c) copper increases and silicon decreases.

(d) copper decreases and silicon increases.

CBSE 2023

14. Pieces of copper and of silicon are initially at room temperature. Both are heated to temperature T. The conductivity of **1**

(a) both increases.

(b) both decreases.

(c) copper increases and silicon decreases.

(d) copper decreases and silicon increases.

CBSE 2023

(d)
Copper decreases and silicon increases

9. The energy required by an electron to jump the forbidden band in silicon at room temperature is about **1**

(A) 0.01 eV
(B) 0.05 eV
(C) 0.7 eV
(D) 1.1 eV

CBSE 2023

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(A) 0.01 eV
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CBSE 2023

(D) 1.1 eV

16. **Assertion (A)** : The resistance of an intrinsic semiconductor decreases with increase in its temperature. **1**

Reason (R) : The number of conduction electrons as well as hole increase in an intrinsic semiconductor with rise in its temperature.

CBSE 2023

16. **Assertion (A)** : The resistance of an intrinsic semiconductor decreases with increase in its temperature. **1**

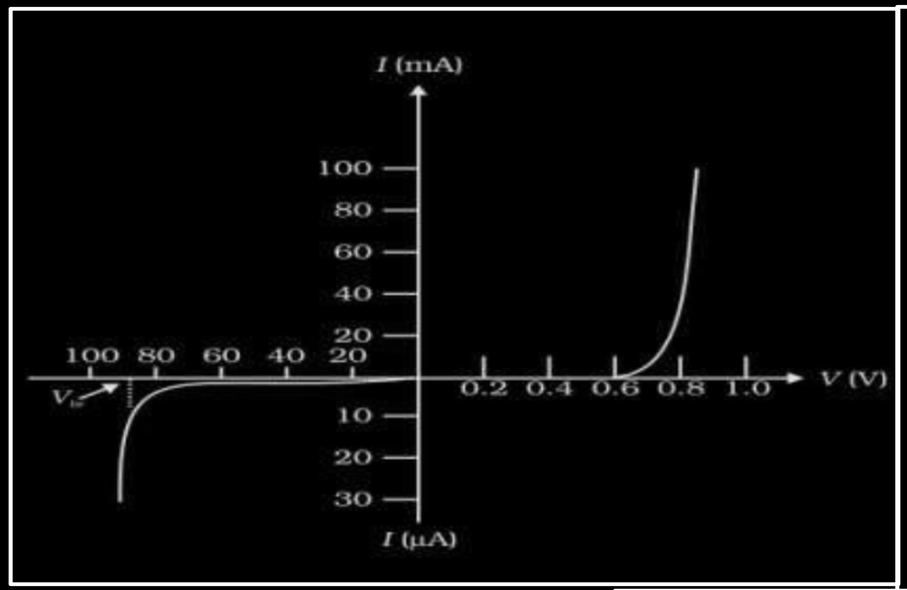
Reason (R) : The number of conduction electrons as well as hole increase in an intrinsic semiconductor with rise in its temperature.

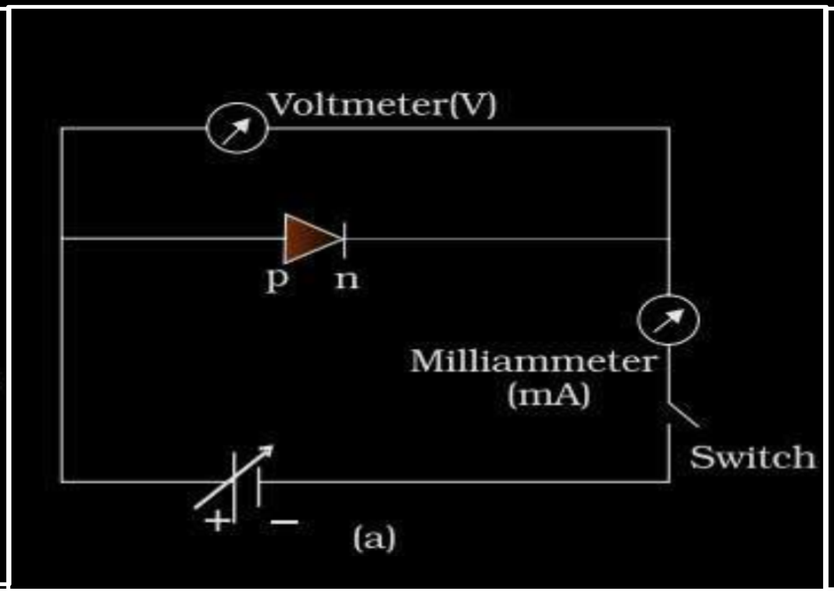
CBSE 2023

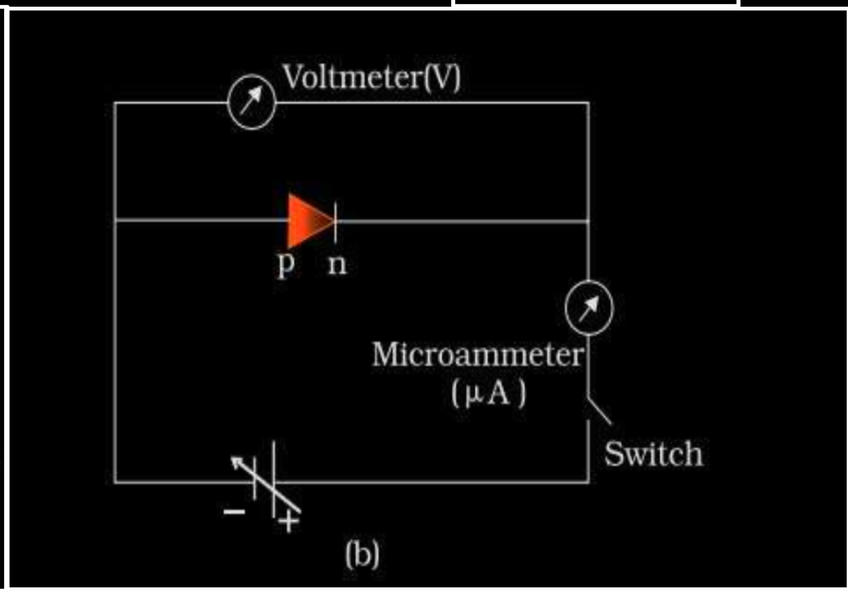
(A) Both Assertion (A) and reason (R) are true and (R) is correct explanation of (A)

31. (a) Draw the circuit arrangement for studying V-I characteristics of a p-n junction diode in (i) forward biasing and (ii) reverse biasing. Draw the typical V-I characteristics of a silicon diode. Describe briefly the following terms : (i) minority carrier injection in forward biasing and (ii) breakdown voltage in reverse biasing. CBSE 2023 **5**

31. (a) Draw the circuit arrangement for studying V-I characteristics of a p-n junction diode in (i) forward biasing and (ii) reverse biasing. Draw the typical V-I characteristics of a silicon diode. Describe briefly the following terms : (i) minority carrier injection in forward biasing and (ii) breakdown voltage in reverse biasing. CBSE 2023 **5**



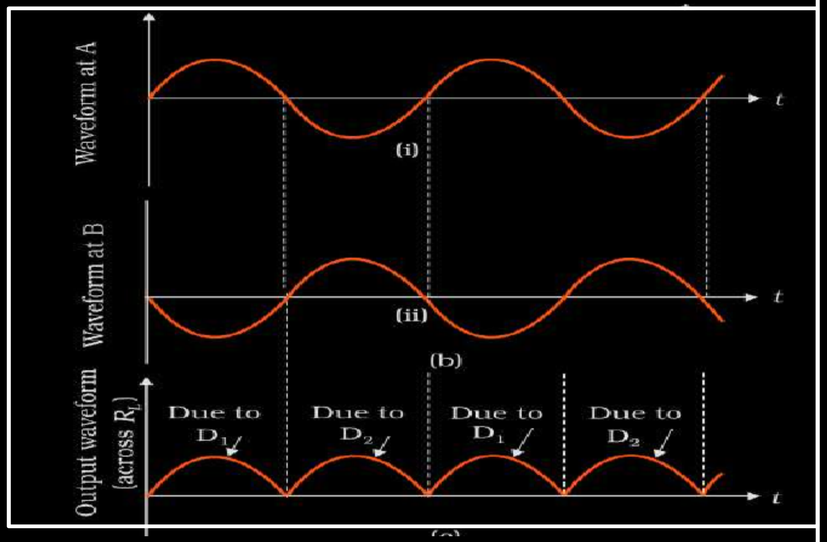




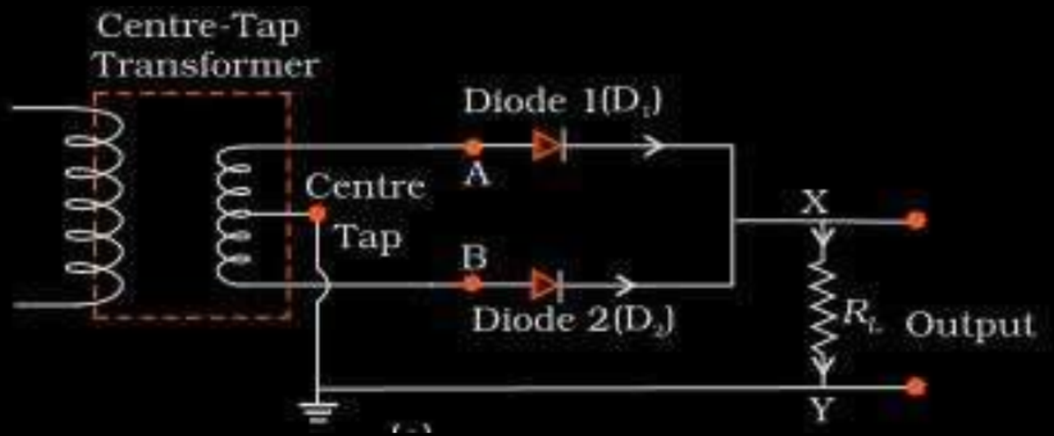
Minority carrier injection: Under forward bias electrons from n-side cross the depletion region and reach p-side. Similarly, holes from p-side cross the junction and reach the n-side.	1
Breakdown voltage: It is the voltage under reverse bias for which reverse current increases sharply	1

(b) Name two important processes involved in the formation of a p-n junction diode. With the help of a circuit diagram, explain the working of junction diode as a full wave rectifier. Draw its input and output waveforms. State the characteristic property of a junction diode that makes it suitable for rectification. CBSE 2023 **5**

(b) Name two important processes involved in the formation of a p-n junction diode. With the help of a circuit diagram, explain the working of junction diode as a full wave rectifier. Draw its input and output waveforms. State the characteristic property of a junction diode that makes it suitable for rectification. CBSE 2023 **5**



(a) Diffusion ½
Drift ½



Suppose the input voltage to A with respect to the centre-tap at any instant is positive. At that instant, voltage at B being out of phase will be negative. So, diode D_1 gets forward biased and conducts (while D_2 being reverse biased is not conducting). Hence, during this positive half cycle we get an output current (and a output voltage across the load resistor R_L). In the course of ac cycle when the voltage at A becomes negative with respect to centre tap, the voltage at B would be positive. In this part of the cycle diode D_1 would not conduct but diode D_2 would, giving an output current and output voltage (across R_L) during the negative half cycle of the input ac. 1 1

Property Junction diode allows current to pass only when it is forward biased. 1

9. During the formation of a p-n junction. **1**

(A) diffusion current keeps increasing.

(B) drift current remains constant.

(C) both the diffusion current and drift current remain constant.

(D) diffusion current remains almost constant but drift current increases till both currents become equal.

CBSE 2023

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CBSE 2023

(D) Diffusion current remains almost constant but drift current increases till both currents become equal.

1. Distinguish between the intrinsic semiconductor and the extrinsic semiconductor. CBSE 2022 C *2*

1. Distinguish between the intrinsic semiconductor and the extrinsic semiconductor. CBSE 2022 C *2*

Intrinsic Semiconductor	**Extrinsic Semiconductor**	
1. These are pure semiconductors having no impurity.	1. These are the semiconductors doped either with pentavalent or trivalent impurity atoms.	1
2. They have low electrical conductivity.	2. Their conductivity is greater than that of intrinsic semiconductors.	1
3. The number density of electrons and holes is equal i.e $n_e = n_h$	3. The number density of electrons and holes is different depending on the type of doping.	

(Any two of the above /any other two difference)

3. Differentiate between drift current and diffusion current in a p-n junction diode. What is the net current at the p-n junction in equilibrium ? CBSE 2022 C *2*

3. Differentiate between drift current and diffusion current in a p-n junction diode. What is the net current at the p-n junction in equilibrium ? CBSE 2022 C *2*

Drift Current: Movement of minority charge carriers (electrons and holes) due to the electric field generated across the junction, constitutes drift current.	½
Diffusion current: Due to the concentration gradient across p-side & n -side, holes diffuse from p-side to n-side & electrons diffuse from n-side to p-side. The current so formed is diffusion current.	½
Net current at p-n junction in equilibrium is zero.	1

2. (a) C and Si both have the same lattice structure, but C is an insulator while Si is a semiconductor. Justify.

CBSE 2022 C

2. (a) C and Si both have the same lattice structure, but C is an insulator while Si is a semiconductor. Justify. CBSE 2022 C

(a) Energy required to liberate an electron (i.e ionisation energy) of C is greater than S_i. Hence number of free electrons for conduction in S_i is more than C.	1

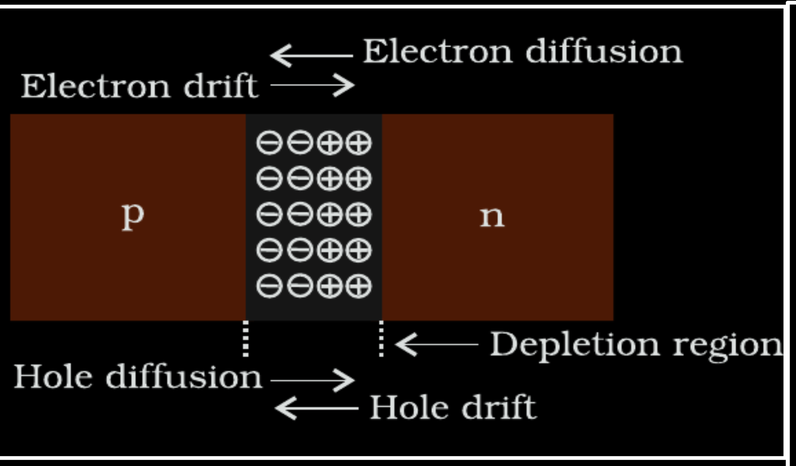
10. (a) Explain the formation of a p-n junction. CBSE 2022 C

(b) Can we take one slab of p-type semiconductor and physically join it to another n-type semiconductor to get a p-n junction ? Explain. *3*

CBSE 2022 C

10. (a) Explain the formation of a p-n junction.

(b) Can we take one slab of p-type semiconductor and physically join it to another n-type semiconductor to get a p-n junction ? Explain. *3*



During the formation of p-n junction due to concentration gradient across p and n sides, holes diffuse from p side to n side and electrons diffuse from n side to p side This motion of charge carriers gives rise to diffusion current across the junction	½
junction and diffusion of holes develops a layer of negative charge on p side of the junction.	½
Due to this space charge region on either side of the junction an electric field is developed. This electric field drifts charge carriers across the junction and sets up drift current in a direction opposite to diffusion current.	½
This process continues until the diffusion current is equal to drift current. Thus p-n junction is formed.	
a) No, Any slab, howsoever flat, will have roughness much larger than the inter-atomic spacing (~2 to 3 Å) and hence continuous contact at the atomic level will not be possible. The junction will behave as a discontinuity for the flowing charge carriers.	1

11. Draw the energy band diagrams (at T > 0 K) for n-type and p-type semiconductors. Using diagram, explain why in n-type semiconductor the conduction band has most electrons from the donor impurities. *3*

CBSE 2022 C

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CBSE 2022 C

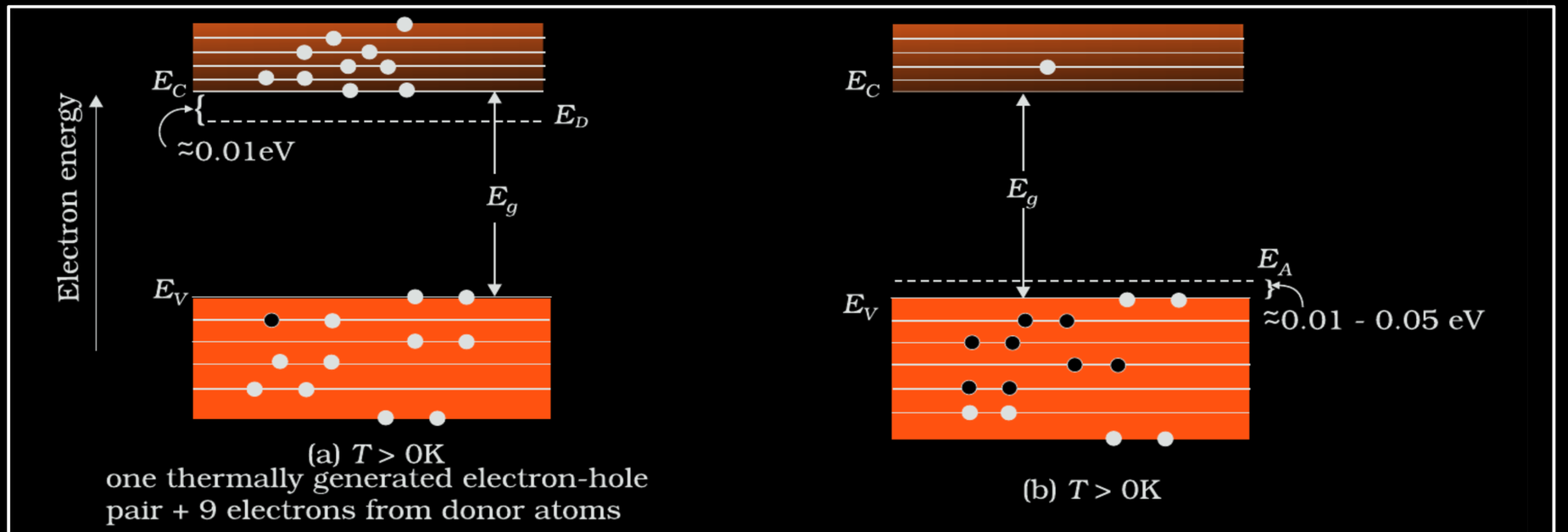

FIGURE 14.9 Energy bands of (a) n-type semiconductor at $T > 0K$, (b) p-type semiconductor at $T > 0K$.

In the energy band of n- type semiconductors, donor energy level E_D is formed slightly below the bottom of E_C of the conduction band. Hence electrons from this level move into the conduction band easily. **Note: If Students explain the cause by using the diagram itself give full credit to this part.**	1

1. (a) In n-type semiconductor, explain how the crystal is electrically neutral although electrons are the majority carriers in it.

CBSE 2022 C

1. (a) In n-type semiconductor, explain how the crystal is electrically neutral although electrons are the majority carriers in it.

CBSE 2022 C

a)	Crystal is electrically neutral as the charge of additional electrons provided by donor impurity is just equal and opposite to that of the ionised cores in the lattice.	1

9. Draw V – I characteristics of a p-n junction diode. Answer the following giving reasons : *3*

(a) Why is the reverse bias current almost independent of applied voltage up to breakdown voltage ?

(b) Why does the reverse current show a sudden increase at breakdown voltage ?

CBSE 2022 C

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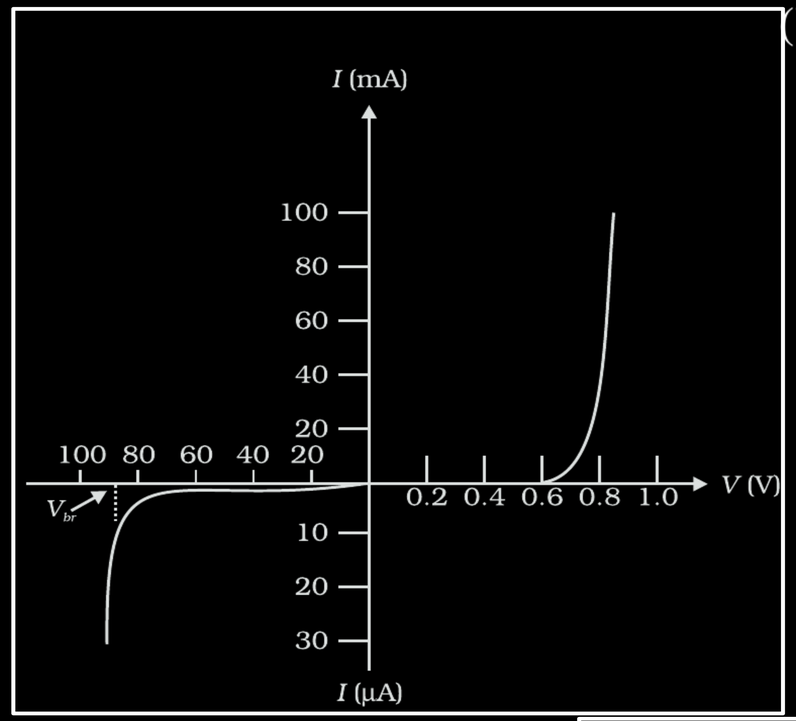
CBSE 2022 C



a) As the applied voltage increases minority charge carrier concentration will not increase. So, it is independent of applied voltage.	1
a) At the breakdown voltage, a large number of covalent bonds break, resulting in the increase of large number of charge carriers. Hence current increases suddenly.	1

11. Explain how the barrier potential is formed in a p-n junction. How is it affected in

(a) forward bias, and

(b) reverse bias ?

CBSE 2022 C

3

11. Explain how the barrier potential is formed in a p-n junction. How is it affected in

(a) forward bias, and

(b) reverse bias ?

CBSE 2022 C *3*

In a p-n-junction diode, the loss of electrons from the n- region and the gain of electrons by the p-region causes a difference of potential across the junction of the two regions. This potential tends to prevent the movement of electrons from the n region to the p region. Hence a potential barrier is created across the junction.	1
(a) In forward bias, the barrier potential is reduced.	1
(b) In reverse bias, the barrier potential increases.	1

3. (a) A p-type semiconductor is electrically neutral although it has holes as the majority carriers. Justify.

CBSE 2022 C

3. (a) A p-type semiconductor is electrically neutral although it has holes as the majority carriers. Justify.

CBSE 2022 C

a) The crystal maintains an overall charge neutrality as the charge of additional charge carriers is just equal and opposite to that of the ionised cores in the lattice.	1

1. Give two differences between a half wave rectifier and a full wave rectifier. 2

CBSE 2022

1. Give two differences between a half wave rectifier and a full wave rectifier. 2

CBSE 2022

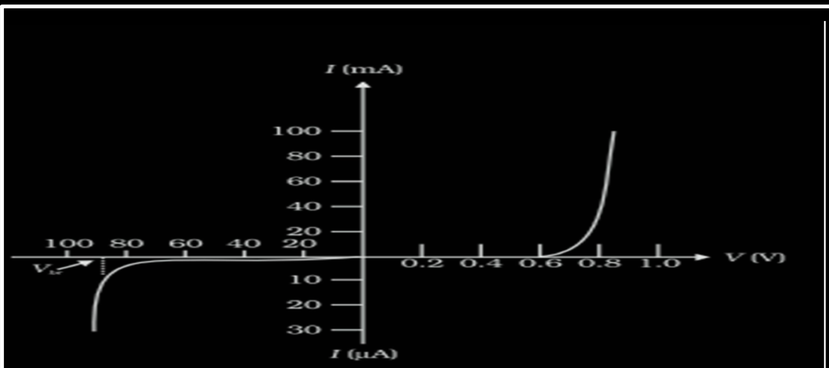
Half Wave Rectifier	Full Wave Rectifier	
1. Only one half of signal is rectified.	1. Both halves of the signal are rectified.	1
2. Frequency of output is same as of input frequency.	2. Frequency of output is double that of the input frequency	1
3. One p-n junction diode is used	3. Two diodes are used.	
(**Note:** Any two of the above or any other two differences)		

11. (i) Draw V-I characteristics of a p-n Junction diode. 3

(ii) Differentiate between the threshold voltage and the breakdown voltage for a diode.

(iii) Write the property of a junction diode which makes it suitable for rectification of ac voltages.

CBSE 2022



1

ii)

Threshold Voltage: Forward bias voltage at which the current increases significantly (exponentially) even for a very small increase in voltage. ½

Alternatively:

Forward bias voltage at which the width depletion layer and barrier potential decreases significantly.

Alternatively:

The voltage at which resistance of junction decreases significantly.

Break down voltage: Reverse bias voltage at which current increases suddenly ½

Alternatively:

Large number of covalent bonds present in the depletion layer break suddenly

iii) Junction Diode conducts when it is forward biased and does not conduct when reverse biased. 1

1. Two crystals C_1 and C_2, made of pure silicon, are doped with arsenic and aluminium respectively. 2

(i) Identify the extrinsic semiconductors so formed.

(ii) Why is doping of intrinsic semiconductors necessary ?

CBSE 2022

1. Two crystals C_1 and C_2, made of pure silicon, are doped with arsenic and aluminium respectively. 2

(i) Identify the extrinsic semiconductors so formed.

(ii) Why is doping of intrinsic semiconductors necessary ?

CBSE 2022

i) C_1 n-type	½
C_2 p-type	½
ii) To have higher and regulated conductivity	1

1. With the help of a circuit diagram, explain briefly how a p-n junction diode works as a half-wave rectifier. 2

CBSE 2022

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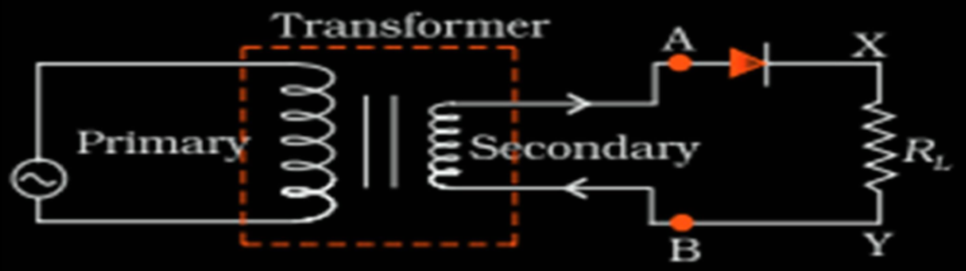
CBSE 2022

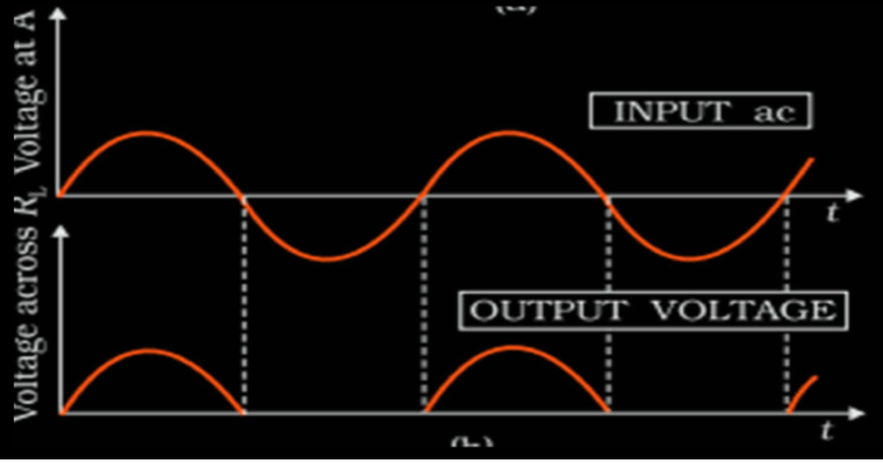


1

- When voltage at A is positive, diode is forward biased and it conducts and when voltage at A is negative, diode is reverse biased and it does not conduct, so output is zero.

Alternatively:

1



2. Explain the formation of the barrier potential in a p-n junction. **2**

CBSE 2022

2. Explain the formation of the barrier potential in a p-n junction. **2**

CBSE 2022

The diffusion of electrons from n-region to p-region and that of the holes from p-region to n-region creates positive charge on the n-side and negative charge on the p-side which causes a difference of potential across the junction.	1
This potential, setup across the junction tends to prevent the movement of electrons from the n-region to p-region .This is called barrier potential.	1

3. Name the extrinsic semiconductors formed when a pure germanium is doped with (i) a trivalent and (ii) pentavalent impurity. Draw the energy band diagrams of extrinsic semiconductors so formed. CBSE 2022 **2**

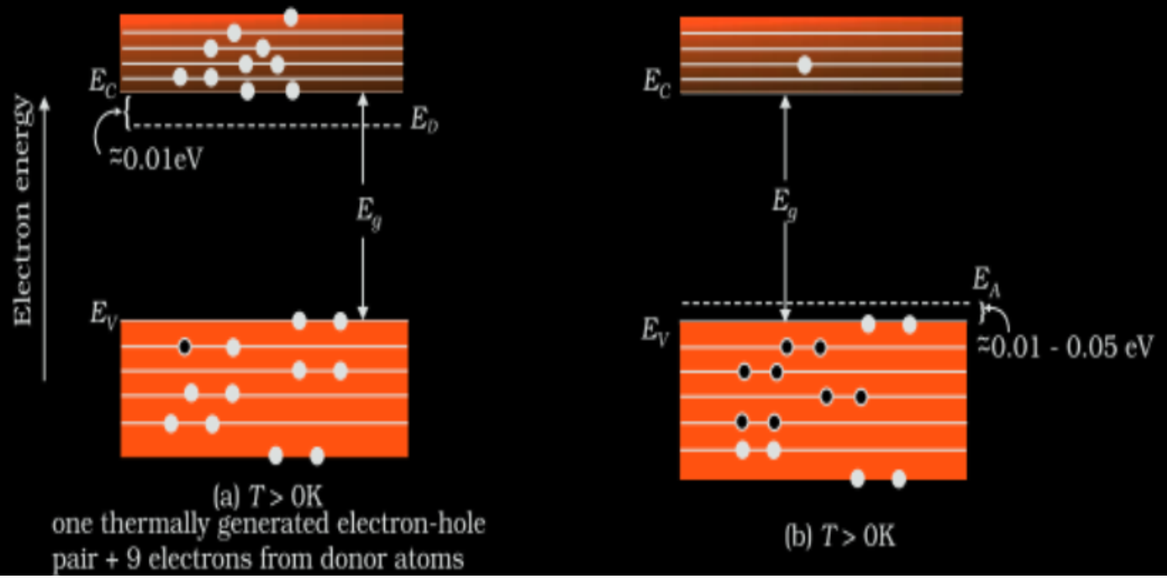
3. Name the extrinsic semiconductors formed when a pure germanium is doped with (i) a trivalent and (ii) pentavalent impurity. Draw the energy band diagrams of extrinsic semiconductors so formed. CBSE 2022 **2**

a)

(i) p-type semiconductor ½

(ii) n-type semiconductor ½

b)



(a) $T > 0K$ one thermally generated electron-hole pair + 9 electrons from donor atoms

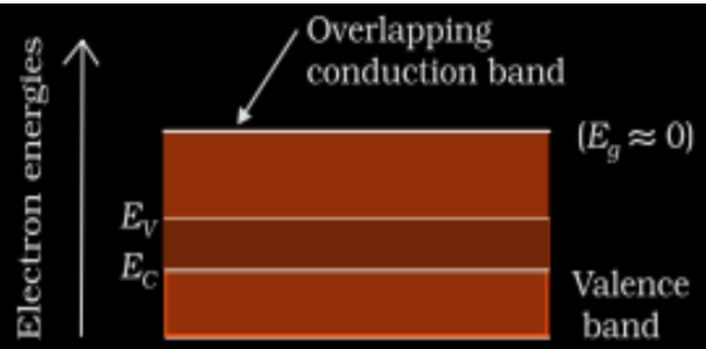
(b) $T > 0K$

½ + ½

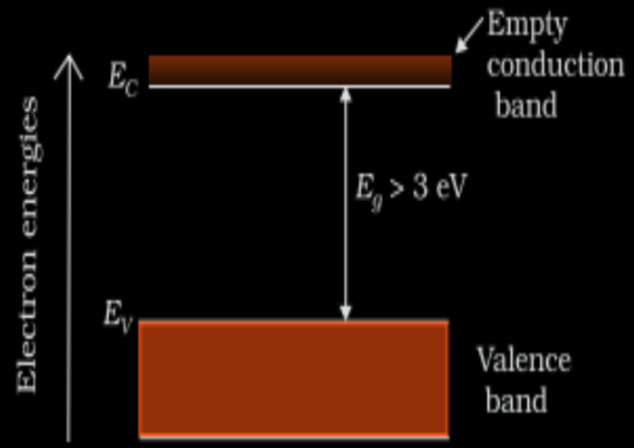
8. Draw the energy band diagrams for conductors, semiconductors and insulators. Which band determines the electrical conductivity of a solid ? How is the electrical conductivity of a semiconductor affected with rise in its temperature ? Explain. CBSE 2022 **3**

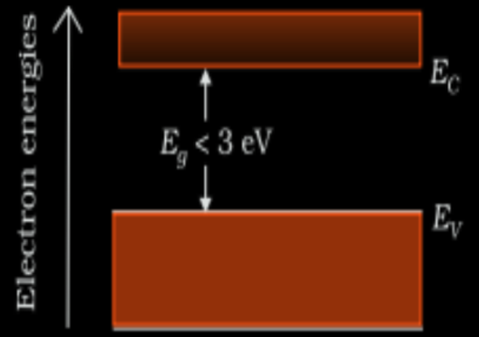
8. Draw the energy band diagrams for conductors, semiconductors and insulators. Which band determines the electrical conductivity of a solid ? How is the electrical conductivity of a semiconductor affected with rise in its temperature ? Explain.

CBSE 2022 **3**



½





½ + ½

Conduction band determines electrical conductivity

½

As temperature of a semiconductor rises, the carrier concentration (electron-hole pair) increases due to breaking of covalent bonds and the conductivity of the semiconductor increases.

1

1. What is meant by doping of an intrinsic semiconductor ? Name the two types of atoms used for doping of Ge/Si. **2**

CBSE 2022

1. What is meant by doping of an intrinsic semiconductor ? Name the two types of atoms used for doping of Ge/Si. **2**

CBSE 2022

Doping is the process of adding some external impurity atoms in an intrinsic semiconductor to increase its conductivity.	1
Dopant atoms may be pentavalent or trivalent.	½+½

2. Distinguish between intrinsic and extrinsic semiconductors. Although in an extrinsic semiconductor $n_e \neq n_h$, yet it is electrically neutral. Why ? **2**

CBSE 2022

2. Distinguish between intrinsic and extrinsic semiconductors. Although in an extrinsic semiconductor $n_e \neq n_h$, yet it is electrically neutral. Why ? **2**

CBSE 2022

	Intrinsic semiconductors are pure semiconductors while extrinsic semiconductors are doped with either trivalent or pentavalent impurities.	1
	Extrinsic semiconductor maintains an overall charge neutrality as the charge of additional charge carriers is just equal and opposite to that of the ionised cores in the lattice.	1

3. Draw energy band diagrams of n-type and p-type semiconductors at temperature T > 0K, depicting the donor and acceptor energy levels. Mention the significance of these levels. CBSE 2022 **2**

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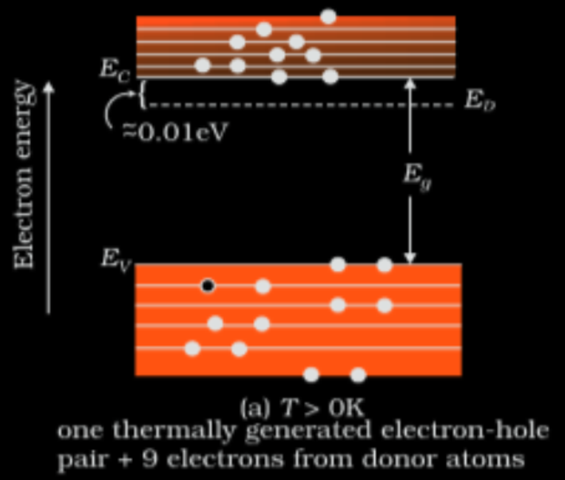
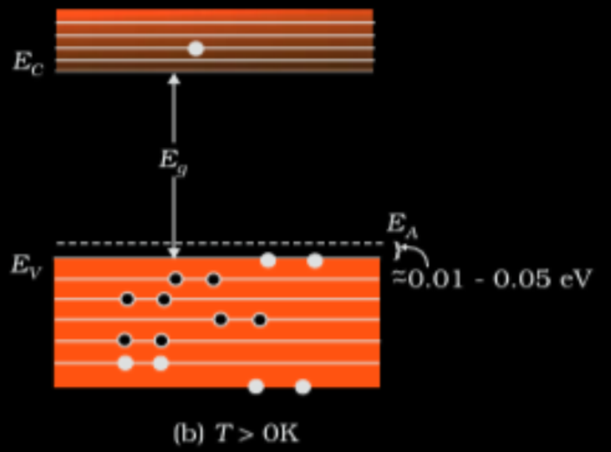




n-type **p-type** ½ + ½

Significance

n-type semiconductors – small energy gap between donor level and conduction band which can be easily covered by thermally excited electrons. ½

p-type semiconductors - small energy gap between acceptor level and valence band which can be easily covered by thermally excited electrons. ½

Alternatively

The conductivity of semiconductor is improved with the creation of donor and acceptor levels.

1. Explain the formation of depletion layer and barrier potential in a p-n junction diode. CBSE 2022 **2**

1. Explain the formation of depletion layer and barrier potential in a p-n junction diode. CBSE 2022 **2**

Formation of depletion region: During formation of p-n junction a hole diffuses from p to n region and electron diffuses from n to p region due to concentration gradient. It leaves behind immobile ions at the junction on both sides. This space charge region at the junction is together called depletion region.	1
Potential Barrier : The immobile ions on p side are negative and on n side are positive. They create a potential difference preventing the further flow of majority charge carriers. This potential is called as barrier potential.	1

1. How is the forward biasing different from the reverse biasing in a p-n junction diode ? Name one device each which works under forward and reverse biasing. CBSE 2022 *2*

1. How is the forward biasing different from the reverse biasing in a p-n junction diode ? Name one device each which works under forward and reverse biasing. CBSE 2022 *2*

In forward biasing the p-side of the junction is connected to the +ve terminal and n-side to the –ve terminal of the battery.	½
In reverse biasing the p-side of the junction is connected to the –ve terminal and n-side to the +ve terminal of the battery.	½
Alternatively	
In forward biasing the width of the depletion layer decreases / resistance of the junction decreases / current increases.	
In reverse biasing the width of the depletion layer increases / resistance of the junction increases / current decreases.	
• Name of the device for forward biasing ----- LED	½
• Name of the device for reverse biasing--------Zener diode	½
(Note : Full credit (½ Marks) for writing diode as rectifier working under any of the two biasing)	

3. How are (a) n-type semiconductor, and (b) p-type semiconductor obtained from an intrinsic semiconductor ? Name the majority charge carriers in each of the semiconductors. *2*

CBSE 2022

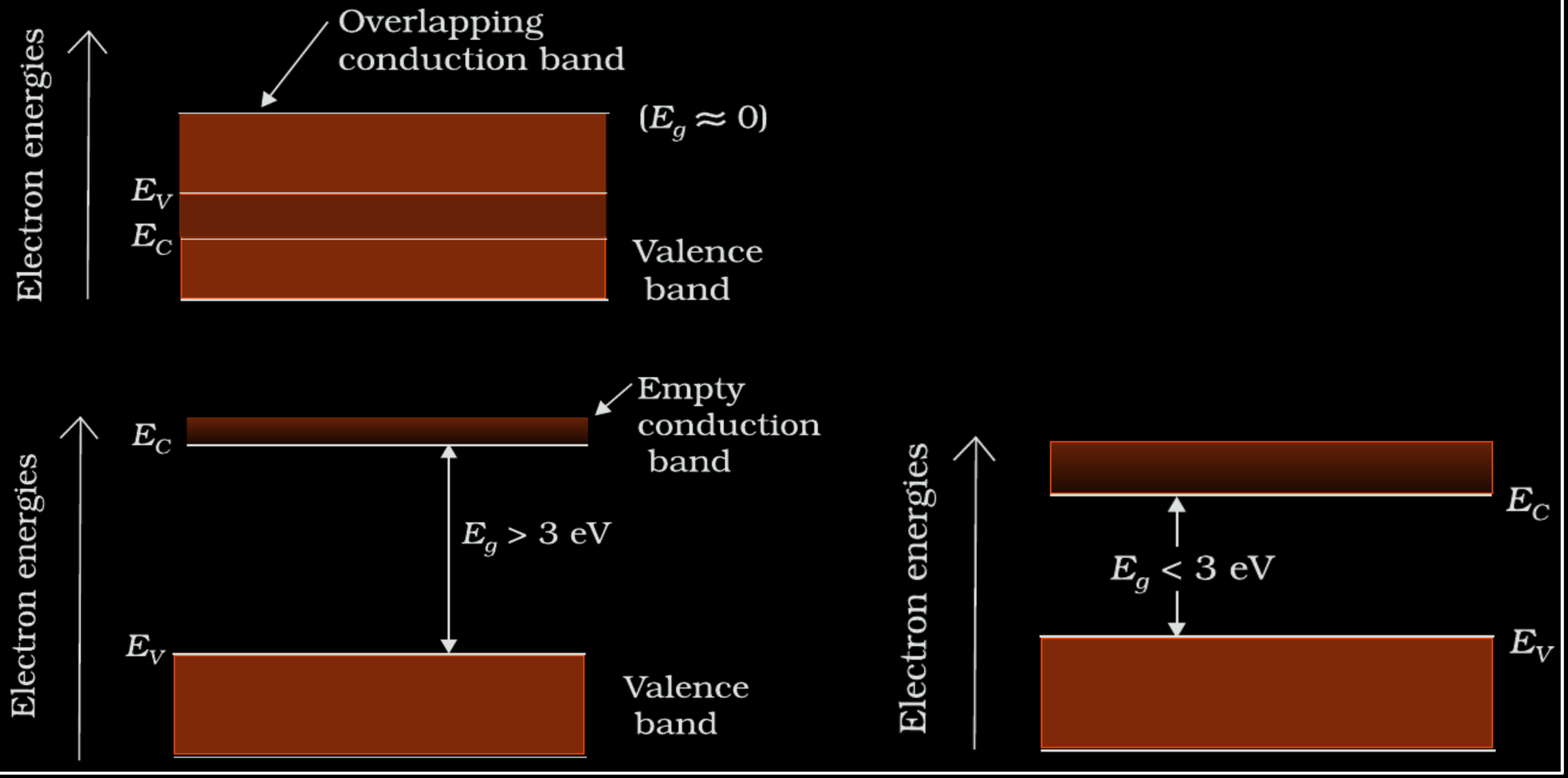
3. How are (a) n-type semiconductor, and (b) p-type semiconductor obtained from an intrinsic semiconductor ? Name the majority charge carriers in each of the semiconductors. CBSE 2022 *2*

• *n*- type semiconductors are obtained when the intrinsic semiconductor are dopped with a pentavalent impurity. • *p*- type semiconductors are obtained when the intrinsic semiconductors are dopped with a trivalent impurity. Majority charge carriers in (i) n-type----electrons (ii) p-type----holes	½ ½ ½ ½

2. What is meant by energy band gap in a solid ? Draw the energy band diagrams for a conductor, an insulator and a semiconductor. CBSE 2022 *2*

2. What is meant by energy band gap in a solid ? Draw the energy band diagrams for a conductor, an insulator and a semiconductor. CBSE 2022 *2*

Energy gap : The gap between the top of the valence band and bottom of the conduction band is called the energy band gap. ½



3. Define barrier potential. Why does the thickness of the depletion layer in a p-n junction diode vary with increase in reverse bias ? CBSE 2022 *2*

3. Define barrier potential. Why does the thickness of the depletion layer in a p-n junction diode vary with increase in reverse bias ? CBSE 2022 *2*

(i)	It is the potential developed across the p-n junction which tends to prevent the movement of majority charge carriers from both sides.	1
(ii)	Under reverse bias, direction of applied voltage is same as the direction of barrier potential. As a result barrier height increase & the depletion region widens due to change in electric field.	1

4. Answer the following, giving reason : $3 \times 1 = 3$

(a) The resistance of a p-n junction is low when it is forward biased and is high when it is reversed biased.

(b) Doping of intrinsic semiconductors is a necessity for making electronic devices.

CBSE 2022

4. Answer the following, giving reason : $3\times1=3$

(a) The resistance of a p-n junction is low when it is forward biased and is high when it is reversed biased.

(b) Doping of intrinsic semiconductors is a necessity for making electronic devices.

CBSE 2022

a) When a *p-n* junction is forward biased, the junction width decreases and as a result, its resistance also decreases.	½
On the other hand, when a *p-n* junction is reverse biased, the junction width increases and as a result, its resistance also increases.	½
b) Conductivity of intrinsic semi-conductors is very low. Hence, no electronic device can be developed using them.	1
Dopping increases conductivity, hence makes intrinsic semiconductor suitable for making electronic devices.	
c) It is easier to observe the change in the current with change in light intensity if a reverse bias is applied.	1
Alternatively	
The fractional change due to photo-effects on the minority charge carriers dominated reverse bias current, is more easily measurable than the fractional change in forward bias current.	

2. Name the device which converts an ac input signal into a dc output signal. Write the principle of working of the device. *2*

CBSE 2022

2. Name the device which converts an ac input signal into a dc output signal. Write the principle of working of the device. *2*

CBSE 2022

❖ Rectifier ❖ The unidirectional property of a diode makes it suitable for reflection. **Alternatively:** The diode conducts when forward biased and does not conduct when reverse biased	1 1

2. Explain the formation of depletion region in a p-n junction. *2*

CBSE 2022

2. Explain the formation of depletion region in a p-n junction. *2*

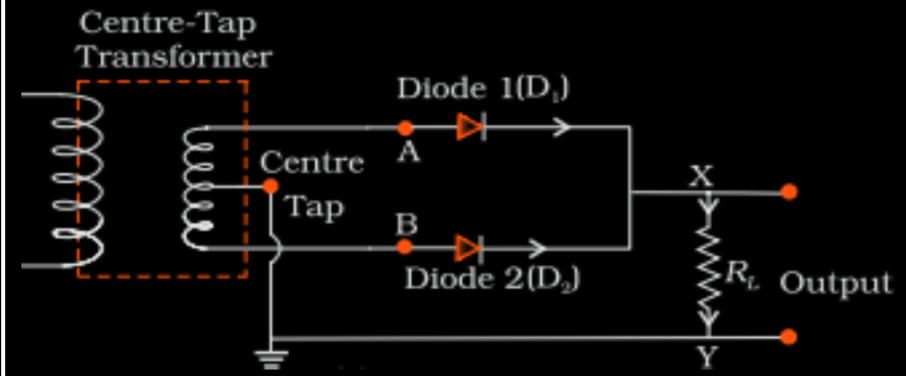
CBSE 2022

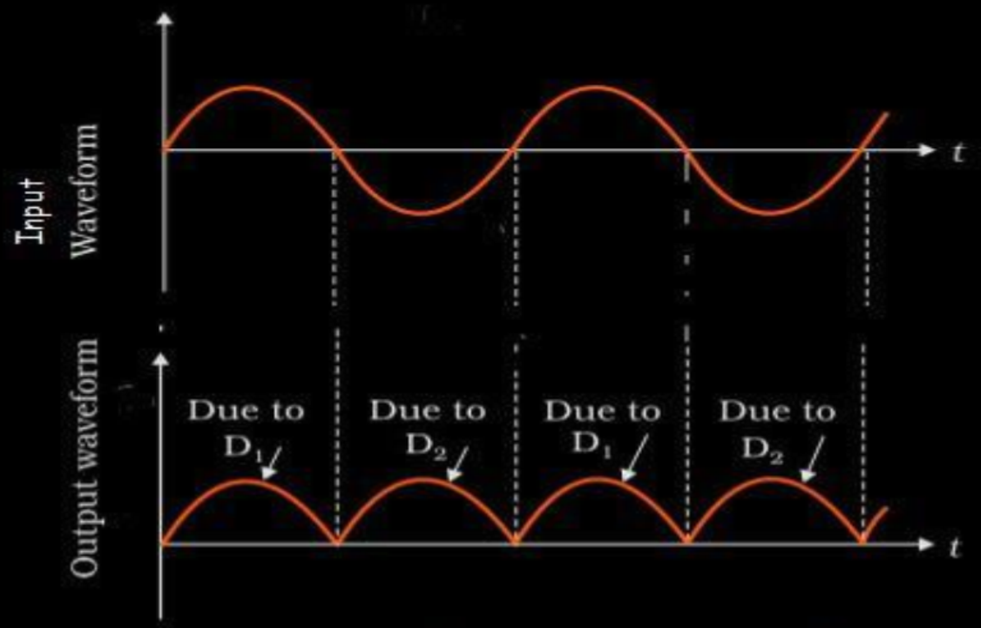
When p-type semiconductor is chipped with n-type semiconductor, e^- from the n-side diffuse towards p-side and holes from p-side diffuse towards n-side leaving behind a layer of immobile +ve ions on n-side and immobile –ve ions on p-side leading to formation of depletion layer. **(Note : Award 1 mark, if a student draws a diagram showing depletion region)**	2

5. With the help of a circuit diagram, explain the working of a p-n junction diode as a full-wave rectifier. Also draw its input and output waveforms. CBSE 2022 *3*

5. With the help of a circuit diagram, explain the working of a p-n junction diode as a full-wave rectifier. Also draw its input and output waveforms. CBSE 2022 *3*

Let input voltage at A w.r.t. the centre tap at any instant is positive, then voltage at B will be negative. So, diode D_1 gets forwarded biased & D_2 gets reverse biased. Hence output current is obtained. When voltage at A becomes –ve ; then voltage at B would be + ve , hence D_1, gets reverse biased & D_2 gets forwarded biased . So output current is again obtained as shown in the figure. 1





½

½

(Note: If the student takes inverted input waveform full credit to be given.)

14. The _____ , a property of materials C, Si and Ge depends upon the energy gap between their conduction and valence bands. CBSE 2020 **1**

14. The _____ , a property of materials C, Si and Ge depends upon the energy gap between their conduction and valence bands. CBSE 2020 **1**

Conductivity/ Resistivity
(Also give full credit if a student writes semiconducting nature)

15. The ability of a junction diode to _____ an alternating voltage, is based on the fact that it allows current to pass only when it is forward biased. **1**

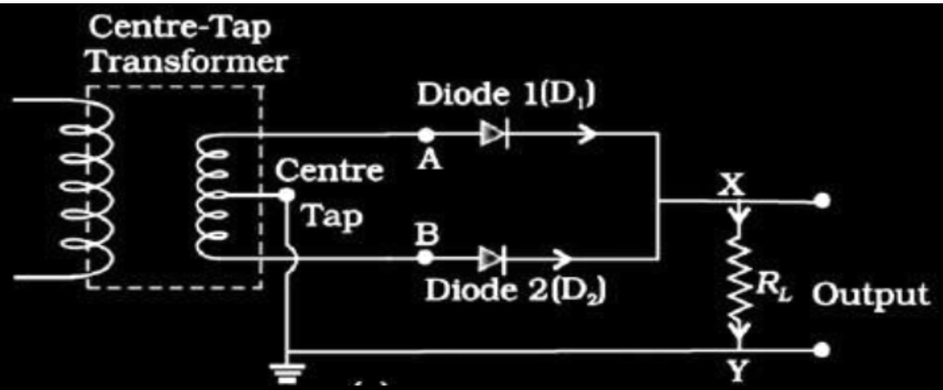
CBSE 2020

15. The ability of a junction diode to Rectify an alternating voltage, is based on the fact that it allows current to pass only when it is forward biased. **1**

CBSE 2020

Draw the circuit diagram of a full wave rectifier. Explain its working showing its input and output waveforms. CBSE 2020 **3**

Draw the circuit diagram of a full wave rectifier. Explain its working showing its input and output waveforms. CBSE 2020 **3**



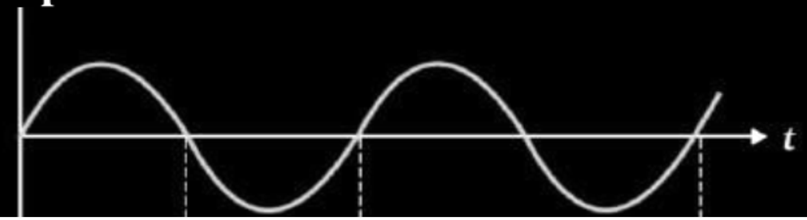
1

During one half cycle of the input a.c. signal, only diode 1 is forward biased and conducts. ½

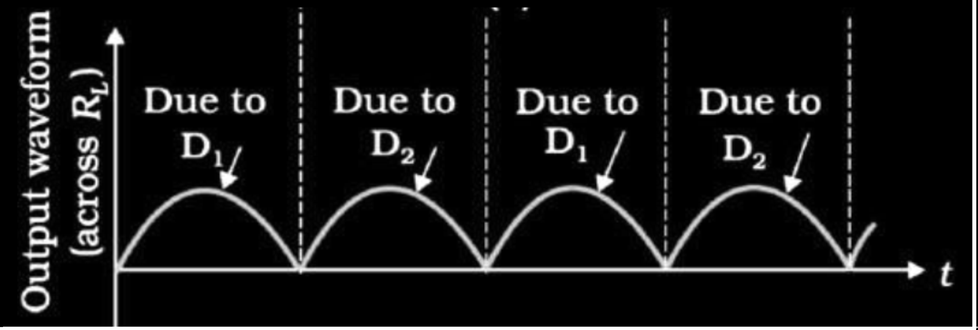
During the next half cycle of the input ac signal only diode 2 is forward biased and conducts.

However, due to the use of the centre tapped transformer, the current in the load flows in the same direction during both these half cycles. The current through the load is therefore unidirectional. ½

Input waveform



Output waveform



20. Can a slab of p-type semi-conductor be physically joined to another n-type semiconductor slab to form p-n junction ? Justify your answer. **1**

OR

In a p-n junction diode the forward bias resistance is low as compared to the reverse bias resistance. Give reason. **1**

CBSE 2020

20. Can a slab of p-type semi-conductor be physically joined to another n-type semiconductor slab to form p-n junction ? Justify your answer. **1**

OR

In a p-n junction diode the forward bias resistance is low as compared to the reverse bias resistance. Give reason. **1**

CBSE 2020

No As there will be discontinuity for the flow of charge carriers / no contact at atomic level. (Any One Justification)	½ ½
OR	
The forward current is large due to majority charge carriers which are very large in number. Hence resistance in forward bias is low. Alternatively: Depletion region decreases or barrier potential decreases.	1

25. Briefly explain how a potential barrier is set up across a p-n junction as a result of diffusion and drift of the charge carriers. CBSE 2020 **2**

25. Briefly explain how a potential barrier is set up across a p-n junction as a result of diffusion and drift of the charge carriers. CBSE 2020 **2**

Diffusion current is set up across the junction due to the concentration difference of the majority charge carriers on the two sides of the junction.	1
This diffusion develops an electric field from n- side to p- side across the junction which creates a drift current in the opposite direction.	½
When diffusion and drift current become equal in magnitude the potential difference across the junction is the barrier potential.	½

27. (a) Explain the formation of energy bands in crystalline solids.
(b) Draw the energy band diagrams of (i) a metal and (ii) a semiconductor. **2**

CBSE 2020

(b) Explain briefly the two processes that occur in p-n junction region to create a potential barrier. *3*

27. (a) Explain the formation of energy bands in crystalline solids.
(b) Draw the energy band diagrams of (i) a metal and (ii) a semiconductor. **2**

CBSE 2020

(b) Explain briefly the two processes that occur in p-n junction region

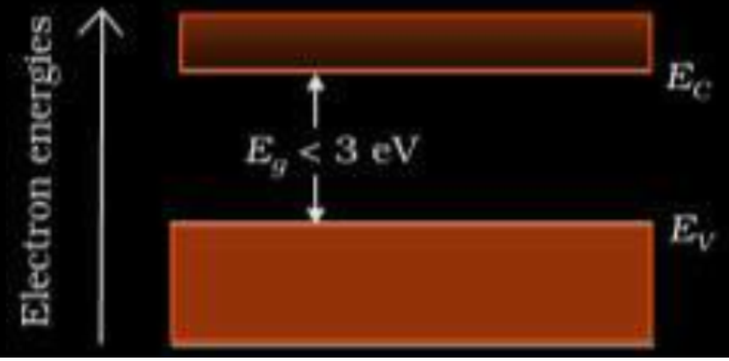
(a) Note: Out of syllabus; marks are distributed in part(b) *3*

(b) [A student may draw both or any one]

(i)

1

(ii)



1

34. (a) Why is an intrinsic semiconductor deliberately converted into an extrinsic semiconductor by adding impurity atoms ?

(b) Explain briefly the two processes that occur in p-n junction region to create a potential barrier. CBSE 2020 *3*

34. (a) Why is an intrinsic semiconductor deliberately converted into an extrinsic semiconductor by adding impurity atoms ?

(b) Explain briefly the two processes that occur in p-n junction region to create a potential barrier. CBSE 2020 *3*

a) To increase the electrical conductivity / to increase the number density of charge carriers	½
b) Diffusion and Drift	½ + ½
Explanation Diffusion: During the formation of p-n junction, due to the concentration gradient across the p and n sides, the motion of majority charge carriers give rise to diffusion current.	½
Drift: Due to the electric field developed at the junction, the motion of the minority charge carriers due to electric field is called drift.	½
With the passage of time, diffusion current decreases whereas drift current increases and balance each other. This, creates a potential barrier.	½

7. At equilibrium, in a p-n junction diode the net current is *1*

(A) due to diffusion of majority charge carriers.

(B) due to drift of minority charge carriers.

(C) zero as diffusion and drift currents are equal and opposite.

(D) zero as no charge carriers cross the junction.

CBSE 2020

7. At equilibrium, in a p-n junction diode the net current is *1*

(A) due to diffusion of majority charge carriers.

(B) due to drift of minority charge carriers.

(C) zero as diffusion and drift currents are equal and opposite.

(D) zero as no charge carriers cross the junction.

CBSE 2020

(C) zero as diffusion and drift current are equal and opposite.

8. In an n-type semiconductor, the donor energy level lies *1*

(A) at the centre of the energy gap.

(B) just below the conduction band.

(C) just above the valance band.

(D) in the conduction band.

CBSE 2020

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CBSE 2020

(B) just below the conduction band

27. Draw V-I characteristics of a p-n junction diode. Explain, why the current under reverse bias is almost independent of the applied voltage up to the critical voltage. *2*

CBSE 2020

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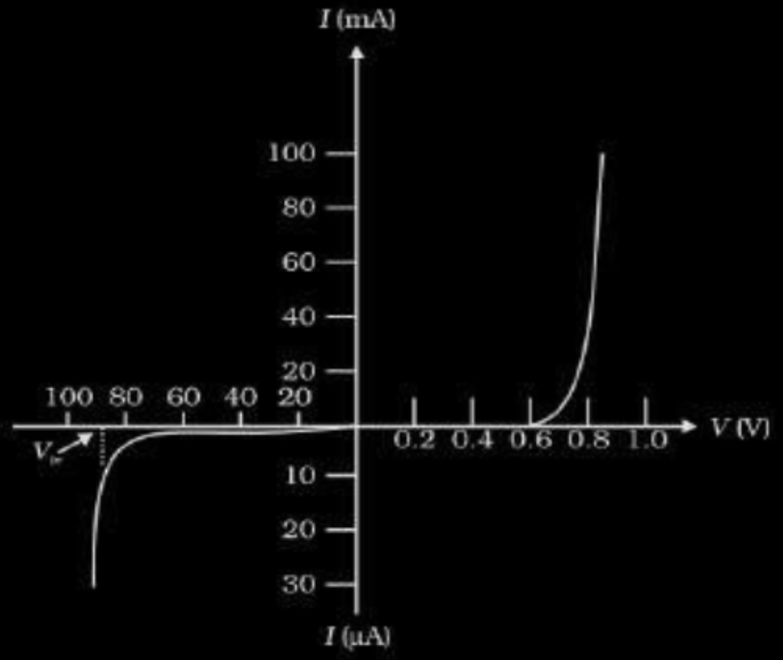
CBSE 2020



1

Since reverse current is due to flow of minority charge carriers across the junction, it is limited due to the concentration of minority carriers on either side of the junction. It is therefore independent of the voltage applied. 1

17. How does an increase in doping concentration affect the width of depletion layer of a p-n junction diode ? CBSE 2020 *1*

17. How does an increase in doping concentration affect the width of depletion layer of a p-n junction diode ? CBSE 2020 *1*

Decreases or reduce

27. Explain the terms 'depletion layer' and 'potential barrier' in a p-n junction diode. How are the (a) width of depletion layer, and (b) value of potential barrier affected when the p-n junction is forward biased ?

CBSE 2020

2

27. Explain the terms 'depletion layer' and 'potential barrier' in a p-n junction diode. How are the (a) width of depletion layer, and (b) value of potential barrier affected when the p-n junction is forward biased ?

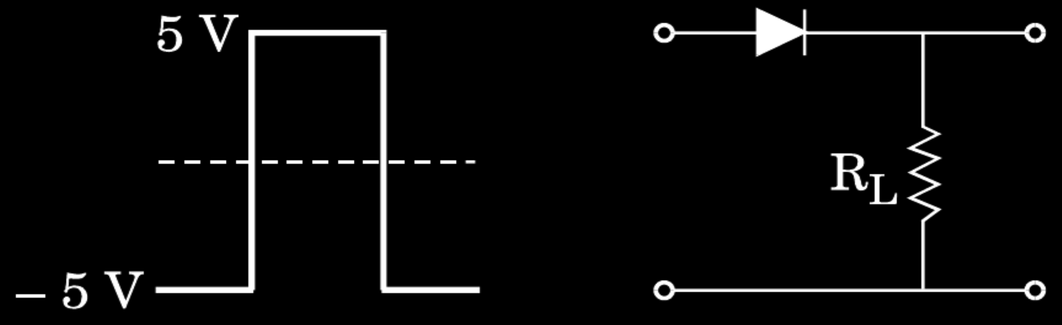
CBSE 2020

2

The small region in the vicinity of the junction which is depleted of free charge carrier and has only immobile ions is called depletion region/ layer.	½
The accumulation of negative charges in p - region and positive charges in n- region set up a potential difference across the junction, which acts as a barrier and is called barrier potential.	½
In forward bias (a) width of depletion layer decreases	½
(b) value of potential decreases	½

Draw the output signal in a p-n junction diode when a square input signal of 10 V as shown in the figure is applied across it.

CBSE 2019



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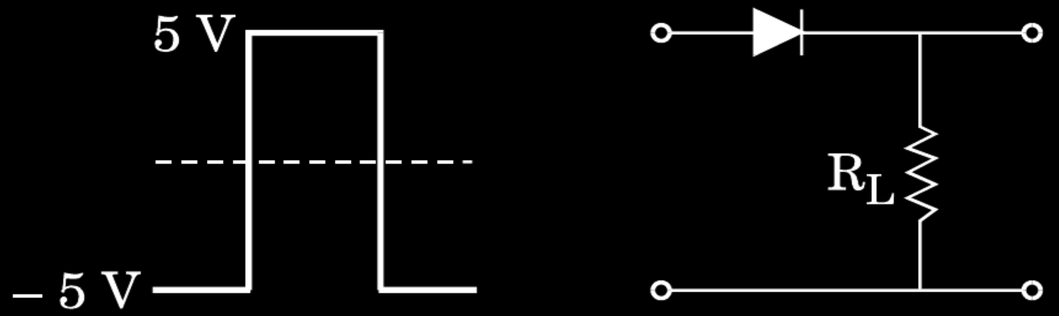
CBSE 2019



Draw the energy band diagram of (i) n-type, and (ii) p-type semiconductors at temperature T > 0 K.

In the case of n-type Si-semiconductor, the donor energy level is slightly below the bottom of conduction band whereas in p-type semiconductor, the acceptor energy level is slightly above the top of valence band. Explain, giving examples, what role do these energy levels play in conduction and valence bands.

CBSE 2019

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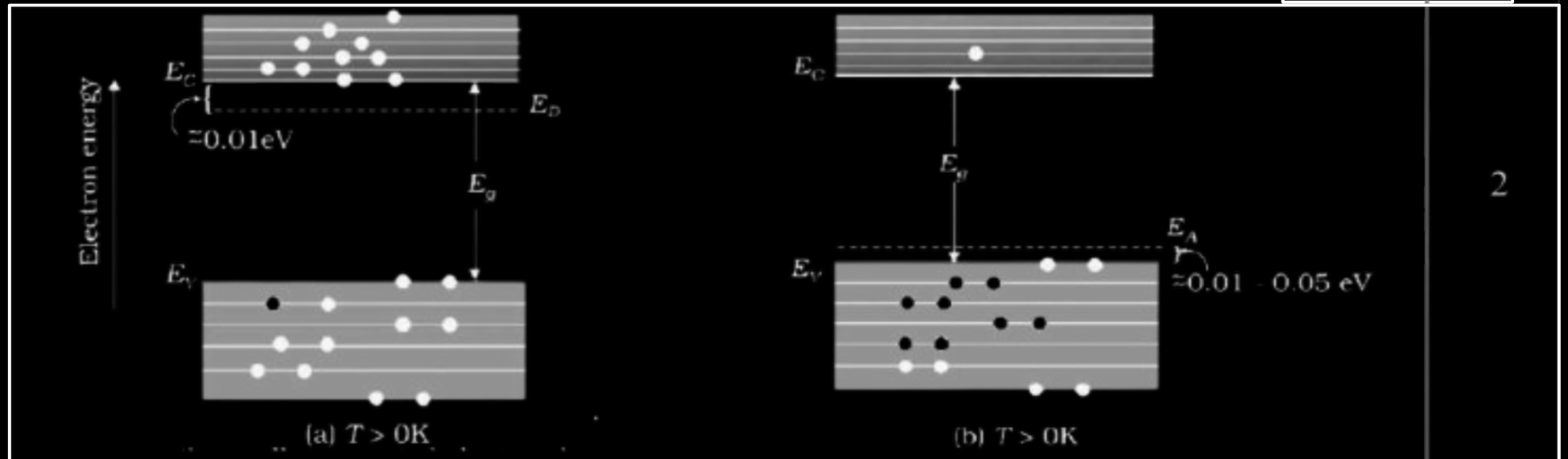
CBSE 2019



2

The donor energy level decreases the energy gap between conduction band and valence band. As a result the conduction band will get more electrons from the donor impurity with very small supply of energy. Whereas in p type semiconductor the holes from acceptor level sinks down into valence band

½

½

(a) Explain how a potential barrier is developed in a p-n junction diode.

(b) Draw the circuit arrangement for studying the V-I characteristics of a p-n junction diode in reverse bias. Plot the V-I characteristics in this case.

CBSE 2019

(a) Explain how a potential barrier is developed in a p-n junction diode.

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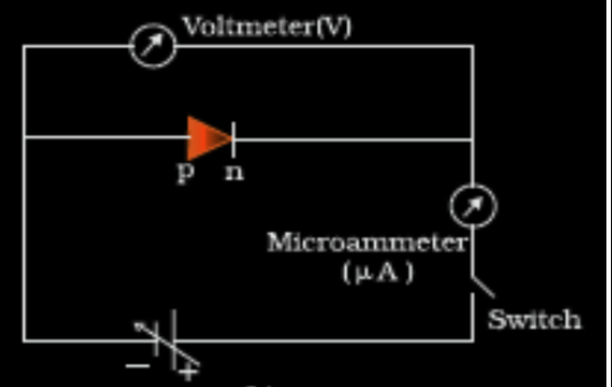
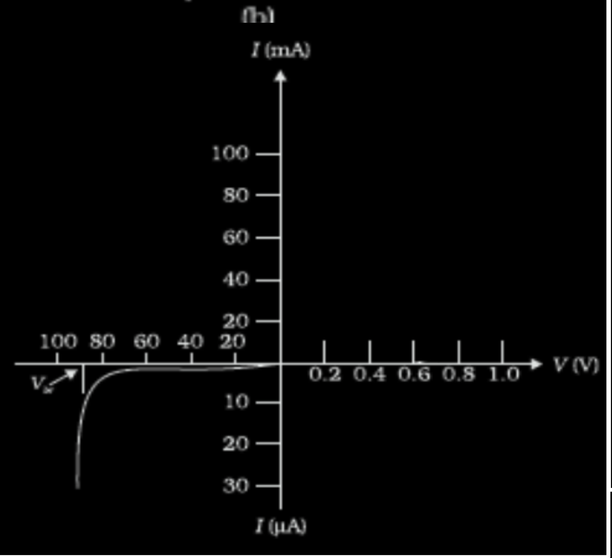
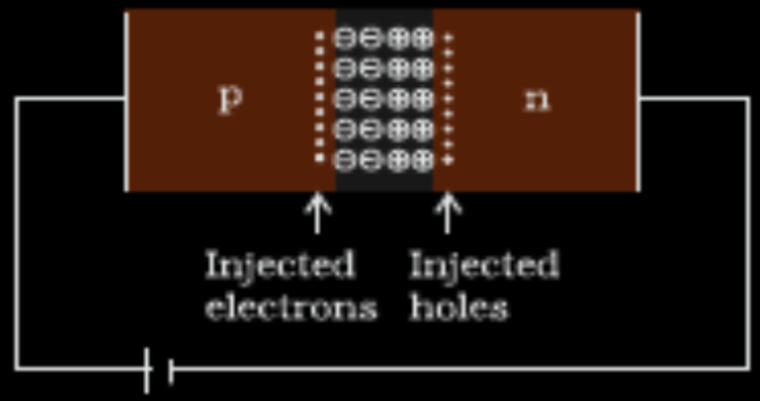
CBSE 2019

b.



c.





½

During the formation of p - n junction diode: due to the concentration gradient across p and n sides of a diode, holes diffuse from p side to n side and electrons diffuse from n side to p side giving rise to development of immobile positive charges on the n side and the negative charges on the p side across the junction. Thus a potential barrier is formed at the junction.

Alternatively: if a student explains with depletion region, award this 1 mark.

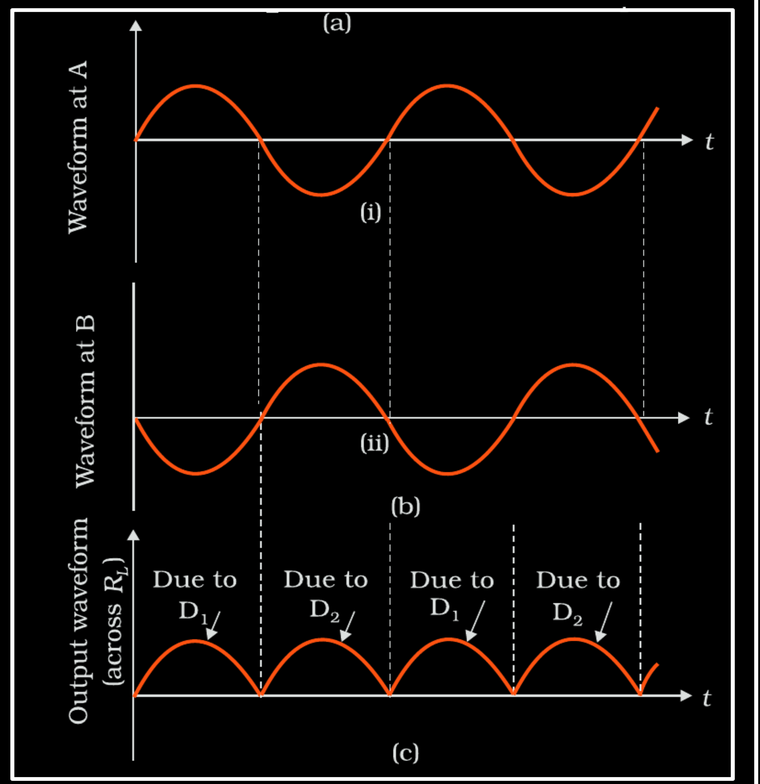
1

Draw the circuit diagram of a full wave rectifier and explain its working. Also, give the input and output waveforms.

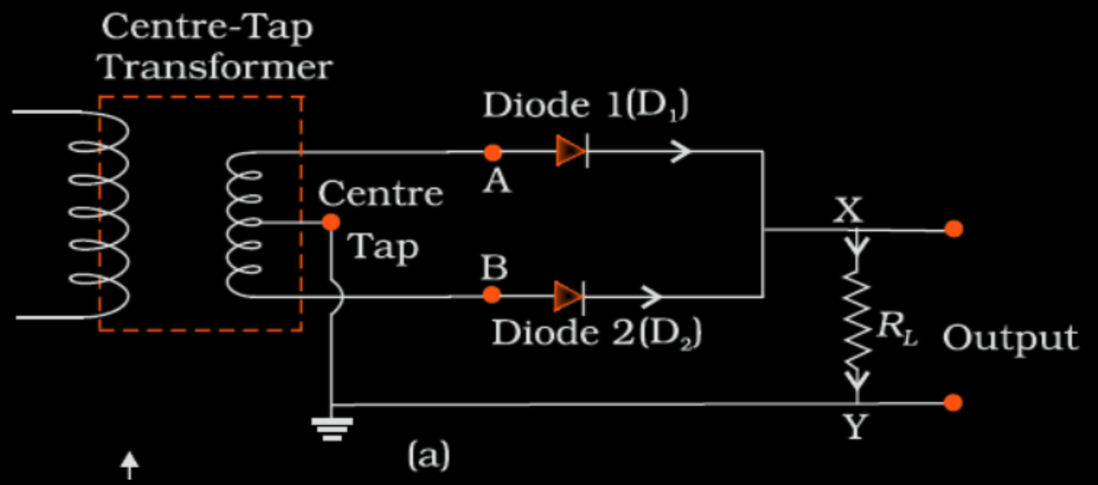
CBSE 2019

Draw the circuit diagram of a full wave rectifier and explain its working. Also, give the input and output waveforms.

CBSE 2019



The circuit diagram of a full wave rectifier is shown below.



Because of the center tap in the secondary of the transformer, diodes 1 and 2 get forward biased in successive halves of the input ac cycle. However the current through the load flows in the same direction in both the halves of he input ac cycle. We therefore, get a unidirectional (rectified) current through the load for the full cycle of the input ac.

1

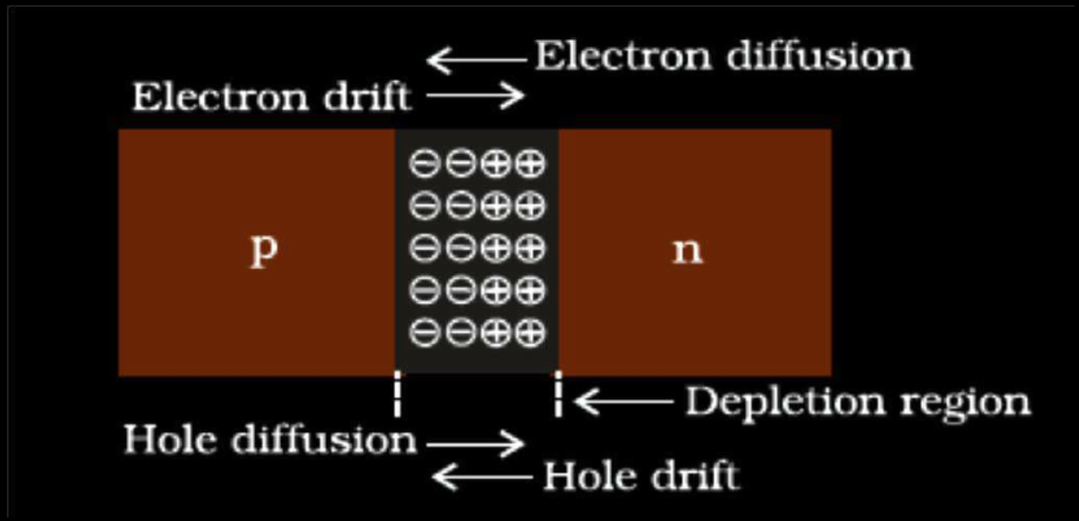
1

Write the two processes that take place in the formation of a p-n junction. Explain with the help of a diagram, the formation of depletion region and barrier potential in a p-n junction.

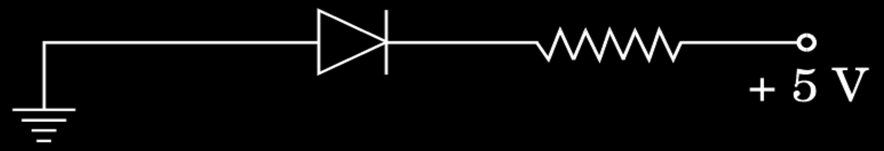
CBSE 2017

Write the two processes that take place in the formation of a p-n junction. Explain with the help of a diagram, the formation of depletion region and barrier potential in a p-n junction.

CBSE 2017

Diffusion	½
Drift	½
	1
Due to the diffusion of electrons and holes across the junction a region of (immobile) positive charge is created on the n-side and a region of (immobile) negative charge is created on the p-side, near the junction; this is called depletion region.	½
Barrier potential is formed due to loss of electrons from n-region and gain of electrons by p-region. Its polarity is such that it opposes the movement of charge carriers across the junction.	½

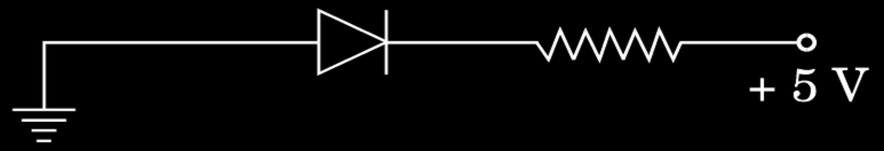
(a) In the following diagram, is the junction diode forward biased or reverse biased ?



CBSE 2017

(b) Draw the circuit diagram of a full wave rectifier and state how it works.

(a) In the following diagram, is the junction diode forward biased or reverse biased ?

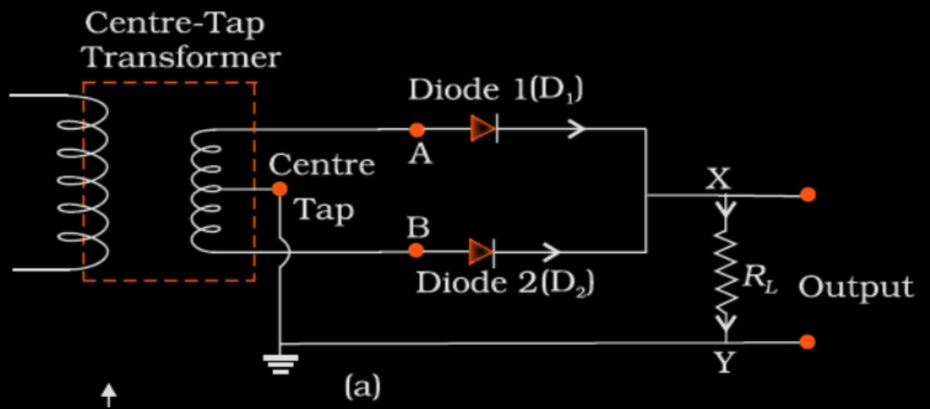


CBSE 2017

(b) Draw the circuit diagram of a full wave rectifier and state how it works.

a) Reverse Biased	1

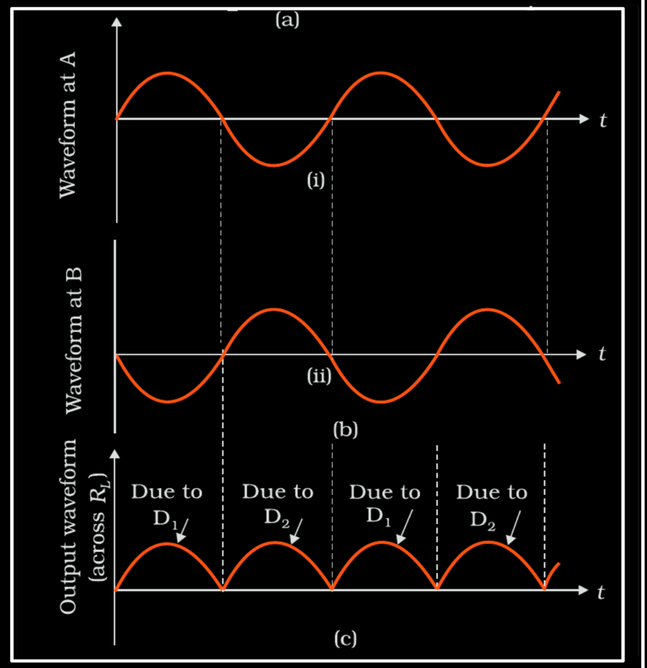
The circuit diagram of a full wave rectifier is shown below.



1

Because of the center tap in the secondary of the transformer, diodes 1 and 2 get forward biased in successive halves of the input ac cycle. However the current through the load flows in the same direction in both the halves of he input ac cycle. We therefore, get a unidirectional (rectified) current through the load for the full cycle of the input ac.

1



(i) Name two important processes that occur during the formation of a pn junction.

(ii) Draw the circuit diagram of a full wave rectifier along with the input and output waveforms. Briefly explain how the output voltage/current is unidirectional.

CBSE 2016

(i) Name two important processes that occur during the formation of a pn junction.

(ii) Draw the circuit diagram of a full wave rectifier along with the input and output waveforms. Briefly explain how the output voltage/current is unidirectional.

CBSE 2016

(i) Diffusion and Drift [Also accept if the student writes a) Appearance of a BARRIER POTENTIAL across the junction. b) Formation of a DEPLETION REGION on either side of the junction.]	½ + ½

Because of (i) the use of the centre tap transformer and (ii) the manner in which the load is connected, the voltage across/current through, the load has the same direction during both halfs of the input wave.	½

(i) Distinguish between a conductor and a semi conductor on the basis of energy band diagram.

CBSE 2016

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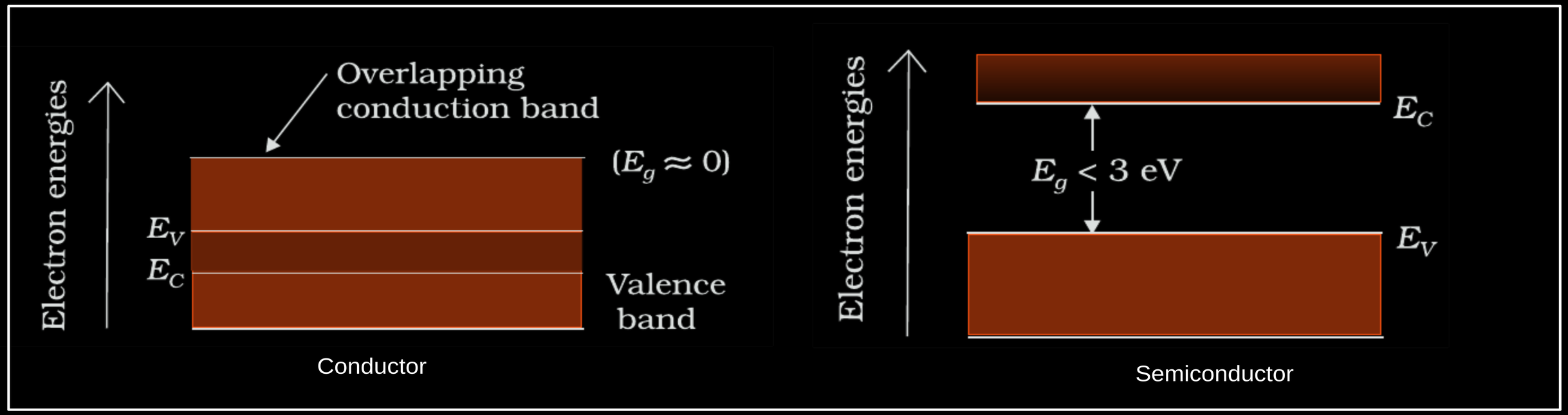
CBSE 2016



(i) Explain with the help of a diagram the formation of depletion region and barrier potential in a pn junction.

(ii) Draw the circuit diagram of a half wave rectifier and explain its working.

CBSE 2016

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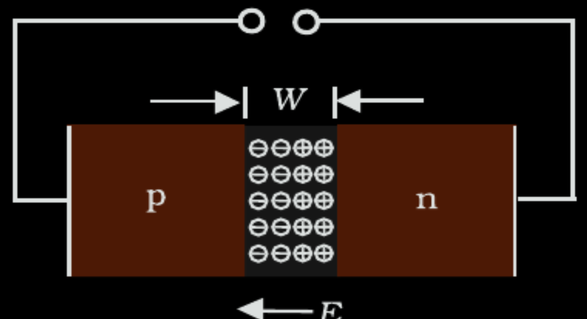
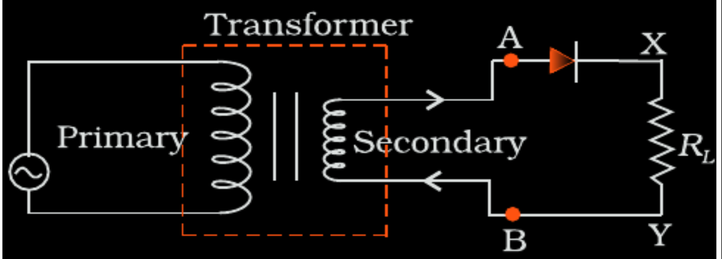
(ii) Draw the circuit diagram of a half wave rectifier and explain its working.

CBSE 2016



(a)



i.Due to diffusion and drift, the electrons and holes move across the junctions, creating a final stage in which a region is created across the junction wall, which gets devoid of the mobile charge carriers. This region is called depletion region; the potential difference across the region is called Barriers potential — ½+½

Working- If an alternating voltage is applied across a diode in series with a load, a pulsating voltage will appear across the load only during that half cycle of the ac input during which the diode is forward biased. — ½

Therefore, in the positive half – cycle of ac input there is a current through the load resistor R_L and we get an output voltage whereas half – cycle.There is no output during the negative half cycle. Thus, the output voltage is restricted to only one direction and is said to be rectified. — ½

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[Note-If the student draws only the input and output wave form, then award ½ marks only]